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Design, Validation and Testing of a thermal interface system for the NVIDIA Jetson Orin Nano in a CubeSat Payload Application

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Thesis for the attainment of the academic degree

Bachelor of Science (B.Sc.)

at the TUM School of Engineering and Design of the Technical University of Munich.

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Submitted:
Munich, 05.10.2026

I hereby declare that this thesis is entirely the result of my own work except where otherwise indicated. I have only used the resources given in the list of references.

Munich, 05.10.2026 Meriam Boussaadia



Zusammenfassung

EventSat ist eine Mission, die vom Lehrstuhl für Raumfahrttechnik der TUM entwickelt wurde. Ziel der Mission ist es, einen 6U-CubeSat mit einer Event-Kamera zu starten, die Objekte im Weltraum anhand der Lichtintensität jedes einzelnen Pixels in Echtzeit erkennt, charakterisiert und identifiziert, selbst unter extremen Umgebungsbedingungen. Dies unterscheidet sie von herkömmlichen Kameras, die Bilder sequenziell Bild für Bild aufnehmen [1].

Das thermische Management ist ein grundlegender Bestandteil des Satellitendesigns und der Validierung. Da die Leistungsfähigkeit der Nutzlasten von ihrer Betriebstemperatur abhängt, ist es eine wesentliche Anforderung, sicherzustellen, dass alle Komponenten innerhalb ihrer spezifizierten Temperaturbereiche bleiben.

Aus der offiziellen Dokumentation von NVIDIA zum thermischen Management geht hervor, dass der NVIDIA Jetson Orin Nano im Vakuum stark überhitzt und sich bei 104.5 $^{\circ}C$ abschaltet [36].

Darüber hinaus bestätigten Vakuumkammerversuche dieses Verhalten und zeigten eine Abschaltung innerhalb von 11 Minuten.

Um die Gerätesicherheit zu gewährleisten, andere Subsysteme vor hohen Temperaturen zu schützen und die Missionsziele zu erfüllen, legten wir für den ungünstigsten Fall fest, dass die Temperatur im stationären Zustand bei einer maximalen Wärmeleistung von 6.77 W 60 $^{\circ}C$ nicht überschreiten darf [37].

Zur Erfüllung dieser Anforderung entwickelten wir einen kundenspezifischen Wärmeverteiler, der die Wärme vom NVIDIA Jetson zur CubeSat-Struktur leitet.

Wir validierten das Modell mithilfe einer stationären thermischen Orbitalsimulation, bei der eine Temperatur von 40.56 $^{\circ}C$ zuzüglich einer Sicherheitsmarge von 5 $^{\circ}C$ ermittelt wurde, sowie durch Vibrationssimulationen.

Darüber hinaus führten wir stationäre und transiente thermische ANSYS-Simulationen in einer Vakuumkammer für drei Konfigurationen durch.

In der ersten Konfiguration betrachteten wir den NVIDIA Jetson und den Wärmeverteiler. In der zweiten Konfiguration ergänzten wir einen Wärmeleitpfad zu einer 6U-CubeSat-Struktur.

In der dritten Konfiguration wurden mehrere Wärmeleitpfade hinzugefügt, indem zwei zu Testzwecken verwendete 3U-CubeSat-Strukturen, vier Stäbe und eine Jetson-Platte berücksichtigt wurden, welche die Unterbaugruppe aus NVIDIA Jetson und Wärmeverteiler trägt. Dabei erreichte der SoC0 im stationären Zustand eine Temperatur von 61.08 $^{\circ}C$, während die Temperatur der Stäbe zwischen 42.78 $^{\circ}C$ und 57.59 $^{\circ}C$ lag. Nach 39807 Sekunden erreichten der SoC0, die CubeSat-Struktur und die Jetson-Platte Temperaturen von 60.83 $^{\circ}C$, 47.25–57.46 $^{\circ}C$ beziehungsweise 57.86–58.09 $^{\circ}C$. Gleichzeitig variierte die Temperatur der Stäbe zwischen 42.52 $^{\circ}C$ und 57.34 $^{\circ}C$.

Wir überprüften das strukturelle und dynamische Verhalten mithilfe statischer, modaler, harmonischer und zufälliger Vibrationsanalysen anhand von fünf verschiedenen geometrischen Konfigurationen. Dabei berücksichtigten wir die Startlastanforderungen von VEGA-C/SSMS in Abhängigkeit von der vertikalen oder horizontalen Ausrichtung des CubeSats sowie der Position der Unterbaugruppe (Jetson-Platte, NVIDIA Jetson Orin Nano und Wärmeverteiler) bei $\pm Z$ sowie $\pm X$ oder $\pm Y$.

Die Materialien und Bauteilgeometrien des Wärmeverteilers und der Jetson-Platte wählten wir auf Grundlage der Simulationsergebnisse aus.

Über alle Konfigurationen hinweg ergab sich in der Modalanalyse eine minimale erste Eigenfrequenz von 244.48 Hz. In den statischen Strukturanalysen und den Zufallsvibrationsanalysen wurde ein minimaler Sicherheitsfaktor von 1.61 bezogen auf die Streckgrenze des Wärmeverteilers von 180 MPa ermittelt. Für die Jetson-Platte betrug der minimale Sicherheitsfaktor 1.58 bezogen auf eine Streckgrenze von 205 MPa.

Die Gesamtverformungen in der statischen Strukturanalyse und die gerichteten Verformungen bei 3σ in den Zufallsvibrationsanalysen waren sehr gering, mit Ausnahme einiger Werte von etwa 0.7 mm, die auf eine Gesamtbewegung der Struktur unter den aufgebrachten Lasten hindeuten.

Auch bei der harmonischen Antwort überschritten die maximalen Spannungen oberhalb der Resonanzfrequenzen die Streckgrenze keines der beiden Materialien, abgesehen von Werten, die wir als Artefakte nachweisen konnten. Die Verformungen waren ebenfalls sehr gering, mit Ausnahme jener, die wir nachweislich auf die aufgebrachte Basiserregung zurückführten.

Nach Abschluss der Analysen ließen wir die Komponenten in der Werkstatt der TUM fertigen und führten thermische Experimente in der Vakuumkammer von BACOeng durch.

Abstract

EventSat is a mission developed by the chair of Spacecraft Systems at TUM. It aims to launch a 6U CubeSat using an Event camera that detects, characterizes, and identifies Objects in space based on light intensity at every pixel in real time, even in extreme environments, unlike conventional cameras that capture images frame by frame [1].

Thermal management is a fundamental part of satellite design and validation. As the performance of payloads depends on their operating temperature, ensuring that all components remain within their specified temperature ranges is an essential requirement.

From the NVIDIA Jetson official thermal management documentation, it is stated that the NVIDIA Jetson Orin Nano overheats drastically in vacuum and shuts down at 104.5 $^{\circ}C$ [36].

Moreover, vacuum-chamber experiments confirmed this behavior and showed a shutdown within 11 minutes.

For device safety, to prevent other subsystems from being subjected to high temperatures, and to meet mission goals, we adopted a worst-case scenario of not exceeding 60 $^{\circ}C$ in steady-state while dissipating a maximum heat load of 6.77 W [37].

To fulfill this requirement, we designed a custom heat spreader to conduct heat from the NVIDIA Jetson to the CubeSat structure.

We validated the model through an orbital steady-state thermal simulation, obtaining a temperature of 40.56 $^{\circ}C$ + 5 $^{\circ}C$ safety margin, and vibrational simulations.

We also performed steady-state and transient thermal ANSYS simulations in a vacuum chamber for three configurations. In the first one, we considered the NVIDIA Jetson and the heat spreader and in the second one we added a conduction path to a 6U CubeSat structure. In the third configuration, after adding multiple conduction paths by considering two testing 3U CubeSat structures, four rods, and a Jetson plate supporting the subassembly NVIDIA Jetson–heat spreader, the SoC0 reached a temperature of 61.08 $^{\circ}C$ at steady-state, meanwhile, the temperature of the rods was in the range of 42.78 $^{\circ}C$ and 57.59 $^{\circ}C$. Within 39807 seconds, the SoC0, CubeSat structure, and the Jetson plate reached 60.83 $^{\circ}C$, 47.25–57.46 $^{\circ}C$, and 57.86–58.09 $^{\circ}C$, respectively. Meanwhile, the temperature of the rods varied between 42.52 $^{\circ}C$ and 57.34 $^{\circ}C$.

We verified structural and dynamic behavior through static, modal, harmonic, and random vibrational analyses, following 5 different geometrical configurations, meeting VEGA-C/SSMS launch load requirements, depending on the CubeSat orientation vertically or horizontally, and the location of the Subassembly (Jetson plate, NVIDIA Jetson Orin Nano, and the heat spreader) at $\pm Z$ and $\pm X$ or $\pm Y$.

We selected the heat spreader and Jetson plate materials and component geometries based on the simulation results.

Across the different configurations, we obtained a minimum first frequency of 244.48 Hz in modal analysis, and a minimum safety factor of 1.61 in the static structure and random vibration analyses, relative to the heat spreader's yield strength of 180 MPa. Meanwhile, the minimum safety factor for the Jetson plate was 1.58 relative to the 205 MPa yield strength. The total deformations in the static structure and 3-sigma directional deformations in random vibrations were very small, except for some values around 0.7 mm, indicating overall movement of the structure under the imposed loads. Also, in the harmonic response, the maximum over-frequency stresses did not exceed the yield strength of either material, except for values we proved were artifacts, and the deformations were very small except for the ones that we proved are associated with imposed base excitation. After obtaining the results, we manufactured the components at the TUM workshop and performed thermal experiments in the BACOeng vacuum chamber.

Acknowledgements

I would like to sincerely thank my supervisor, Vincenzo Messina, for his continuous support, guidance, and constructive feedback throughout this project. His technical expertise, encouragement, positive energy, and availability were instrumental in shaping the direction of this work and overcoming challenges encountered.

I am particularly very grateful for his mentorship and for the valuable learning experience he provided throughout this project.

I would also like to extend my sincere thanks to Jaspar Sindermann and Dina Sbouii for their guidance and support throughout the experiments conducted during this project, and to the TUM Workshop team, whose expertise and assistance contributed significantly to the manufacturing process of the heat spreader, Jetson plate, and the modification of the 3U CubeSat structure.

Working on this project has been a very positive and rewarding experience for me. This project taught me a lot and boosted my interest and enthusiasm for Spacecraft systems, namely satellites.

I would also like to acknowledge the student resources and licenses that supported this project through the university infrastructure, namely MATLAB, ANSYS, SolidWorks, and Microsoft Excel, as well as open-source resources like NASA's GMAT and ARDUINO.

Finally, I would like to mention the use of Claude, Gemini, and ChatGPT for debugging MATLAB codes, that I used in this thesis, as well as Grammarly, which helped me to improve the writing quality.

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Glossary

ABEX Alabama Burst Energetics Explorer. xvi, 6

BMS Battery management system. xvi, 5

CAD Computer-aided-design. xvi, 13, 18, 19, 39, 40, 42, 44

CCD Charge-Coupled Device. xvi, 5

CLK Clock. xvi

CM5 Compute Module 5. xvi, 5

CMB cosmic microwave background. xvi, 9

CMOS complementary metal- oxide- semiconductor. xvi, 16

CNC Computer numerical control. ix, xvi, 2, 13, 16, 32, 39, 42

COTS Commercial Off-The-Shelf. xvi, 1, 4, 5, 7, 8, 72

CPU Central Processing Unit. xvi, 1, 16, 20, 38

CS Chip Select signal. xvi

DLR German aerospace cenetr. xvi, 4

DO Data Output. xvi

DVFS Dynamic Voltage and Frequency Scaling. xvi, 39, 41

ECSS European Cooperation for Space Standardization. xvi, 31

FEM Finite element method. xvi, 1, 6

FPGA Field-programmable gate array. xvi, 5

GMAT General mission analysis tool. vi, xvi, 1, 2, 8–10

GND Ground. xvi

GPU graphics processing unit. xvi, 1, 16, 38

GRMS Root mean square acceleration. xvi, 30

GUI Graphical User Interface. xvi, 21, 23

IGBT insulated gate bipolar transistor. xvi, 6

IR Infrared radiative. xvi, 10

JGM-2 Joint gravity model 2. xvi

LEO Low earth orbit. xvi, 4, 6

MATLAB Matrix laboratory. vi, xvi, 1, 2, 6, 8–10, 33, 38–42, 44

MIST Miniature student satellite. xvi, 7

MPFL mechanically pumped fluid loop. xvi, 11

NASA-STD NASA Technical Standard. xvi, 54

ODE Ordinary differential equation. xvi, 5

PCB Printed Circuit Board. xv, xvi, 7, 14, 16, 19, 23, 25, 35, 46, 47, 69, 71

PDE partial differential equations. xvi, 6

Peek polyether Ether ketone. xvi, 5

PLA Polylactic Acid. xvi, 14, 19, 22, 32

PSD Power spectral density. xvi, 6, 55, 56

PSD-G Power Spectral Density in Gravitational Acceleration. xvi, 30, 48, 55–57

PU Polyurethane. xvi, 14, 15, 19, 21, 23, 39

RAAN Right Ascending node. xvi, 10

RC Resistance capacitance. xvi, 6

sat satellite. xvi

SCK Serial Clock. xvi

SO Serial Output signal used in SPI communication, corresponding to the slave-to-master data line, equivalent to the MISO signal in modern SPI notation. xvi

SoC System on Chip. iii, v, ix, xv, xvi, 5, 14, 16, 17, 20, 34–44

SPENVIS Space Environmental Information System. xvi, 6

SSH Secure shell. xvi, 21–23

SSMS Small Spacecraft mission service. iii, v, xvi, 2, 6–8, 13, 29–31, 48, 49, 58, 72

SSO Sun- Synchronous orbit. xvi, 10

STK Systems Tool Kit. xvi, 6

SysML System modeling language. xvi, 6

TIM Thermal interface material. xvi, 14, 18, 19, 21, 22, 24, 34, 37, 39, 41, 71

TJ Junction Temperature. xvi

TUM Technical university Munich. iii–v, xvi, 1, 2, 13

TVAC thermal vacuum. xvi, 5

VCC Voltage at the common collector. xvi

VDD Positive power supply voltage. xvi, 38, 40, 42

VEGA Vettore Europeo di generazione Avanzata. iii, v, xvi, 2, 6, 8, 13, 29–31, 48, 72

1 Introduction

EventSat is a 6U CubeSat mission developed by the Chair of Spacecraft Systems at TUM.

A CubeSat is a class of Nanosatellites. Its baseline is a 1U CubeSat, which is 10 x 10 x 10 cm^3. CubeSats allow deployment of commercial off-the-shelf (COTS) units [89] and are compatible with standardized launch containers, which reduce development costs and accelerate deployment timelines.

This mission uses an Event camera to track stars and space debris. This Event camera is connected to a processing computer, namely the NVIDIA Jetson Orin Nano. Using its central processing unit (CPU) and graphics processing unit (GPU), the NVIDIA Jetson performs clustering and data detection on incoming information from the Event Camera.

To introduce the NVIDIA Jetson Orin Nano for this mission, we needed to resolve several points to ensure efficient operation.

The NVIDIA Jetson tended to overheat quickly, leading to a software shutdown at 104.5 $^{\circ}C$ within 11 minutes in the BACOeng rough vacuum chamber. To address this limitation, we implemented a passive Cu 2.0060 heat spreader that conducts heat from the NVIDIA Jetson Orin Nano chip to the CubeSat structure. To validate the solution, we performed thermal and vibrational simulations.

Thermal analysis is an essential factor in nanosatellite failure. To prevent this, we used GMAT to propagate the orbit and MATLAB to compute the worst-case sink temperature, which we then used as a boundary condition in ANSYS Mechanical. In addition to this lumped-parameter model, we implemented a 3D finite-element thermal model FEM in ANSYS Mechanical, including steady-state analyses for orbital conditions. This simulation ensures that target hardware remains within its operational limits. We also performed vacuum-chamber simulations, namely steady-state and transient thermal analyses, to compare experimental results with simulation results and assess the error between them.

Nevertheless, thermal analysis alone is not sufficient, as vibrational simulations are also important before manufacturing the piece.

In fact, before reaching the harsh space environment, satellites are initially subjected to intense launch conditions, namely static accelerations and dynamic random vibrations.

To ensure structural integrity, we performed static structural and random vibration analyses. To prevent resonance failure, we performed a modal analysis. Finally, to check for sinusoidal excitation, we evaluated the harmonic response.

1.1 Motivation

To enable the use of new space technology, namely the Event Camera, which records changes in light intensity at every pixel in real time, we introduced the EventSat mission, which relies on the onboard processing unit NVIDIA Jetson Orin Nano.

For a successful EventSat mission, we use the NVIDIA Jetson Orin Nano to process data from an event-based camera, improving space situational awareness, as well as space surveillance and tracking.

Nevertheless, after vacuum testing, the NVIDIA Jetson tended to shut down after 11 minutes. This duration is insufficient for efficient event-camera operation, leading to limited data capture, information loss, and eventually mission failure.

This high temperature does not only affect the Event camera or the module itself alone, nevertheless, it affects the neighboring subsystems that could overheat and violate their operating temperature limits after being subjected to this heat load imposed by the NVIDIA Jetson. To avoid this behavior in real-world applications, we set a worst-case operating temperature limit of 60 $^{\circ}C$ for our module to ensure its safety, prevent other subsystems from overheating, and meet mission requirements.

This thesis investigates the feasibility of integrating the NVIDIA Jetson Orin Nano in the EventSat mission.

To do this, we must implement a thermal solution that survives launch vibrations, considering that active cooling is impractical for CubeSats.

In space, convection is prohibited. As a result, heat generated by the NVIDIA Jetson Orin Nano can leave only via conduction and radiation.

To address the shutdown behavior at 104.5 $^{\circ}C$ and keep the NVIDIA Jetson below 60 $^{\circ}C$, we implemented a passive-cooling copper heat spreader that must also withstand vibrations.

These requirements lead us eventually to the pivotal question of this thesis: whether the NVIDIA Jetson Orin Nano remains below the operational temperature limit of 60 $^{\circ}C$ after adding the passive thermal solution, which must also survive the launch loads.

1.2 Work approach

To answer our problem statement question, we followed the following work approach.

In the orbital simulation (section 4.1), we defined the NVIDIA Jetson Orin Nano constraints and described how we retrieved the sink temperature using GMAT and MATLAB. This gave us the allowable thermal resistance for our thermal problem.

After framing the requirements of our thermal solution, the design and the material must respect those constraints. For this reason, in section "Copper heat spreader design" (section 3.2) we compared whether Cu 2.0060 or A5083 is more suitable for our design due to material limitations at the TUM workshop. This comparison was based on orbital thermal simulation results (section 4.1). We then confirmed our choice by supporting the thermal results with vibrational analysis results (section 4.4) and CNC-manufactured the passive heat spreader. Also, after a qualitative mechanical comparison between A5083 and Cu 2.0060, we chose to manufacture an A5083 Jetson plate, which acts as a support for the NVIDIA Jetson and heat spreader subassembly.

To assess the behavior of the NVIDIA Jetson Orin Nano in the vacuum chamber, we performed experiments under three configurations (A, B, and C), each with different components (section 3.4). We also supported these experiments with thermal simulations (section 3.3.2) to assess the deviation between our simulation results and real-life experimental results (section 4.3).

In vibrational analysis, we considered five different configurations. In each configuration, we performed four types of simulations, namely static structure with altering the launch direction each time, modal, random vibrations, and harmonic response following SSMS VEGA-C user manual.

In this thesis, orbital thermal simulations showed that the NVIDIA Jetson Orin Nano remained within the defined thermal requirements after adding the passive heat spreader. This result allowed us to eliminate one of the materials for the heat spreader which led to thermal requirements violation, and vibrational analysis results confirmed our choice. To hold the NVIDIA Jetson-heat spreader assembly in the CubeSat structure, we made a Jetson plate. Since the Von Mises stresses did not exceed the yield strength of the heat spreader and Jetson plate materials, and deformations were very small, except for those reflecting CubeSat structure movement under the applied loads, we confirmed our material choice for both components.

Regarding the vacuum chamber simulations, adding multiple conduction paths, namely the 3U testing CubeSat structure (made for vibrational analysis), the rods, and the Jetson plate, significantly decreased the equilibrium temperature.

As relevant limitations, we could not perform the vibrational tests in Oberpfaffenhofen due to operational problems in the shaker facility, and we performed the thermal testing in rough vacuum conditions in the BACOeng vacuum chamber. Also, due to poor CNC surface finish, we picked effective emissivity values for the heat spreader and the Jetson plate in our thermal simulations.

1.3 Thesis structure

To support the thesis objective, we divided the thesis into state of the art, methodology, results discussion, and conclusion sections.

In the state of the art (section 2), we covered the background of this research and the resources that showed how these problems were previously solved.

In the methodology part (section 3), we highlighted the problem-solving processes and the steps followed to validate the thermal solution, including orbital simulations, heat spreader design and material selection, ANSYS thermal simulations inside the vacuum chamber and in orbital conditions, experiments inside the BACOeng vacuum chamber, and vibrational simulations, including static structure, modal, random vibrations, and harmonic response.

In the results section (section 4), we presented the simulation and experimental results, discussed them, and compared the simulated thermal results with the experimental ones.

In the conclusion part (section 5), we summarized the work done throughout this thesis, addressing the limitations and the future work in the context of the EventSat mission.

2 State of the art

Because of their short mission lifespan, CubeSats provide inexpensive access to space compared to traditional non-CubeSats. They benefit from rapid development cycles, typically 1–2 years to become fully operational.

They rely primarily on commercial-off-the-shelf (COTS) components, can reach orbit as secondary payloads on existing rockets, and can be developed within universities [59].

These factors support a cost-efficient implementation. COTS have some strict requirements to be mission-efficient. In our case, deploying the NVIDIA Jetson Orin Nano requires a thermal management system.

The following sections cover thermal management operation inside CubeSats, the new trend of using commercial-off-the-shelf (COTS) on board, numerical modeling and orbital thermal analysis, and vibrational qualification.

2.1 Thermal management inside CubeSats

Thermal management is a fundamental requirement in spacecraft to keep payloads within their operational limits and fulfill their mission. In orbit, thermal management consists of 2 methods: active and/or passive.

Passive methods control spacecraft temperature without relying on electrical power (e.g., multilayer insulation, heat spreaders, etc.) and suit small-area deployments, especially CubeSats.

In contrast, active methods are precise and accurate (e.g., thermoelectric coolers, resistance heaters, cryocoolers, etc.), rely on electrical power, and are usually used for missions that require more rigorous temperature control than a computing payload typically needs [71] [72].

Convection transfers energy through fluid motion. Because the space environment lacks this medium, heat transfer occurs by conduction to an adjacent structural sink through physical contact and by radiative exchange to other surfaces and/or deep space.

The governing equations for these mechanisms are (equations 2.1 2.2) [55]:

$$Q_{\text{cond}} = -k\, A_1 \left(\frac{\Delta T}{\Delta x}\right) \tag{2.1}$$

$$Q_{\text{rad}} = \varepsilon\, \sigma\, A_2 \left(T_s^4 - T_{\text{sink}}^4\right) \tag{2.2}$$

with:

- $\frac{\Delta T}{\Delta x}$: Temperature gradient
- A_1: Cross-sectional area normal to the heat flow
- A_2: Radiating area
- $\sigma = 5.670 \times 10^{-8}\ \frac{\text{W}}{\text{m}^2 \cdot \text{K}^4}$: Stefan–Boltzmann constant
- ε: Emissivity

In the German Aerospace Center (DLR), some researchers developed a hybrid passive and active thermal management system for a COTS-based lithium-ion modular battery system intended to operate in a Satellite mission in low Earth orbit (LEO).

In the charging state, this electrical power subsystem has an operating temperature window between 0 $°C$ and 40/45 $°C$. During discharge, the temperature must remain between -20 $°C$ and 60 $°C$.

Passively, they used Aluminum thermal wedges and structural end plates to conduct heat from the cells, with polyether Ether ketone (Peek) holders providing electrical isolation.

Actively, they applied four integral heaters controlled by the battery management system (BMS) to keep cells above the operational limit of 0 $°C$, during cold conditions [42].

In another study for the "DARKNESS" mission, which goal was deploying a skipper-charge-coupled device (CCD) CubeSat observatory to search for dark matter (DM) from low earth orbit, researchers applied a hybrid active/ passive Ricor K508N cryocooler thermal system to keep skipper-(CCDs) at -103 $°C$ to minimize thermal dark current and maintain high "readout precision" under internal and external heat loads. Actively, it cooled the detectors. Whereas passively, copper brackets and straps transferred heat from the device to three external radiator panels, radiating it into space [4].

For COTS specifically, some researchers at the University of Porto in Portugal proved that a 25 µm flexible pyrolytic graphite passive thermal strap, which connected a power amplifier to its aluminum casing heat sink to conduct heat away, is a low-cost thermal control solution for COTS power amplifiers and field-programmable gate arrays (FPGAs) in a CubeSat mission by studying its thermal management in ANSYS Icepak. The results showed a decrease in temperature from 131 $°C$ to 103 $°C$ [50].

Spacecraft thermal design relies essentially on passive thermal management methods. Nevertheless, active methods are used to complement the passive ones [72].

2.2 Commercial Off-The-Shelf (COTS) Computing hardware in small satellites, thermal and vibrational qualification challenges

The use of commercial off-the-shelf (COTS) computing modules, such as the NVIDIA Jetson Orin Nano, in CubeSat payloads demonstrates a growing trend toward onboard data processing, known as “edge computing”, in small satellite missions. This shift is driven by the need to reduce downlink bandwidth demands and enable quicker responses to observed events [68] [74]. However, COTS are made and tested for terrestrial applications [71].

Ground-based thermal vacuum (TVAC) experiments on a Raspberry Pi Compute Module 5 (CM5) performing real-time 1080p video encoding show that the system-on-chip (SoC) temperature stabilized 7–8 $°C$ above ambient below 60 $°C$. However, ambient temperatures exceeding 60 $°C$ caused the temperature difference to widen to 10–11 $°C$, pushing the SoC close to the 80 $°C$ thermal-throttling limit and raising power from 1.38 W to 1.56 W [20].

SONATE-2, a 6U LEO CubeSat mission, team tested NVIDIA Jetson Xavier NX in vacuum and performed radiation and vibrational tests. Results did not indicate the presence of any damage [54].

NVIDIA stated that "the customer is responsible for designing and implementing a system-level thermal solution to keep the module’s components within specified temperature limits" [37].

As a limitation of the previous literature, many researchers have studied thermal management for different COTS systems, but no literature addresses thermal management of the NVIDIA Jetson Orin Nano in vacuum.

2.3 Thermal modeling approaches in spacecraft

To implement a thermal model, lumped-parameter and finite-element models solve the heat equation at different fidelity levels. A lumped-parameter model solves ordinary differential equations (ODEs) for a few discrete isothermal nodes connected by conduction and radiation (equations 2.3 and 2.4), which is suitable for early-stage design.

$$Q_{\text{cond}} = -k\, A_1 \left(\frac{\Delta T}{\Delta x} \right) \tag{2.3}$$

$$Q_{\text{rad}} = \varepsilon \, \sigma \, A_2 \left(T_s^4 - T_{\text{sink}}^4 \right) \tag{2.4}$$

In contrast, the finite element method solves the partial differential equations (PDEs) over thousands of elements and solves the heat equation spatially.

Alabama Burst Energetics Explorer (ABEX) is a 12U CubeSat designed to observe Gamma-Ray bursts in low Earth orbit (LEO). It uses a model-based system integrating SysML, STK, SPENVIS, MATLAB, Simulink, and Thermal Desktop to assess thermal control requirements. SysML provides centralized spacecraft parameters, while STK and SPENVIS generate orbital and radiation data. MATLAB calculates environmental heat loads and energy balance to estimate heater power and radiator area requirements. Simulink uses a lumped-parameter thermal network to determine a suitable operating envelope. Then Thermal Desktop performs a 3D finite element method (FEM) analysis to evaluate transient temperature distributions and thermal behavior at mounting interfaces [51].

To perform an orbital simulation, some parameters (e.g., orbit characteristics, optical properties, mission timeline,...) must be specified to solve the power balance around a spacecraft [39].

Designing a high-power insulated-gate bipolar transistor (IGBT) module requires thermal simulations. Researchers performed a finite-element method by dividing the IGBT into four sections and running a transient step-response finite-element method (FEM) simulation across a matrix of operating boundary conditions to obtain thermal impedance curves. Then they curve-fitted these results mathematically into single-layer Foster resistance-capacitance (RC) exponential equations to establish "lookup functions" for how thermal resistance (R_{th}) and capacitance (C_{th}) dynamically respond to boundary variations [24].

2.4 Vibrational qualification

During launch, the satellite is subjected to quasi- static accelerations and dynamic loads, including random and sinusoidal loads vibrations.

SSMS VEGA-C user manual frames these loads and requirements for successful vibrational analysis. A typical requirement for modal analysis is to have a first natural frequency above a specific frequency specified in the manual based on the case (Mini S/C, Micro S/C, Nano S/C, etc...) [21].

Structural qualification is based on 4 types of analyses:

- Static structure: It evaluates the Von Mises stresses and total deformations under quasi-static launch accelerations [16].
- Modal: This analysis gives the natural frequencies, mode shapes, participation factor calculations, and effective mass excited in each mode. A pre-stressed modal analysis can be performed by linking it to the static structure one [17].
- Random vibrations: This analysis relies on applying power spectral density (PSD). From this, 1-sigma and/or 3-sigma directional deformations and Von Mises stresses can be retrieved [19].
- Harmonic response: it reflects the response to sinusoidal excitation over a frequency range and allows dynamic amplification at resonance to be checked [18].

In a CubeSat university project, researchers reviewed previous literature to assess whether the CubeSat's Aluminum structure meets the stiffness requirements set by Soyuz, Ariane 5, and the P-Pod deployer, and performed vibrational analysis, including static structure, modal analysis, and random vibrations. In the static structure analysis, they imposed a 10 g launch acceleration. Then they performed a non-pre-stressed modal analysis to identify natural frequencies and mode shapes. In other words, they did not link the modal analysis to the previously performed static structure simulation. Finally, they performed a random vibration analysis, applying a frequency-domain power spectral density PSD so that the stress peaked at the fundamental natural frequencies, and a cumulative fatigue analysis by extracting stress peaks and evaluating them using Miner's cumulative damage rule and material S-N curves [85].

In another project, the researchers performed only modal analysis without external loads. Then, they analyzed random vibrations and harmonic response separately. They performed a harmonic response analysis to validate the modal analysis, noting that other literature performed modal analysis without harmonic response and that it is "quintessential" to include it. Then, they checked for peak deformation amplitudes and stresses at resonance [26].

In Finite Element Analysis of Stresses in the Miniature student satellite (MIST) CubeSat due to Dynamic Loads During Launch by CARL JÄRMYR ERIKSSON, the sine vibration analysis was not considered since the minimum natural frequency exceeded 100Hz [44]. Nevertheless, this condition was not specified in the SSMS launch manual.

2.5 Summary

Implementing COTS, namely the NVIDIA Jetson in CubeSats, ensures fast and cost-efficient space access.

Nevertheless, their use in CubeSat applications is limited by thermal constraints, which dictate the maximum temperature range that should never be exceeded to ensure effective operation. On their small printed circuit Board (PCB), they tended to overheat, leading to shutdown at a hardware-defined temperature.

Previous literature demonstrates how active and passive thermal management systems could help to maintain the target devices within their operational limits.

In the literature we reviewed, thermal control relies heavily on passive approaches, as is evident in hybrid methods, where active methods complement passive ones.

Vibrational analysis does not follow a single path. Instead, it is performed differently. To validate the tested components structurally, researchers performed static, modal, random vibration, and harmonic response analyses, combining them, performing some analyses alone, or sometimes abandoning one or more of them.

Also, some researchers tested COTS in a space environment without implementing a thermal management system for the NVIDIA Jetson Orin Nano specifically, and some proved that the NVIDIA Jetson survived vibrations without any damage.

3 Methodology

The objective of this thesis is to design a passive heat spreader whose role is to extract heat from the NVIDIA Jetson and conduct it to the CubeSat structure in a way that the module temperature stays within the operating limits and hence prevents an unwanted shutdown, which would lead to an interruption in the on-Board computing during operation and to data loss. The heat spreader should keep the NVIDIA Jetson below 60 $^{\circ}C$, which provides a large margin below the documented software shutdown temperature of 104.5 $^{\circ}C$. To meet the heat-spreader design goal, we performed several complementary simulations. First, we ran an orbital thermal simulation using GMAT and MATLAB to extract the worst-case sink temperature, which we used as a boundary condition in ANSYS thermal simulations to check the steady-state Jetson temperature and whether it stays within operating limits.

Secondly, since an orbital testing experiment was not possible, the design was verified within the BA-COeng vacuum chamber considering three different configurations (sections 3.4 and 4.3):

- Configuration A: NVIDIA Jetson Orin Nano 8GB + heat spreader
- Configuration B: Configuration A + aluminum structure
- Configuration C: NVIDIA Jetson Orin Nano/NX + heat spreader + Including 2 aluminum structures + Jetson support plate + 4 rods

The experimental results were later compared with corresponding ANSYS transient thermal simulations to evaluate the accuracy of the numerical model and its deviation from the experimental results.

Third, we performed a vibrational qualification analysis to validate the heat spreader and Jetson plate (support) across five possible configurations, since the launch direction is still undefined. In fact, we conducted four coupled ANSYS mechanical analyses: static structure, modal, random vibrations, and harmonic response.

They were superimposed as follows:

Static structure –> Modal –> Random vibrations

Static structure –> Modal –> Harmonic response

We extracted the data used in vibrational analysis from the SSMS VEGA-C User's manual.

Note: The SSMS VEGA-C manual does not require applying a pre-stressed modal analysis. However, in this thesis, we adopted this approach to account for stiffness changes due to the applied static load.

3.1 Orbital simulation

In this section, we demonstrated which parameters we needed to perform the orbital simulation and which requirements constrained our thermal solution.

3.1.1 Jetson operating limits and heat load

In order to function properly, all COTS have an operating margin that should not be violated, which is the case for our NVIDIA Jetson Orin Nano (Table 3.1).

The thermal requirements stipulate that the NVIDIA Jetson Orin Nano operates at a maximum of 60 $^{\circ}C$ in steady-state worst-case conditions to ensure reliable performance while dissipating a maximum of 6.77 W [37], a 104.5 $^{\circ}C$ software thermal shutdown, and a 105 $^{\circ}C$ hardware thermal shutdown limits at which the module's thermal protection forces throttling leading to an immediate power-down to avoid damage [76]. These requirements drive the design of our passive heat spreader.

Table 3.1 Thermal constraints and power of the NVIDIA Jetson Orin Nano

Parameter	Numerical value	unit
Heat flow	6.77	W
Software thermal Shutdown	104.5	$^{\circ}C$
Hardware thermal Shutdown	105	$^{\circ}C$
Operating temperature	60	$^{\circ}C$

By keeping the chip temperature below 60 $^{\circ}C$, we ensure enough buffer against the 104.5 $^{\circ}C$ limit and prevent the other subsystems from being subjected to high temperatures.

3.1.2 Sink temperature derivation

As stated in "Schmid" thesis [88] to obtain the equilibrium sink temperature, we performed a radiative energy balance around the satellite (equation 3.1):

$$Q_{\text{in}} = Q_{\text{out}} \tag{3.1}$$

where:

$$Q_{\text{in}} = Q_{\text{albedo}} + Q_{\text{IR,earth}} + Q_{\text{CMB}} + Q_{\text{solar}} \tag{3.2}$$

$$Q_{\text{albedo}} = S_0 \, \alpha_{\text{albedo}} \, F_{\text{sat}\to\text{earth}} \, A_{\text{coms}} \, \varepsilon_{\text{coms}} \tag{3.3}$$

$$Q_{\text{IR,earth}} = F_{\text{face,nadir}\to\text{earth}} \, \sigma \, T_{\text{earth}}^4 \, A_{\text{earth}} \, \varepsilon_{\text{earth}} \, \cos\theta_{\text{earth}} \tag{3.4}$$

with A_{coms} being the surface area of the top face and the view factor:

$$F_{\text{sat}\to\text{earth}} = \left(\frac{r_{\text{earth}}}{r_{\text{earth}} + h_{\text{sat}}} \right)^2 .$$

The third contribution is the cosmic microwave background (CMB) radiation:

$$Q_{\text{CMB}} = A_{\text{eff}} \, \sigma \, T_{\text{CMB}}^4 \tag{3.5}$$

where $\sigma = 5.670374419 \times 10^{-8}$ is the Stefan–Boltzmann constant and A_{eff} is the effective radiating area exposed to deep space. This area is obtained by summing the outer panel areas weighted by their emissivities, excluding the nadir face which interfaces with earth.

The final contribution is the direct solar irradiance:

$$Q_{\text{solar}} = \sum_{i \in \{+X,+Y,+Z\}} S_0 \, \alpha_i \, A_i \, \max(0, \; \hat{n}_i \cdot \hat{s}) \tag{3.6}$$

where $\hat{n}_i$ is the outward normal of face i, $\hat{s}$ is the unit sun vector in the satellite frame, and A_i is the corresponding face area.

The only mechanism for heat leaving the satellite is thermal radiation from the outer surfaces:

$$Q_{\text{out}} = \sigma \, A_{\text{eff}} \, T_{\text{sat}}^4 \tag{3.7}$$

Using the energy balance $Q_{\text{in}} = Q_{\text{out}}$ and substituting Eq. (3.7), the satellite temperature becomes

$$T_{\text{sat}} = \left(\frac{Q_{\text{in}}}{\sigma \, A_{\text{eff}}} \right)^{1/4} . \tag{3.8}$$

Using a lumped-parameter model in GMAT and MATLAB, we define the sink temperature. It represents the CubeSat's outer-surface equilibrium temperature. GMAT propagates the satellite's orbit over 31 days.

It targets a sun-synchronous circular orbit (SSO) at an altitude of 450 km and a 97.4° inclination, which fits our EventSat's mission design.

The propagation includes Earth's gravity, point-mass effects from the Sun and Moon, solar radiation pressure, and atmospheric drag. At each time step, GMAT reports the Sun vector and the eclipse entry and exit events in "May2026.script" in GMAT [30].

These orbital data are then processed in the "EquillibriumTemp.m" MATLAB script [30], where the incident solar, Albedo, and Earth-IR heat loads on each satellite face are computed from the Sun vector, face orientation, and surface properties, namely absorptance and emissivity. It also accounts for eclipse periods.

The satellite is assumed to be in observation mode, with the optics face pointed towards Zenith. Assuming radiative equilibrium, where absorbed heat equals radiated heat, an equilibrium surface temperature is calculated at each time step. We took the maximum value during the 31-day window as the worst-case sink temperature. Eclipse durations and solar incidence angles vary seasonally in a Sun-synchronous orbit (SSO). After a sensitivity analysis across all months, we considered a worst-case scenario in May, following the methodology established in an earlier thesis by Schmid [88]. We modified the original simulation script, specifically the right ascension of the ascending node (RAAN), to account for the orbital conditions of 31 days in May 2026 and retrieve the sink temperature. We set the right ascension of the ascending node (RAAN) to 118.3°. This value is based on a rate of 0.9856°/day over the 120-day interval from the initial epoch of January 2026 (0 + 0.9856°/day *120) (table 3.2).

Table 3.2 Orbital simulation parameters for GMAT and MATLAB

Parameter	Value	Unit
Orbital altitude	450	Km
Inclination	97.4	°
Propagation duration	31	Days
Earth gravity model	JGM2, order 4	–
Attitude	Observation mode, optics face pointed to Zenith	-
Rate of RAAN precision	0.9856	°/day
RAAN	118.3	°
Solar panel	emissivity: 0.88	-
	absorptance: 0.91	-
Structure	emissivity: 0.85	-
	absorptance: 0.93	-
PCB	emissivity: 0.85	-
Optics	emissivity: 0.9	-
	absorptance: 0.95	-
Communications	emissivity: 0.9	-
	absorptance: 0.25	-
Albedo	0.3	-
IR flux	237	W
Space temperature	2.7	K
Solar irradiance S_0	1367	W/m^2
Earth radius r_{earth}	6378	Km

3.2 heat spreader design

To maintain the NVIDIA Jetson at an adequate temperature, we adopted a passive cooling solution. We chose a passive cooling solution because it does not rely on electrical power or moving parts, like in active

solutions, which are more complex to design and manufacture and are eventually more prone to failure and integration problems [5].

We take, for example, "axial grooved" heat pipes as an instance of passive cooling solutions. Although their manufacturing process is straightforward, their design is constrained by the need to optimize wick geometries, ensure material-fluid compatibility, and integrate appropriate control systems. For successful manufacturing, cleanliness, proof pressure testing, ultra-high fluid purity, etc. are required [2].

As a further example of active cooling instances, Cryocoolers are constrained by size, weight, and power limitations, making their integration into CubeSats challenging. They also require thermal isolation and efficient heat rejection to stay within cryogenic temperatures. The compressor's mechanical vibration could affect the pointing of the entire CubeSat [71].

Also, mechanically pumped fluid loops (MPFLs) include pumps, sealed tubing, heat exchangers, etc. This is what implies more power consumption and makes them hard to design, manufacture, and deploy in small volumes [72].

Because satellites rely mostly on passive thermal solutions, which do not require power and are easier to implement [90], we chose a passive thermal solution, namely a heat spreader.

The implemented solution must fulfill some requirements, including not exceeding the operational temperature limit of 60 $^{\circ}C$ and remaining within the resistance budget defined by (equation 3.9)

$$R = \frac{T_{operating} - T_{sink}}{Q} \tag{3.9}$$

Thermal and vibrational constraints drive our heat spreader's geometric design.

The choice lay between using CU-ETP 2.0060/C11000 and Aluminum A5083. In our case, we do not have any weight constraints. To select the appropriate material, we performed a qualitative analytical analysis by comparing material properties (Table 3.3) [49] [48] [31] [22] [86] and simulations.

Table 3.3 Material properties comparison of Cu-ETP 2.0060/C11000 and Aluminum A5083

PropertyMaterial	Cu-ETP 2.0060	A5083	unit
Density	8940	2660	[kg/m^3]
Specific heat capacity C_p	394	900	[J/kg·$^{\circ}$C]
Thermal conductivity	390	110–140	[W/m·$^{\circ}$C]
Young's modulus	127	71	[GPa]
Yield strength	180	205	[MPa]
Poisson ratio	0.394	0.33	[-]
Emissivity	0.1	0.15	[-]

Cu-ETP 2.0060 is roughly 3.55 times more conductive than A5083, while A5083 is more likely to survive vibrations (Table 3.3).

For the same geometrical model, from a deflection point of view (equation 3.10) [27]

$$\delta = \frac{FL}{AE}, \tag{3.10}$$

where:

• F: applied force, • L: length of the beam, • A: cross-sectional area, and • E: Young's modulus,

since $E_{Cu,2.0060} > E_{A5083}$, Cu 2.0060 experiences lower axial and bending deflection compared to A5083 ($\delta_{Cu,2.0060} < \delta_{A5083}$) and since $\rho_{Cu,2.0060} > \rho_{A5083}$, consequently, Cu 2.0060 adds more inertial mass and experiences self-weight stress, leading to a lower natural frequency since the natural frequency formula is given by (equation 3.11) [84]

$$f_n = \frac{1}{2\pi}\sqrt{\frac{k}{m}} \tag{3.11}$$

with the mass $m = \rho.V$ and the stiffness $k \propto E$. However, Aluminum A5083 experiences a higher natural frequency and lower dynamic amplification across fasteners and joints.

And since the dynamic displacement amplitude is given by (equation 3.12) [84]

$$X(\omega) = \frac{F_0}{\sqrt{(k - m\omega^2)^2 + (c\omega)^2}}, \tag{3.12}$$

where •F_0: amplitude of the harmonic force, • k: stiffness, • m: mass, • c: damping coefficient, and • ω: excitation angular frequency,

and Cu 2.0060 has a lower resonance frequency, then it is more susceptible to higher amplification.

We resolved this trade-off by performing dedicated thermal and vibrational simulations to select the material best suited to our case and requirements.

In the following table (table 3.4), we presented our numerical model thermal constraints and boundary conditions

Table 3.4 Thermal requirements and heat load

Constraints	Numerical values	Unit
Shutdown temperature	104.5	℃
Operating temperature	60	℃
Heat load	6.77/2=3.385	W
Sink temperature	section 4.1	℃

These limits establish the maximum allowable thermal resistance along the conduction path (table 3.4).

Geometrically, the heat spreader features an L-shaped bend dictated by the NVIDIA Jetson module's higher physical emplacement relative to the CubeSat structure frame, into which heat will be conducted. The first design draft had a thickness of 2 mm for the L-shaped part, which was impractical to manufacture because the manufacturer stated that it might bend during milling, resulting in incorrect dimensions.

As a result, we considered two solutions: splitting the piece into 3 separate parts or increasing the thickness to at least 4 mm. The second option was more rational, as it would keep the heat conduction path uninterrupted and ensure better structural performance (figure 3.1).

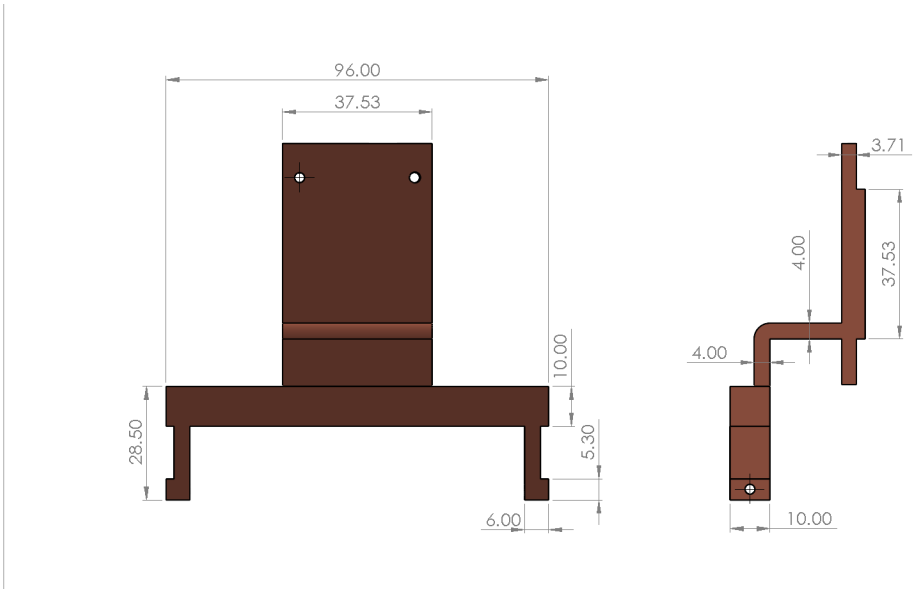


Figure 3.1 2D drawing of the heat spreader

From a thermal perspective, increasing the thickness means increasing the cross-sectional area available for conduction, hence decreasing the resistance, as thermal resistance is defined by (equation 3.13) [55]

$$R = \frac{L}{K\,A} \tag{3.13}$$

where $A = w \times t$ denotes the cross-sectional area, such that w is the width and t is the thickness, L is the length, and k is the thermal conductivity.

On the other hand, from a structural perspective, thickness is proportional to the second moment of area given by (equation 3.14) [40]

$$I = \frac{w\,t^3}{12} \tag{3.14}$$

and the natural frequency of a structural element scales approximately as (equation 3.15)

$$f_n \propto \sqrt{\frac{E\,I}{m\,L^4}} \tag{3.15}$$

where • E: Young's modulus, • m: mass, and • L: beam length [69].

Hence, increasing thickness raises the natural frequency, reducing the likelihood of resonance failure.

After performing simulations and meeting both the thermal and vibrational requirements specified in the SSMS VEGA-C manual for the chosen material, we selected the primary prototype heat spreader for manufacture at the TUM workshop.

Our 6U CubeSat structure allows only four possible mounting positions for the module, which defined the final heat spreader design.

We will qualify the final mounting position based on the Event camera and the heat it should receive by conduction from the rods and other payloads' behavior.

Before initiating CNC machining, we 3D-printed the prototype (figure 3.2a) to verify that the geometry matches the chip area and that the whole piece fits the 6U CubeSat structure, since no fully conforming CAD model of the NVIDIA Jetson Orin Nano 8GB and the NVIDIA Jetson Orin Nano/NX was available. Hence, we carefully measured the dimensions directly from the module in the laboratory using a sliding Caliper. After confirming it fits, we produced the piece (figure 3.2b).



(a) 3D- printed Heat spreader prototype



(b) CNC machined Cu 2.0060 heat spreader

Figure 3.2 3D printed and CNC-manufactured heat spreader model

3.3 Thermal simulation

In this section, we describe the approach we used to perform the different thermal simulations.

The orbital simulation file "orbitalconditions.wbpj" and the vacuum chamber different simulations "Heatpipe in vacuum chamber nw simulation.wbpj" and "Heatpipe in vacuum chamber nw simulation1.wbpj" files are in GITLAB [30].

3.3.1 ANSYS Orbital simulation

Because the ANSYS student edition 2026R1 has a 120k node limit, we used a half-symmetry model with a global mesh element size of 5 mm. We set the boundary conditions as follows (Table 3.5).

Table 3.5 Simulation's boundary conditions

Boundary conditions	Numerical values	Unit
Heat load	3.385	W
Sink temperature	23.93	℃
Near screw region conductance (solar panel-structure)	2834.15	$W/m^2.$
Remaining area of the Solar panel touching the structure	126.47	$W/m^2.K$
TIM conductivity	8370	$W/m^2.K$

As stated in "Schmid" thesis, we based the thermal contact resistance between the components on the supplier's specified total interface conductance of 6.652 $fracWK$, with 50% distributed over the entire contact surface and 50% concentrated near the regions of screws. The areas having a near-screw influence were defined analytically by the formula $A = \pi \times D^2$, where D is the diameter of the M3 fasteners that join the solar panel to the satellite structure at the 14 joint locations. Since the theoretical pressure circles do not completely overlap with the actual structural geometry, we calculated an effective thermal conductance of 2834.15 $\frac{W}{m^2.K}$ for the near-screw regions in order to ensure thermal equivalence. We then applied the remaining 50% of the total conductance to the remaining contact area, giving a full-surface contact conductance of 126.47 $\frac{W}{m^2.K}$. The solar panel PCB was assigned orthotropic thermal properties, with a longitudinal thermal conductivity of 48.01 [W/(m.K)] and a transverse thermal conductivity of 0.46 $\frac{W}{m.K}$. The satellite aluminum EN AW 6082 structure was modeled using isotropic thermal properties, assuming a constant thermal conductivity of 200 $\frac{W}{m.K}$ [5]. The input heat flow was distributed uniformly over the component footprint. Lastly, a Thermal Interface material (TIM) was added at the joint between the heat pipe and the satellite structure, resulting in a contact thermal conductance of 8250 $\frac{W}{m^2.K}$ [88].

3.3.2 Simulation inside the vacuum chamber

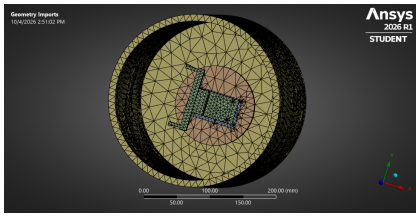
To assess the behavior of the NVIDIA Jetson Orin Nano inside the vacuum chamber, we simulated three different configurations to assess the temperature evolution in SoC0:

• **Configuration A:** In this configuration, we considered the subassembly heat spreader-NVIDIA Jetson, and we placed the subassembly on a PU packaging foam to prevent the BACOeng chamber base from overheating.

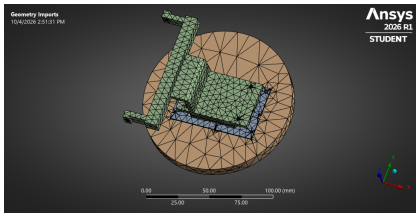
• **Configuration B:** In this configuration, we mounted the subassembly heat spreader-NVIDIA Jetson to the 6U CubeSat structure, which we held stationary using a 3D-printed PLA support.

• **Configuration C:** In this configuration, our goals were, namely, assessing the NVIDIA Jetson temperature evolution in the presence of two conduction paths to the 3U testing CubeSat structures made for vibrational testing and retrieving the temperature evolution along the CubeSat rods, 3U testing CubeSat structures, and the Jetson plate. To perform this simulation, we considered the Jetson plate, on which we mounted the subassembly heat spreader-NVIDIA Jetson, which we bonded to the two 3U CubeSat structures. We also considered the four CubeSat rods that pass through the holes in the Jetson plate.

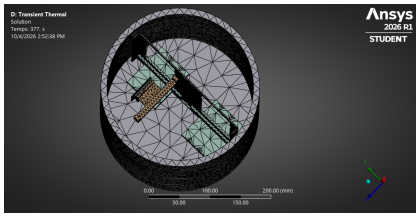
Note: All the simulations (Figure 3.3) included the vacuum chamber cover.



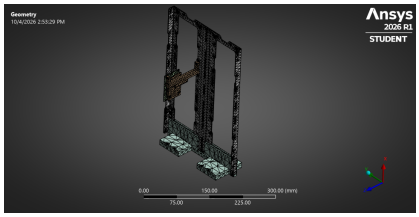
(a) Configuration A geometry inside the vacuum chamber



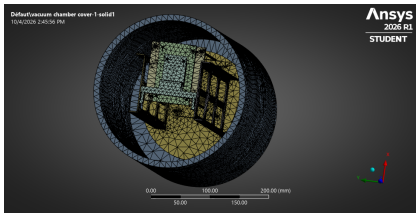
(b) Configuration A geometry with the PU packaging foam



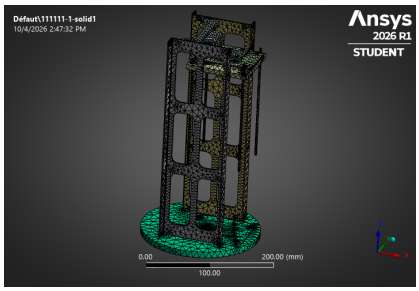
(c) Configuration B geometry inside the vacuum chamber



(d) Configuration B geometry with the PLA support



(e) Configuration C geometry inside the vacuum chamber



(f) Configuration C geometry with the PU support

Figure 3.3 Simulation geometries of Configurations A and B and a half-geometrical simulation for configuration C

In ANSYS, we set the following material properties [32] [86] [73] [96] [81] [28] [49] [48] (table 3.6)

Table 3.6 Material properties comparison for various components.

	Density [kg/m^3]	**Isotropic thermal conductivity** [W/(m·K)]	**Specific heat at constant pressure (C_p)** [J/(kg·℃)]	**Emissivity** [-]
Vacuum chamber	8000	15	500	0.15
Vacuum chamber cover	1180	0.19	1400	0.9
Cu 2.0060	8940	390	394	0.1
A5083 Jetson plate	2660	120	900	0.15
CubeSat structure	2700	167	896	0.2
Polyurethane (PU) packaging foam	40	0.035	1400	0.9
Rods (Aluminum 6061)	2700	167	896	0.2

Note: Even though our vacuum chamber cover is Acrylic, we picked an emissivity of 0.9 because we covered the top cover of the chamber in a way that it becomes dark inside.

In the meshing process, we used global mesh sizes of 10 mm, 15 mm, and 10 mm for configurations A, B, and C, respectively. We then used a finer 5 mm mesh for the heat spreader and the NVIDIA Jetson Orin Nano in configurations A and B. However, we set the value to 7 mm for configuration C. For this approach, we got the following nodes and elements (Table 3.7)

Table 3.7 Mesh statistics of the three configurations.

	Configuration A	Configuration B	Configuration C
Nodes	74689	101026	104901
Elements	37898	49059	49666

We did not reach the 120k-node limit to ensure fast execution. Transient thermal analysis simulation takes a long time to converge. Even with these settings, configuration B took 3 hours and 22 minutes to converge in the transient thermal analysis.

To define radiation in ANSYS, we adopted an open surface-to-surface enclosure approach, although it is a closed-cavity problem. In our case, a perfectly enclosed surface-to-surface enclosure produced a convergence error in ANSYS Mechanical. Since the bodies tested inside the vacuum chamber are small compared to the chamber volume, the chamber walls dominate the radiative environment, so the net exchange is governed by the bodies' emissivities and the wall temperature, which remained nearly isothermal throughout the experiments. Hence, chamber walls dominate the radiative environment, making it reasonable to treat the different configurations' geometries as radiating toward an effective ambient temperature equal to that of the wall [78].

For the PCB, NVIDIA defined the orthotropic thermal conductivity properties as follows: the in-plane thermal conductivity as k = 55 $\frac{W}{m.^\circ C}$ and the out-of-plane thermal conductivity as k = 2 $\frac{W}{m.^\circ C}$ [37]. For density and specific heat at constant pressure (C_p), we defined 1850 $\frac{Kg}{m^3}$ and 900 $\frac{J}{kg^\circ C}$, respectively [62]. For the chip part, NVIDIA states that the Jetson Orin Nano module is based upon a low-power variant of the NVIDIA Orin system-on-chip (SoC) like almost all conventional CMOS GPU/CPU SoCs, that integrated circuit is manufactured on a silicon die (table 3.8) [35] [65].

Table 3.8 Material properties of PCB and silicon chip

Property	**PCB**	**Chip (Si)**	**Unit**
Density	1850	2330	kg/m^3
Specific heat C_p	900	700	J/(kg·°C)
Thermal conductivity (in-plane)	55	–	W/(m·°C)
Thermal conductivity (out-of-plane)	2	–	W/(m·°C)
Thermal conductivity (isotropic)	–	148	W/(m·°C)

Normally, the emissivity of Cu 2.0060 is around $\varepsilon_{\text{smooth}} = 0.03$ [94]. In this thesis, because of the limited CNC machining finish at the workshop, we chose an effective value of $\varepsilon_{\text{eff}} = 0.1$ by trial and error in our simulations in order to reduce the error obtained between the simulation and the experimental results [57].

Calculating the equivalent number of smooth-layer interfaces n, we get (equation 3.16)

$$n = \frac{\ln(1-\varepsilon_{\text{eff}})}{\ln(1-\varepsilon_{\text{smooth}})} = \frac{\ln(1-0.1)}{\ln(1-0.03)} = \frac{\ln(0.9)}{\ln(0.97)} \approx 3.5 \text{ bounces} \tag{3.16}$$

is physically reasonable for CNC machining tool-mark grooves that vary in the range of 2-5 [95]. This is what makes $\varepsilon_{\text{eff}} = 0.1$ reasonable for Ra = 1-2 µm machining marks [63].

Also, for the aluminum CubeSat structure and the Jetson plate, we used the same methodology to prove that $\varepsilon_{\text{eff}} = 0.15$, instead of $\varepsilon_{\text{smooth}} = 0.0346$ [52], is suitable in this specific case.

After several experiments, we found that the NVIDIA Jetson Orin Nano in vacuum had an initial temperature T_0 higher than the other parts, which were at ambient temperature.

For configurations A and B, we adopted a 2-stage approach. Firstly, we conducted a steady-state temperature analysis in which we fixed the NVIDIA Jetson temperature to 40.54 $^\circ C$ and 36.6 $^\circ C$, respectively, as in real-life experiment. This step served as an initial boundary condition, namely the initial temperature of SoC0 at the start of the transient thermal simulation, for the subsequent transient thermal analysis. In other words, the goal was to begin the transient analysis with a Jetson temperature different from that of

the other components and matching the experimental SoC0 temperature (sections 4.3.1 and 4.3.2). For simplicity, we set the ambient temperature to T = 25 $^{\circ}C$ in both stages, since the vacuum chamber walls are nearly isothermal. For configuration A, we set a transient thermal duration of 1700 seconds with an initial time step of 17 seconds, a minimum time step of 1.7 seconds, and a maximum time step of 170 seconds. However, in configuration B, we used a transient thermal duration of 32096 seconds with an initial time step of 80 seconds, a minimum time step of 8 seconds, and a maximum time step of 100 seconds (Table 3.12).

Table 3.9 Transient thermal analysis for configurations A and B

Parameter	Config A	Config B	Unit
Total duration	1700	32096	s
Initial time step	17	80	s
Minimum time step	1.7	8	s
Maximum time step	170	100	s

In order to ensure the compilation of the simulation, we set natural convection and conduction boundary conditions (table 3.10) [55]. We modeled the vacuum chamber with a height of H = 380 mm and a radius of R = 130 mm. Heat was rejected from the outer walls of the chamber and the upper cover by natural convection. We set the ambient air value to T_{∞}= 25 $^{\circ}C$.

we define the areas $A_{side} = 2\pi RH$ and $A_{top} = A_{bottom} = \pi R^2$ and the Rayleigh number (equation 3.17)

$$Ra = \frac{g\,\beta\,|(T_w - T_{\infty})|\,L_C^3}{\nu\,\alpha} \tag{3.17}$$

with g being the gravitational acceleration, • T_w: wall temeprature, • T_{∞}: ambient temperature, • L_C: characteristic length, • υ: kinematic viscosity of air, • α: diffusivity of air, and • β: volumetric thermal expansion coefficient. We evaluated the air properties at $T_f = \frac{T_w + T_{\infty}}{2}$

We also determined the average Nusselt number $\overline{Nu}_L$ (equation 3.18)

$$\overline{Nu}_L = \left(0.825 + \frac{0.387\,Ra_L^{1/6}}{\left[1 + \left(\frac{0.492}{Pr}\right)^{9/16}\right]^{8/27}} \right)^2 \tag{3.18}$$

Where Ra_L is the Rayleigh number and Pr is the Prandtl number. As a consequence, we get our convective heat transfer coefficient h_{conv} (equation 3.19)

$$h_{\text{conv}} = \frac{\overline{Nu}_L\,K}{L_c} \tag{3.19}$$

where k is the thermal conductivity, and by altering the characteristic length L_C each time, namely L_C= H for the chamber walls, and equals $\frac{A_s}{P}$= $\frac{R}{2}$ at the top and bottom, with A_s being the surface area and P the perimeter.

we also determined the wall temperature T_w by solving the overall energy balance (equation 3.20)

$$Q = (h_{\text{side}}\,A_{\text{side}} + h_{\text{cover}}\,A_{\text{top}} + U_{\text{bottom}}\,A_{\text{bottom}})\,(T_w - T_{\infty}) \tag{3.20}$$

In the chamber base, the heat transfer occurred based on thermal resistance (equation 3.21)

$$U_{\text{bottom}} = \left(\frac{1}{h_{\text{contact}}} + \frac{t_{\text{table}}}{k_{\text{table}}} + \frac{1}{h_{\text{conv,bottom}}} \right)^{-1} \tag{3.21}$$

with $h_{contact}$ = 200 $\frac{W}{m^2k}$; $k_{table} = 0.04$ m ; k_{table}= 0.15 $\frac{W}{mk}$

Following those rules (equations 3.19 and 3.21), we got the convection values [30] that we added to our simulations inside the vacuum chamber as boundary conditions (table 3.10).

The NVIDIA Jetson chip has a volume of $V_{\text{Jetson}} = 37.8 \times 38.01 \times 2.63mm^3$ in the CAD model.

As the power of the NVIDIA Jetson Orin Nano is not constant, we took an average power of the fluctuating results from the last run of each performed experiment to define the internal heat generation as P/V_{Jetson}, with P being the heat dissipated by the Jetson (table 3.10).

Table 3.10 Simulation data for ANSYS configurations

Parameter	Config A	Config B	Config C	Unit
Average power	4.79	4.89	3.72	W
Internal power generation	1.3e^{-03}	1.3e^{-03}	9.98e^{-04}	W/mm^3
Convection from chamber side walls	3.33e^{-06}	3.33e^{-06}	2.82e^{-06}	W/mm^2·°C
Convection from top lid	8.95e^{-06}	8.95e^{-06}	8.30e^{-06}	W/mm^2·°C
Conduction from bottom face to table	2.53e^{-06}	2.53e^{-06}	2.48e^{-06}	W/mm^2·°C

To define the thermal conductance between components (table 3.11), we used the following relation [64] (equation 3.22)

$$h = h_c + h_r \tag{3.22}$$

We calculated the contact conductance h_c using the Yovanovich plastic contact correlation [25] (equation 3.23)

$$h_c = 1.25 \times k_s \frac{m}{\sigma} \left(\frac{P}{H_{\text{mic}}} \right)^{0.95} \tag{3.23}$$

where the harmonic mean thermal conductivity k_s (equation 3.24) is

$$k_s = \frac{2k_1k_2}{k_1 + k_2} \tag{3.24}$$

where k_i is the thermal conductivity of a component.

The combined RMS surface roughness σ [58] (equation 3.25) is

$$\sigma = \sqrt{\sigma_1^2 + \sigma_2^2}. \tag{3.25}$$

where σ_i is the roughness of the component, m= 0.1 is the mean absolute asperity slope, P is the contact pressure and H_{mic} is the microhardness

We calculated the radiative conductance h_r as [29] (equation 3.26)

$$h_r = 4\varepsilon_{\text{eff}}\sigma_{\text{SB}}T^3, \tag{3.26}$$

where the effective emissivity [70] (equation 3.27) is

$$\varepsilon_{\text{eff}} = \left(\frac{1}{\varepsilon_1} + \frac{1}{\varepsilon_2} - 1 \right)^{-1}. \tag{3.27}$$

the Stefan–Boltzmann constant is $\sigma_{\text{SB}} = 5.67 \times 10^{-8}\ \text{W m}^{-2}\,\text{K}^{-4}$,

We also assumed that the mean interface temperature as T = $298.15\,\text{K}$ as a conservative value, since the objects inside the vacuum chamber exceed 25 $°C$. By adopting this value, we increase the thermal contact resistance. Also, ε_1 and ε_2 are the emissivities of the touching objects.

For the TIM interfaces, we used a thickness of t_{TIM} = 0.5 mm, and we got the conductance h_{TIM} by applying

$$h_{\text{TIM}} = \frac{k_{\text{TIM}}}{t_{\text{TIM}}} \tag{3.28}$$

where k_{TIM} is the thermal conductance.

For the interfaces between the modules themselves, and between the modules and the rods, we assume a thermal contact conductance provided by EnduroSat team, who state that the corresponding contact areas are $500\ W/(m^2 \cdot K)$ as defined in the CAD.

In the following table (table 3.11), we define the different thermal conductances [30] across the different configurations where:

Table 3.11 Thermal conductances for the different configurations

	Configuration A	Configuration B	Configuration C	Unit
Packaging foam–chamber	1.55×10^{-5}	–	–	$\mathrm{W/(mm^2\,°C)}$
PCB*–PU*	1.37×10^{-5}	–	–	$\mathrm{W/(mm^2\,°C)}$
Chamber cover–chamber	1.78×10^{-6}	–	–	$\mathrm{W/(mm^2\,°C)}$
PLA*–chamber	–	6.1×10^{-6}	–	$\mathrm{W/(mm^2\,°C)}$
CubeSat structure–PLA	–	4.33×10^{6}	–	$\mathrm{W/(mm^2\,°C)}$
CubeSat structure–PU*	–	–	1.17×10^{-6}	$\mathrm{W/(mm^2\,°C)}$
TIM* interfaces	0.00828	0.00825	0.0278	$\mathrm{W/(mm^2\,°C)}$
PCB–Jetson plate	–	–	4.62×10^{-4}	$\mathrm{W/(mm^2\,°C)}$
Jetson plate–rods	–	–	5×10^{-4}	$\mathrm{W/(mm^2\,°C)}$

• PCB*: Printed circuit Board • PU*: Polyurethane • PLA*: Polylactic Acid • TIM*: Thermal interface material

For configurations A and B, we adopted an internal heat generation of $1.3e^{-03}\ W/mm^3$.

In configuration B, we retrieved the steady-state temperature of the NVIDIA Jetson Orin Nano, which had an initial temperature of the ambient, ignoring the fact that it heats up fast once initiated.

This plot (figure 3.4) illustrates a schematic representation for configurations A and B

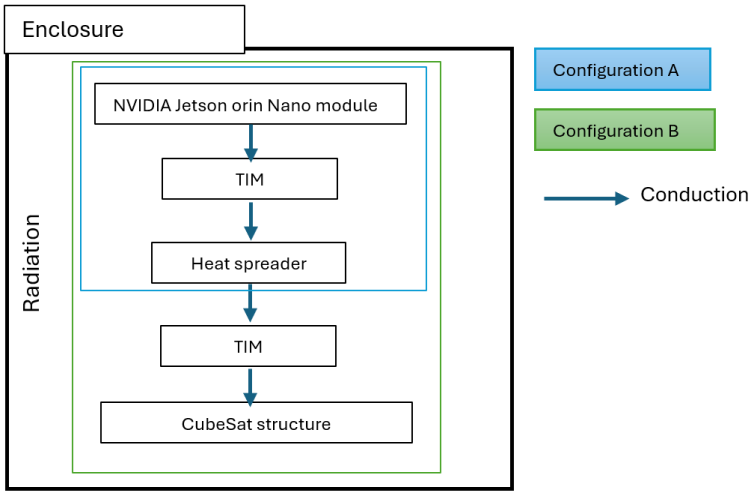


Figure 3.4 Thermal path in configurations A and B

To further assess the NVIDIA Jetson's temperature evolution and the conduction rate through the rods, Jetson plate, and the testing 3U CubeSat structures, and to evaluate how much this temperature could influence the Event Camera, we included this time the Jetson plate, 4 CubeSat rods, and two CubeSat structures.

In this configuration, we first retrieved the steady-state temperature, then performed a 2-stage simulation, as in configurations A and B, but with a fixed NVIDIA Jetson temperature of 36.47 $°C$ (section 4.3.3) and an ambient temperature of 25 $°C$.

In the transient thermal simulation, we set a duration of 39807 seconds with an initial time step of 100 seconds, a minimum time step of 10 seconds, and a maximum time step of 1000 seconds.

Table 3.12 Transient thermal analysis for configuration C

Parameter	Config C	Unit
Total duration	39807	s
Initial time step	100	s
Minimum time step	10	s
Maximum time step	1000	s

We also defined an internal heat generation value of 9.98e^{-04} W/mm^3, since in the experimental results the NVIDIA Jetson had an average power of 3.71 W.

Note: in the laboratory the air conditioning was working. However, we didn't consider forced convection in those simulations.

3.4 Experimental validation

To validate the simulation results, we conducted multiple experiments inside the BACOENG vacuum chamber, which provided a rough vacuum of -28 inHg [23]. In the earliest runs, the NVIDIA Jetson shuts down at $104.5°\mathrm{C}$. To prevent software damage, we adjusted the thermal management configuration in a way that the system initiates a controlled shutdown when the critical thermal zones (CPU, SoC, etc.) reach a temperature of $90°\mathrm{C}$, while the hardware emergency shutdown remains at $104.5°\mathrm{C}$ as specified by NVIDIA [75]. To conduct this experiment, each configuration had different components.

In configuration A, we used the NVIDIA Jetson Orin Nano 8GB and the heat spreader. In the second one (Configuration B), we added the aluminum CubeSat structure as a new conduction path.

In the third configuration (Configuration C), we used the NVIDIA Jetson Orin Nano/NX, the heat spreader, 2 aluminum 3U CubeSat testing structures, 4 rods, Arduino, K-type, and SD card. The following table lists the relevant components used inside the vacuum chamber for configurations A, B, and C (Table 3.13).

Configuration no.	Setup
A	NVIDIA Jetson Orin Nano 8GB, heat spreader
B	NVIDIA Jetson Orin Nano 8GB, heat spreader, and the CubeSat structure
C	NVIDIA Jetson Orin Nano/NX, heat spreader, 2 CubeSat structures, 4 rods, Arduino, K-type thermocouple, and SD card

Table 3.13 Experimental setup for Configurations A, B, and C.

Note: In each configuration, we ran several trials to compare results and assess accuracy, even though the NVIDIA Jetson's initial temperature varied slightly each time and in configuration C, we used a different power bank "Model: BI-B633" for the NVIDIA Jetson "Model: D65" because the power bank used in configurations A and B was powering the Arduino. The following table represents the characteristics of the power banks, that powered the NVIDIA Jetson Orin Nano 8GB for configurations A and B and NVIDIA Jetson Orin Nano/NX for configuration C (table 3.14).

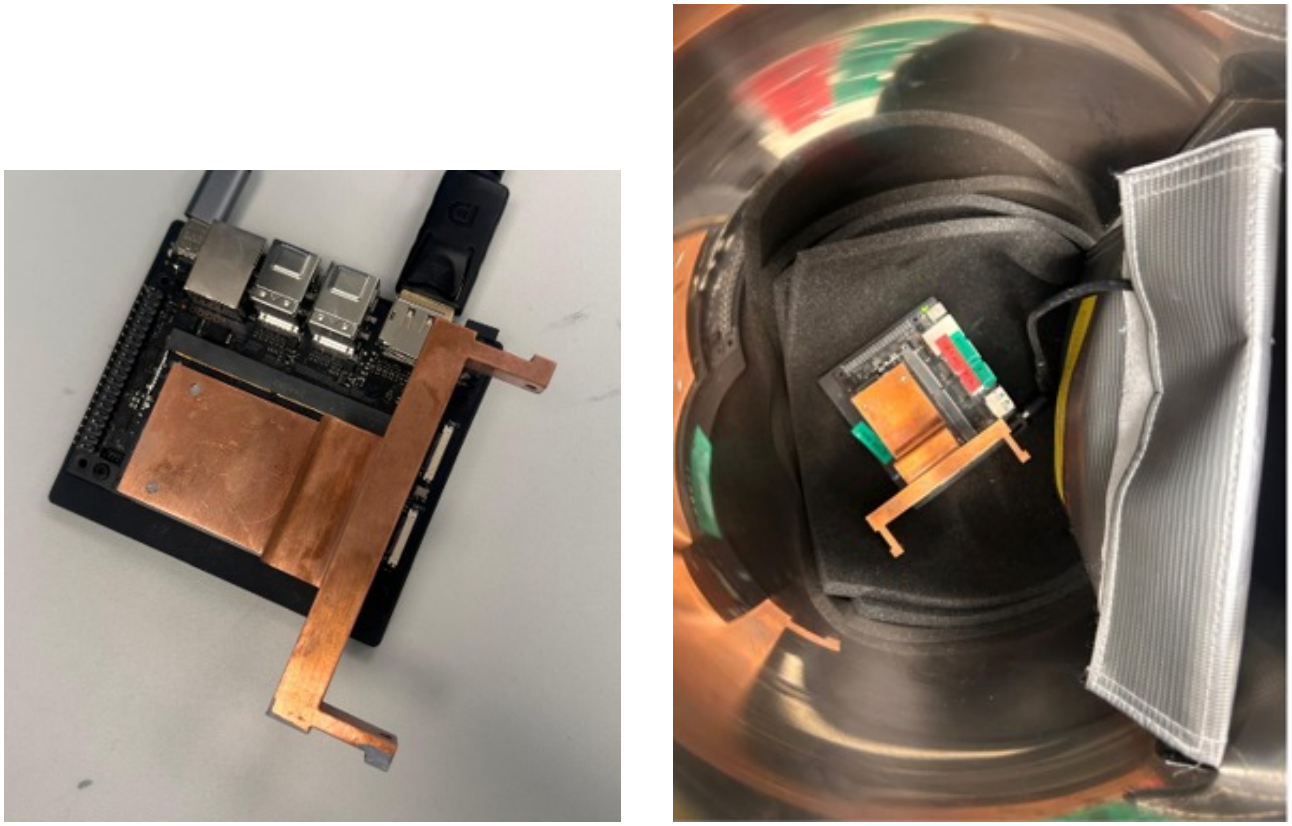
Table 3.14 Power bank specifications for the different configurations A, B, and C

Power bank	Configuration A	Configuration B	Configuration C
Nominal capacity [mAh]	20000	20000	25000
Nominal energy [Wh]	74	74	92.5

3.4.1 First experiment

This setup (Figure 3.5) included the NVIDIA Jetson Orin Nano 8GB and the Cu 2.0060 heat spreader, mounted to it with M3 screws. We then placed the NVIDIA Jetson-heat spreader subassembly on a (PU) packaging foam to prevent heat conduction to the chamber base.

To power the Jetson inside the vacuum chamber, we included a power bank "Model: D65".



(a) Configuration A assembly

(b) Configuration A assembly inside the vacuum chamber

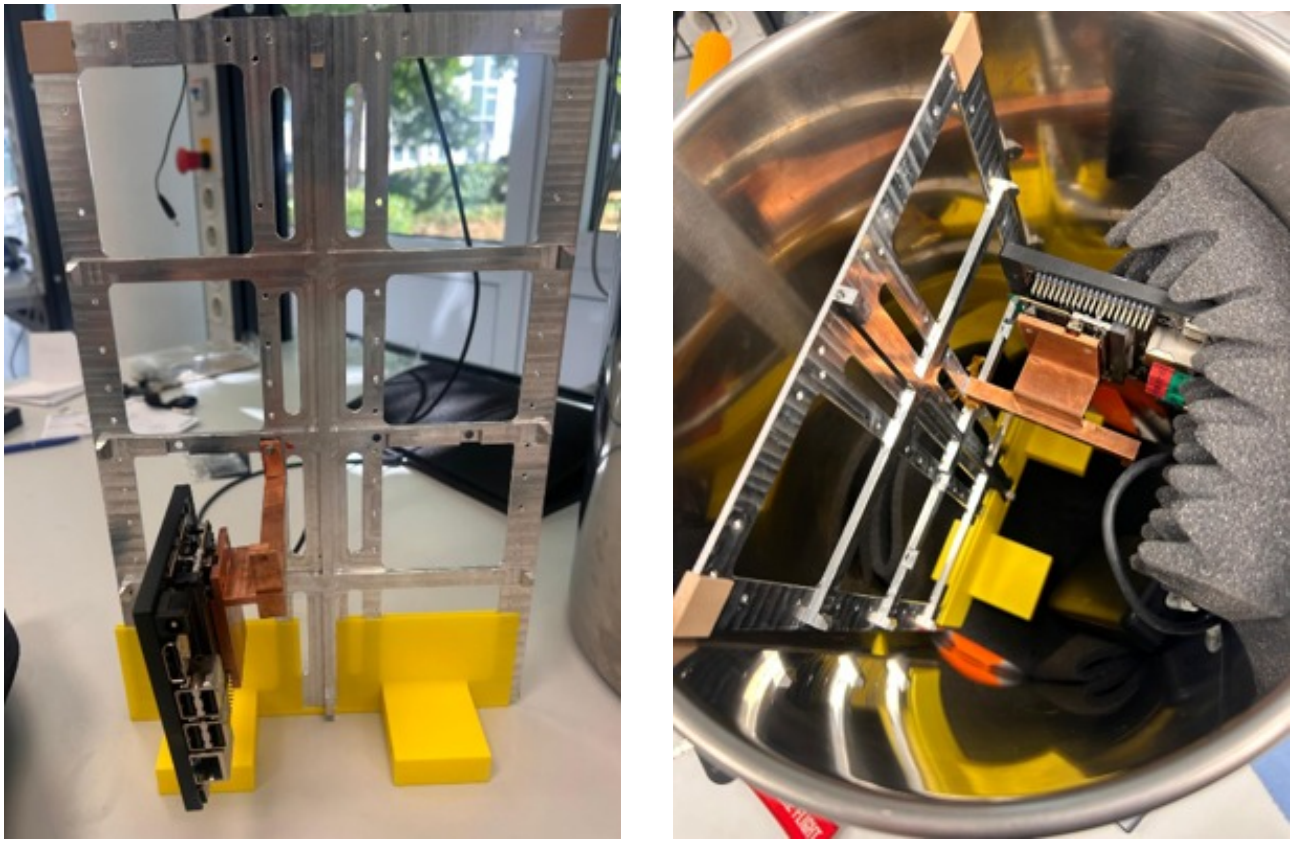
Figure 3.5 first experiment setup inside the vacuum chamber

The ultimate goal of this experiment is to assess the thermal behavior of the NVIDIA Jetson Orin Nano 8GB in the presence of the heat spreader. We repeated this experiment 5 times to assess the relevance of the results. In the first two attempts, we ran the experiment under atmospheric conditions for about 4 minutes to start the Jetson Power GUI recording before placing it in the vacuum chamber. For more accurate results, we conducted the last 3 runs directly inside the vacuum chamber and captured data over SSH.

In the first 4 Runs, the TIM had a thermal conductivity of k = $13.9\ \frac{W}{m.K}$, which was reduced by a factor of 3 in the fifth run due to the use of Stanol T12, with a thermal conductivity of k = $4.185\ \frac{W}{m.K}$.

3.4.2 Second experiment

To perform this experiment, the CubeSat structure shouldn't touch the vacuum chamber walls. To do this, we 3D-printed a PLA support to hold the configuration B setup statically. This setup included the Jetson-heat spreader assembly, which we mounted via an M3 screw to the CubeSat structure. To prevent heat conduction to the vacuum chamber's stainless steel base, we 3D-printed a PLA support (Figure 3.6). We used a thermal interface material (TIM) at the contact points: heat spreader-NVIDIA Jetson Orin Nano and heat spreader-CubeSat structure. To power the Jetson inside the vacuum chamber, we included a power bank "Model: D65".



(a) Configuration B setup **(b)** Configuration B setup inside the vacuum chamber

Figure 3.6 Configuration B experiment setup

This experiment aims to assess the effect of conduction through the CubeSat aluminum structure on the Jetson's temperature evolution. We initiated Runs 1 and 4 in the atmosphere and transferred to vacuum in about 4 minutes, whereas we initiated Runs 2 and 3 in vacuum at -28 inHg.

In Runs 1–3, we used a high-conductivity TIM (K = 13.9 W/m. K), while Run 4 used a lower-conductivity TIM (K = 4.185 W/m. K).

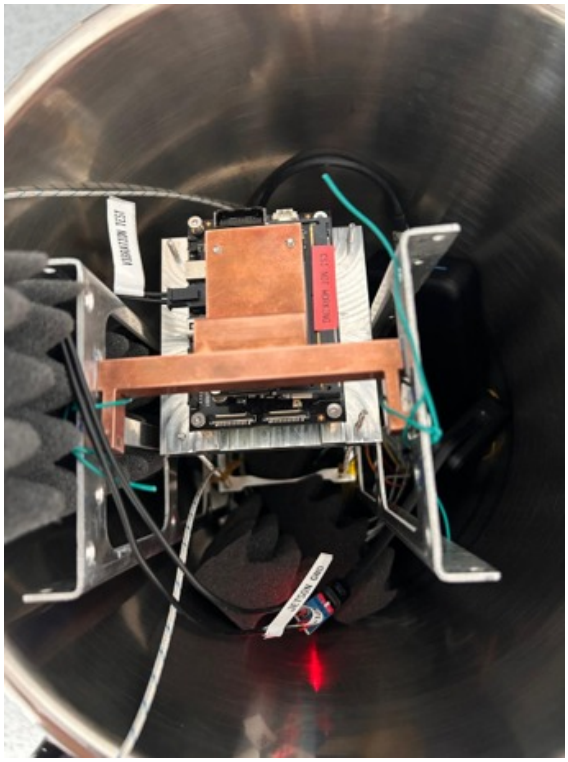
A mechanical baffle arises in the process: the threaded mounting feature on the copper spreader progressively wore out across cycles, making the screw connection to the structure less secure in later Runs, that's why, in Runs 3 and 4, we held the spreader end by Kapton tape rather than a direct screw joint, giving a much lower clamping pressure than the bolted connections used in Runs 1–2. This raises the thermal contact resistance at that interface in later runs, independent of the TIM change.

Note: We ran Run 4 in the atmosphere for the first 4 minutes to record the temperature evolution directly on UBUNTU, since logging via SSH showed a broken connection after approximately 7 hours in the previous runs. However, our goal was to keep the NVIDIA Jetson Orin Nano running as long as possible using the power bank to retrieve the steady-state temperature.

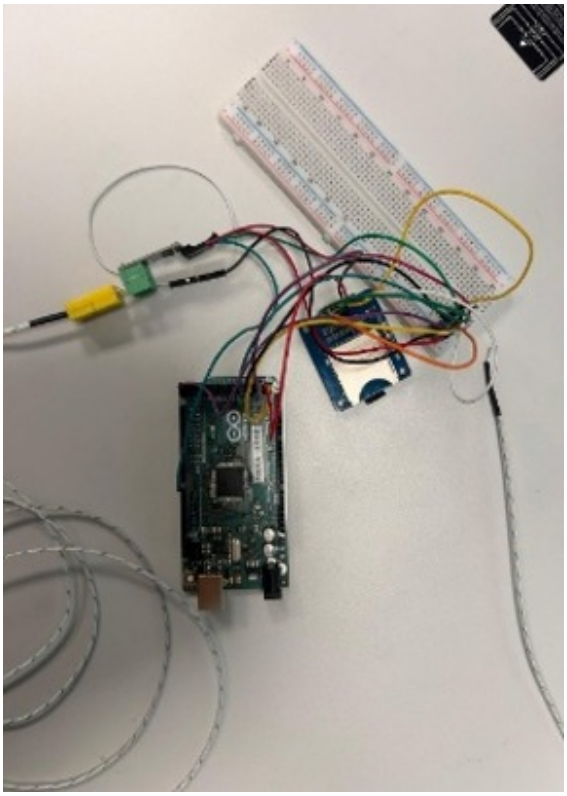
Since Run 4 is the longest, we compared this Run's results to our simulation.

3.4.3 Third experiment

We began this experiment under atmospheric conditions since the NVIDIA Jetson Orin Nano/NX model didn't have WIFI, which prevented us from logging in via SSH. We also used a Python script [30] to record the temperature evolution because this model has no Power GUI.



(a) Configuration C setup inside the vacuum chamber



(b) : setup of the Arduino + MAX6675 + SD card to measure rods' temperature in configuration C

Figure 3.7 Components and parts to conduct the Configuration C experiment

To conduct this experiment (Figure 3.7a), we mounted the NVIDIA Jetson Orin Nano/NX and the heat spreader assembly on the aluminum Jetson plate using spacers to prevent the PCB from touching the Jetson plate and M3 screws. We mounted the Jetson plate on the CubeSat structure rods. We also fixed the rods with a structural ring and mounted the heat spreader ends to the testing 3U CubeSat aluminum structures designed for vibrational testing.

To hold the structures statically inside the vacuum chamber, we used (PU) packaging foam on the sides and at the base of the chamber to prevent thermal conduction to the chamber structure. To measure the rods' temperature, we used a K-type MAX6675 thermocouple with an Arduino Mega 2560 as the main controller for the temperature-logging system, powered via a power bank, and an SD card module to store temperature evolution data [30] (Figure 3.7b) (Table 3.15). To power the Jetson inside the vacuum chamber, we included a power bank "Model: BI-B63" and for the Arduino, we used the power bank "Model: D65".

Table 3.15 Mounting configuration of the peripheral devices connected to the Arduino Mega 2560

Component	Signal	Arduino Pin
MAX6675 Thermocouple Module	SCK	52
	SO	50
	CS	4
	VCC	5V
	GND	GND
SD Card Module	CLK	52
	DO	50
	DI	51
	CS	53
	VCC	5V
	GND	GND

We taped the thermocouple end to one of the rods (figure 3.7a). At the chip-heat spreader and heat spreader interfaces, we used a high-conductivity TIM (K = 13.9 W/m.K). This experiment aims to assess the steady-state temperature of the rods and the NVIDIA Jetson Orin Nano after adding multiple conduction paths.We performed this experiment only once due to boot issues on the Jetson.

3.5 Vibrational simulation

To perform the vibrational test, we needed a support to hold the NVIDIA Jetson Orin Nano–heat spreader subassembly inside the 3U CubeSat designed for vibrational testing. This support had to fit inside the CubeSat's rails (rods) and match the heat spreader end dimensions so they would contact the CubeSat structure. We call this support the "Jetson plate" because it is a plate without a cover. This support must survive vibrations and minimize heat conduction through the CubeSat rods. For this reason, we chose aluminum A5083 because of its material properties (section 3.2) and the limited options available at the workshop.

To validate the Copper heat spreader and the Jetson plate against launch loads, we performed vibration analyses. To perform these analyses, we defined the materials' properties (tables 3.16 and 3.17). For the NVIDIA Jetson Orin Nano, we provided an approximate definition, as the mechanical properties are not publicly available. For the chip, we considered silicon, and FR-4 for the PCB as follows [45] [79]

Component	Property	Value
Chip (Silicon)	Density	2330 kg/m^3
	Young's modulus	165000 MPa
	Poisson ratio	0.22
PCB (FR-4)	density	1850 kg/m^3
	Young's modulus (out of plane)	7400 MPa
	Young's modulus (in plane)	16900 MPa
	Poisson ratio (out of plane)	0.11
	Poisson ratio (in plane)	0.39
	Shear modulus (in plane)	3300 MPa
	Shear modulus (out of plane)	7600 MPa

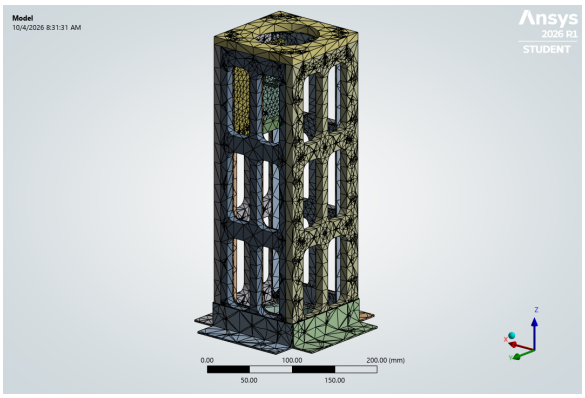
Table 3.16 Mechanical property assumptions for the NVIDIA Jetson Orin Nano chip and PCB.

This table (table 3.17) is a complementary table for the materials' properties we used in vibrational simulations [60] [93] [66] [22]

Table 3.17 Material properties used in structural simulations

Material	Density [kg/m^3]	Young's Modulus [GPa]	Poisson's Ratio
Stainless steel 304 (screws)	7950	193	0.28
Aluminium 6061-T6 (outside structure and front/rear interfaces (3U CubeSat))	2700	68.9–70	0.33
Aluminium EN AW-6060 (L-brackets)	2700	69.5–70	0.33
Aluminium 6082 (top and bottom plates for the 3U CubeSat)	2700	69–71	0.33

To obtain relevant results and consider all possible mounting cases, we followed five different configurations (Figures 3.8 and 3.9), switching between mounting axes X/Y and Z and moving the subassembly (NVIDIA Jetson Orin Nano, Heat spreader, and Jetson plate) along them.

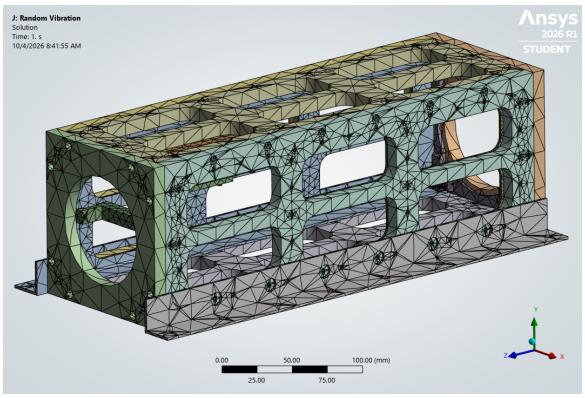


(a) Configuration 1 geometry

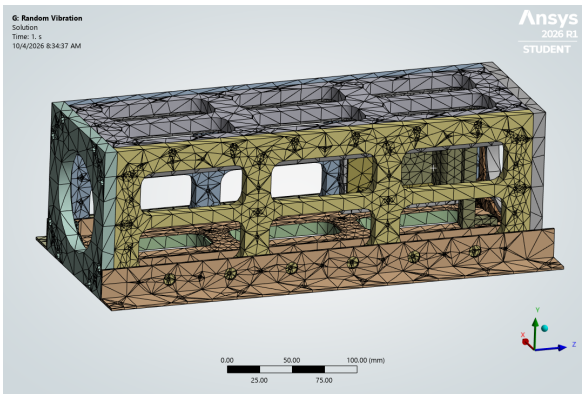


(b) Configuration 2 geometry

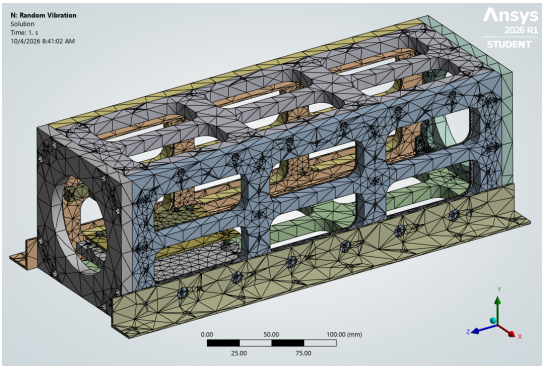
Figure 3.8 Vertical configurations 1 and 2



(a) Configuration 3 geometry



(b) Configuration 4 geometry



(c) Configuration 5 geometry

Figure 3.9 Horizontal configurations 3, 4, and 5

We performed four analyses, namely static structural, Modal, harmonic response, and random vibrations, linked in the following way

Static structure –> Modal –> Random vibrations

Static structure –> Modal –> Harmonic response

to keep the boundary conditions used in static structure analysis throughout the dynamic analyses.

In static structure analysis for each configuration, we applied 10 g quasi-static acceleration multiplied by a qualification factor of 1.25 in $\pm X, \pm Y, and \pm Z$. For every configuration, we applied modal analyses (sec-

tion 3.5.2). We also performed one random vibration analysis (section 3.5.3) and one harmonic response analysis (section 3.5.4) for each pre-stressed modal analysis.

For meshing, each configuration had its own mesh characteristics because the vertical and horizontal configurations differed in element size, since our L-bracket supports differ in size (Table 3.18).

In the meshing process, we defined a coarse mesh for the whole assembly, with a local sizing of 7–8 mm for the components of interest (heat spreader and Jetson plate), where we evaluate stresses, deformations, and frequency-related responses, to keep the ANSYS Student Edition computational cost adequate.

	1	**2**	**3**	**4**	**5**	**Unit**
Whole assembly element size	50	50	45	45	45	mm
Heat spreader and Jetson plate element size	7	7	8	8	8	mm
Nodes	118193	118193	119105	118741	118883	–
Elements	59586	59586	59592	59235	59314	–

Table 3.18 Meshing settings for the different configurations

3.5.1 Static structure

Static structure analysis assesses the structure under quasi-static launch accelerations. It aims to solve the equation (equation 3.29) [61]

$$[K]\, u = F \tag{3.29}$$

[K]: global stiffness matrix
u: displacement vector
F: equivalent nodal forces

Using ANSYS Mechanical, we performed a 2-step nonlinear static structural analysis (Table 3.19) to evaluate the structural integrity of the passive copper heat spreader and the Jetson plate, manufactured to support the NVIDIA Jetson Orin Nano on the CubeSat rails.

(a) Analysis settings for substep 1

Step controls	
Number of steps	2
Current step number	1
Step end time	1 s
Auto time stepping	On
Define by	Substeps
Initial substeps	5
Minimum substeps	5
Maximum substeps	50

(b) Analysis settings for substep 2

Step controls	
Number of steps	2
Current step number	2
Step end time	2 s
Auto time stepping	On
Define by	Substeps
Carry over time step	Off
Initial substeps	10
Minimum substeps	10
Maximum substeps	100

Table 3.19 Analysis settings for the two steps

For each configuration, we applied boundary conditions (Fixed support, bolt pretension, and acceleration). For the L-shaped frames, we applied the fixed-support constraint at the mounting-holes level for each configuration. These frames were designed to mount the structure on the shaker table. At step 1, we applied a bolt pretension (equation 3.30) to the different screws and spacers [7], whose materials were

stainless steel 304 with a yield strength of 205 MPa [92]. For simplicity, Ansys supports modeling the screws as smooth cylinders [15].

$$0.75 \times \text{Proof strength} \times A_{\text{t}} = 657.47 \text{ N} \tag{3.30}$$

We defined the proof strength as (equation 3.31)

$$\text{Proof strength} = 0.85 \times \text{Yield strength} \tag{3.31}$$

and the tensile-stress area is given by (equation 3.32)

$$A_{\text{t}} = \frac{\pi}{4}\left(\frac{\text{Pitch diameter} + \text{Minor diameter}}{2}\right)^2 \tag{3.32}$$

In step 2, we set the bolt pretension to "lock" (table 3.20).

Table 3.20 Bolt pretension application across the different substeps

Steps	defined by	Preload [N]
1.	load	657.47
2.	lock	N/A

We also applied an acceleration of 10g multiplied by a qualification factor of 1.25, considering the gravitational acceleration as g= $9.8067\ m/s^2$ in the $\pm X, \pm Y, and \pm Z$ axes, each at a time, since static structural analysis depends on the direction [14] (Table 3.21).

Steps	Time [s]	X [$\mathrm{mm/s^2}$]	Y [$\mathrm{mm/s^2}$]	Z [$\mathrm{mm/s^2}$]
1	0	0	0	0
1	1	0	0	0
2	2	$\pm$ 12.5 g	0	0

Steps	Time [s]	X [$\mathrm{mm/s^2}$]	Y [$\mathrm{mm/s^2}$]	Z [$\mathrm{mm/s^2}$]
1	0	0	0	0
1	1	0	0	0
2	2	0	$\pm$ 12.5 g	0

Steps	Time [s]	X [$\mathrm{mm/s^2}$]	Y [$\mathrm{mm/s^2}$]	Z [$\mathrm{mm/s^2}$]
1	0	0	0	0
1	1	0	0	0
2	2	0	0	$\pm$ 12.5 g

Table 3.21 Accelerations applied in $\pm X$, $\pm Y$, and $\pm Z$ for the different configurations

To facilitate convergence, we applied automatic time-stepping with an initial 5 time steps in step 1. In contrast to step 1, step 2 used automatic time-stepping with 10 initial sub-steps (minimum 10, maximum 100) because the addition of 12.5g acceleration to the already pre-loaded assembly required finer stepping to ensure convergence of the Newton-Raphson solver (Table3.19) and we enabled large-deflection controls to account for nonlinearities [16].

For connections, we applied frictional contacts with a conservative friction coefficient of $\mu = 0.2$ to the contact zones of threaded screws-unthreaded holes (e.g., Screw- Jetson PCB and Screw-CubeSat structure), and bonded contact for the rest.

Normally, for a dry contact between steel and dry aluminum, the friction coefficient μ varies between 0.45 and 0.61 [43]. As a limitation, we did not have the friction coefficient for our NVIDIA Jetson board. As an approximation, in a g-10/FR-4 laminate manufacturer's datasheet, they stated a friction coefficient of μ in 0.25-0.30 [56]. Therefore, we set all frictional contacts to $\mu = 0.2$ as a conservative value. Subsequently, we evaluated equivalent von Mises stresses and total structural deformations for the Heat spreader and the Jetson plate at t=2 seconds.

3.5.2 Modal analysis

To perform modal analysis, we could consider a free-free and a hard-mounted configuration. In this case, we considered only the hard-mounted configuration, which includes boundary constraints inherited from static structural analysis because the launcher does not support a free-free modal analysis.

We conducted a pre-stressed modal analysis to determine the dynamic behavior and natural frequencies of the whole structure, including the heat spreader and the Jetson plate.

Note: The SSMS VEGA-C manual does not specify performing a pre-stressed modal analysis. However, we used this approach to capture stress-stiffening effects that alter the structure's effective stiffness and, hence, the natural frequencies and mode shapes.

Modal analysis aims at solving the following equation (equation 3.33)

$$[M]\ddot{x} + [K^*]x = 0 \tag{3.33}$$

with a harmonic motion of (equation 3.34)

$$x(t) = \phi \sin(\omega t) \tag{3.34}$$

This leads to solving the eigenvalue equation (equation 3.35)

$$[K^*]\,\phi_i = \omega_i^2\,[M]\,\phi_i \tag{3.35}$$

which aims to find the eigenfrequencies. [8] [9].

[K*]: pre-stressed stiffness matrix

ϕ_i: eigenvector denoting the ith mode shape

ω_i : natural circular frequency with $f_i = \frac{\omega_i}{2\pi}$ denoting the natural frequency

[M]: mass matrix

Natural frequencies reveal the frequencies at which the structure would vibrate without external intervention. Knowing them helps in avoiding resonances that could lead to a negative outcome.

For each configuration, we performed six pre-stressed modal analyses related to the static structural analysis in different directions: $\pm X, \pm Y, and \pm Z$. Modal analysis is inherently linear. Before linking the pre-stressed modal analysis to the static structural analysis, we replaced the frictional contacts with bonded contacts, even though pre-stress linking in ANSYS results in nonlinear contact linearization. We followed this approach to avoid additional variability.

We extracted 30 modes ranging from 0 to 2000 Hz to capture structural modes. Based on the SSMS VEGA-C user manual, the first natural frequency must exceed 115 Hz (table 3.22) [10].

Table 3.22 Analysis settings for modal analysis

Options	
Max modes to find	30
Limit search to Range	yes
Range minimum	0 Hz
Range maximum	2000 Hz

From the modal analysis, we extracted the natural frequencies and cumulative effective mass fraction to assess which frequencies dominate in each direction. We also retrieved the mode shapes.

Note: We selected one representative modal solution per configuration as the basis for further simulations, namely random vibrations and harmonic response, because the modal analysis results differ only at the decimal level (section 4.4.2).

3.5.3 Random vibrations

Since many physical excitations are random and cannot be described as a function of time, the reaction to them is non-deterministic.

Using the mode superposition method, we linked random vibration analysis to the pre-stressed modal analysis and performed it to evaluate the structural response under random vibration launch-phase environments using PSD-G loads from the SSMS VEGA-C user, natural frequencies, mode shapes, and modal effective mass fractions from the modal analysis.

We applied a worst-case scenario combining SSMS top-central and Tower positions (GRMS=8.8g), with a damping ratio of 0.01, which ANSYS supports and considers a default value [11].

NASA also specifies this value as suitable when test data is unavailable [3].

Furthermore, a qualification factor of 2.25 is used for the PSD-G loads that we retrieved from Table "4.3.3.3a-Random levels–SSMS Top central and Tower positions" in the SSMS VEGA-C user manual [21].

Multiplying those random levels by the qualification factor 2.25, we get these PSD-G accelerations (Table 3.23):

Frequency [Hz]	**G acceleration [G^2/Hz]**
20	1.5975×10^{-2}
150	1.5975×10^{-2}
200	1.2667×10^{-2}
270	1.2667×10^{-2}
300	0.225
400	0.225
450	1.2667×10^{-2}
1000	7.65×10^{-2}
2000	6.5925×10^{-2}

Table 3.23 Random levels applied in ANSYS Mechanical after applying the qualification factor

3.5.4 Harmonic response

SSMS VEGA-C manual states that: "The objective of the sine vibration tests is to verify the spacecraft secondary structure dimensioning under the flight limit loads multiplied by the appropriate safety factors".

Harmonic response resolves the deterministic sinusoidal excitation problem (equation 3.36)

$$\left(-\omega^2[M] + [K^*] + j\,\omega[C]\right)\, u(\omega) = F(\omega) \tag{3.36}$$

with

$$u(\omega) = \sum_i \phi_i\, q_i(\omega) = \sum_i \phi_i\, H_i(\omega)\, F_i(\omega) \tag{3.37}$$

representing the response (equation 3.37) and

$$H_i(\omega) = \left(\omega_i^2 - \omega^2 + 2\,j\,\zeta_i\,\omega_i\,\omega\right)^{-1} \tag{3.38}$$

representing the modal frequency response (equation 3.38) linked to the pre-stressed modal analysis to evaluate the assembly's steady-state dynamic response to sinusoidal excitation.

Note: $[K^*]$, $[C]$, $[M]$, and ϕ_i denote the pre-stressed stiffness matrix, damping matrix, mass matrix, and mode shape, respectively, ω: excitation angular frequency, and $F(\omega)$: harmonic force vector [12] [38].

By linking them together, we preserve fastener clamping stiffness and base boundary conditions inherited from the static structure.

We applied harmonic excitation (Table 3.24) as a continuous sine sweep from 5 to 125 Hz in the 3 different axes X, Y, and Z, to capture resonant peaks accurately, and a damping ratio of 0.01, which is supported by ECSS spacecraft structural dynamic guidance, since they normally take values in the range of 1-2% [46].

The sine levels (Table 3.25) follow the SSMS VEGA-C qualification profile from Table "4.3.3.2a-Sine levels in the SSMS manual" [21].

Step controls	
Options	
Frequency spacing	Linear
Range minimum	5 Hz
Range maximum	125 Hz
Cluster number	10
Cluster results	Yes
Damping controls	
Damping defined by	Damping ratio
Damping ratio	1×10^{-2}

Table 3.24 Harmonic response analysis settings

Frequency [Hz]	**Acceleration [$\mathrm{mm/s^2}$]**	**Phase angle [°]**
5	24517.25	0
69.99	24517.25	0
70	12258.125	0
110	12258.125	0
125	12258.125	0

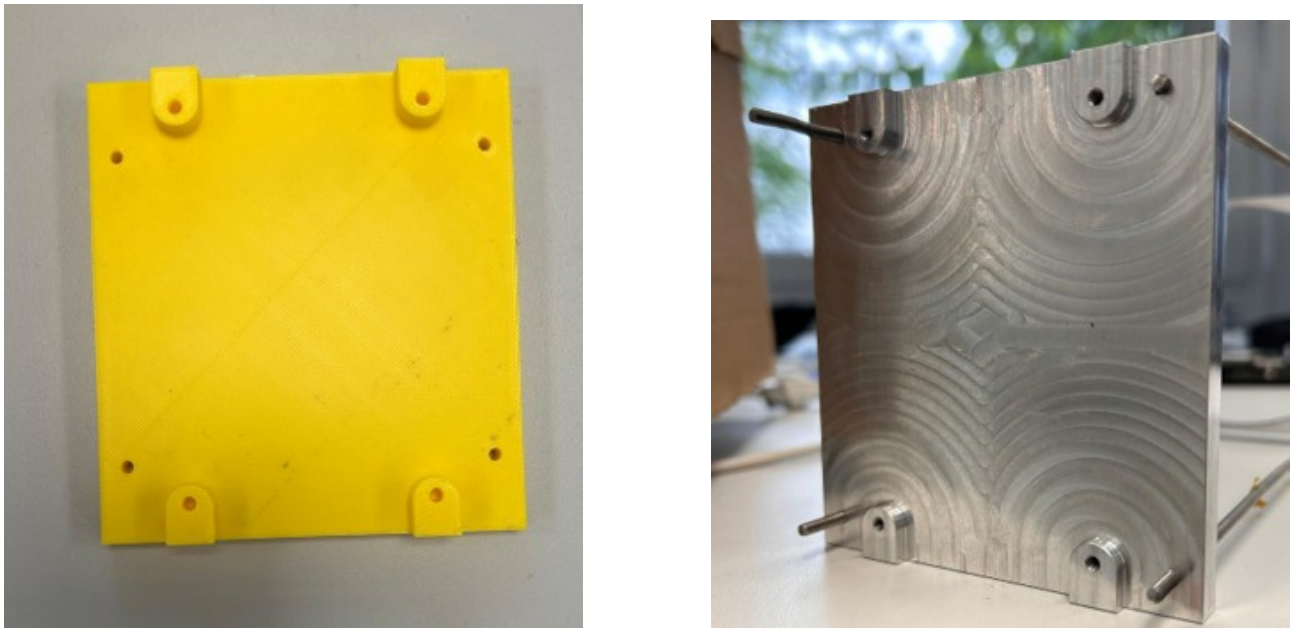
Table 3.25 Qualification sine levels

Note: In the manual, it is stated that 2.5g belongs to the range 5-70 Hz and 1.25g belongs to 70-110 Hz. Nevertheless, ANSYS does not allow entering two different accelerations for the same frequency. For this reason, we entered 2.5 g at 69.99 Hz.

In GitLAB, "VAconfig1.wbpj" contains configurations 1 and 4, "VAconfig2.wbpj" contains configuration 2, and "VAconfig5.wbpj" contains configurations 3 and 5 simulations [30].

3.6 Manufacturing process

As we did for the Cu 2.0060 heat spreader, an initial PLA model was 3D printed to make sure it fits the testing pod made for vibrational experiments, which is a 3U CubeSat [87] (figure 3.10a). After confirming the dimensions, we CNC-machined the Jetson plate (figure 3.10b).



(a) 3D printed PLA Jetson plate

(b) CNC machined Jetson plate

Figure 3.10 3D-printed and CNC-machined Jetson plates

Note: The Jetson plate's mounting holes inside the CubeSat structure must be unthreaded. However, since the subassembly (Jetson plate + NVIDIA Jetson Orin Nano + Heat spreader) does not fit properly inside the 3U testing CubeSat structure, we had to add threaded holes to mount the Jetson plate statically to the 3U testing CubeSat structure with M3 screws.

4 Results

In this section, we present the results of the different simulations and experiments, which we compare and discuss.

4.1 ANSYS orbital simulation

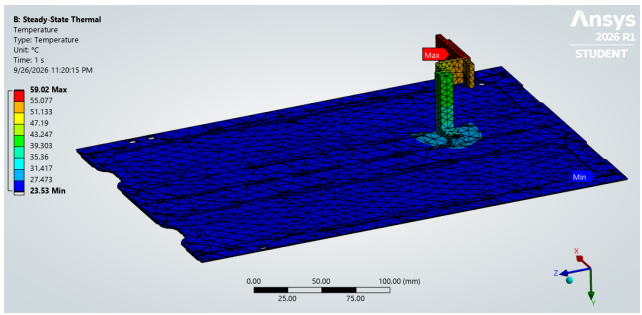
The equilibrium temperature script covering the month of May 2026 in MATLAB [30] produced a peak equilibrium sink temperature of 23.93 $°C$, which is 1.08 $°C$ higher than that of January 2026 [88]. After a sensitivity analysis across all the months of 2026, this temperature seems to be the worst-case scenario. This value leads to a resistance requirement of 5.33 K/W. Using the $T_{sink} = 23.93°C$ as a boundary condition, we implemented the simulation "orbitalconditions" once with A5083 and once with Cu 2.0060 [30].

After retrieving the results, we must implement a safety margin of 5 $°C$ [80]. Using aluminum A5083 for our heat spreader model, we got a temperature of 59.02 $°C$ (figure 4.1a) (table 4.1).

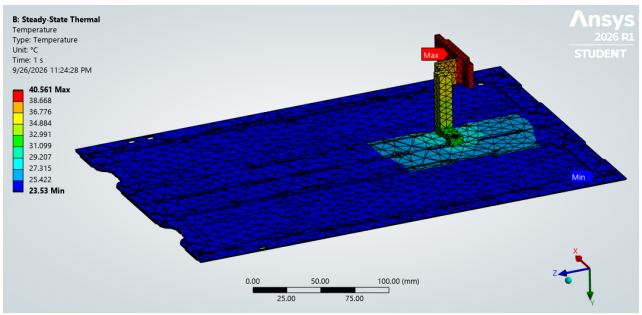
Nevertheless, the simulation with the Cu 2.0060 model gave a maximum temperature of T = 40.475 $°C$ (figure 4.1b) (table 4.1).

	Simulation results [°C]	+5°C margin [°C]
Cu 2.0060	40.475	45.475
A5083	59.02	64.02

Table 4.1 Simulation results and required safety margin



(a) Steady- state temperature using the aluminum A5083 heat spreader



(b) Steady-state temperature using the Cu 2.0060 heat spreader

Figure 4.1 Orbital simulation results for the aluminum A5083 and Cu 2.0060 heat spreaders

After applying the safety margin, the steady-state temperature for the A5083 heat spreader tended to exceed the thermal constraints, violating our operational limits. In contrast, Cu 2.0060 showed a margin of 14.525 $°C$ below the limit.

As a result, the total resistance calculated from this simulation result was 2.44 $°C$/W, corresponding to 46% of the allowable resistance of 5.33 $°C$/W.

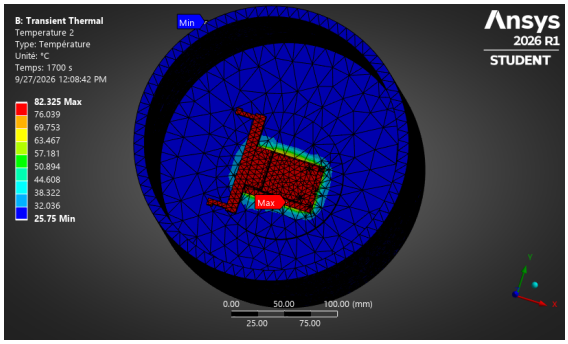
These thermal results are not sufficient to validate our choice. Instead, we still need to perform vibrational simulations and thermal testing to assess whether the current design and material choice fulfill our vibrational requirements.

4.2 Vacuum chamber simulations

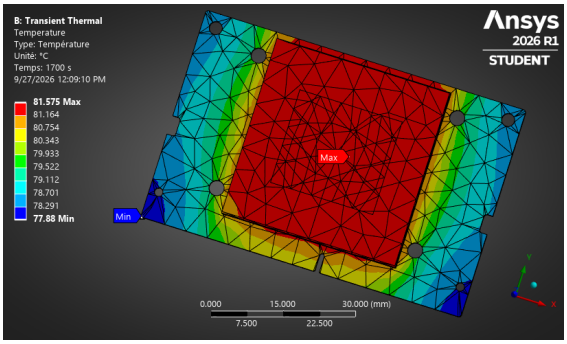
To assess the influence of the heat spreader and the added conduction paths to the structure, rods, and Jetson plate on the NVIDIA Jetson, we performed steady-state and transient thermal simulations for configurations A, B, and C. It's important to know that in those simulations we chose Stannol T12 as the thermal interface material (TIM) with a thermal conductivity of k= 4.185 $\frac{W}{mK}$, except for configuration C, where we applied a thermal interface material (TIM) with a thermal conductivity of k= 13.9 $\frac{W}{mK}$.

4.2.1 Configuration A

Within 1700 seconds (28 minutes and 20 seconds), the NVIDIA Jetson Orin Nano 8GB reached 81.76 $°C$ at the SoC0 level (figure 4.2). We chose a time interval of 1700 seconds to match our experiments' duration after adjusting the shutdown temperature to a controlled 90 $°C$, allowing us to compare our simulation results with those of the experiment (section 4.3.1).



(a) Configuration A meshed geometry inside the vacuum chamber

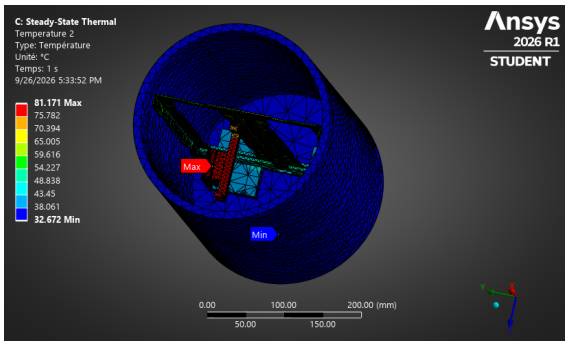


(b) temperature evolution across the Jetson module

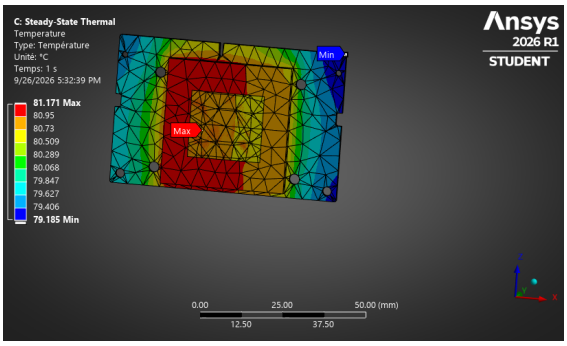
Figure 4.2 Transient thermal simulation of configuration A within 1700 seconds inside the vacuum chamber

4.2.2 Configuration B

To assess the NVIDIA Jetson's temperature evolution after adding the conduction path to the CubeSat structure, we performed steady-state temperature and transient thermal simulations. In steady-state analysis, the NVIDIA Jetson achieved a value of 81.17 $°C$ with a margin of roughly 23.33 $°C$ to the shutdown temperature stated by NVIDIA 104.5 $°C$ and a margin of 8.83 $°C$ to the forced shutdown temperature of 90 $°C$ that we altered (figure 4.3).



(a) Configuration B meshed geometry inside the vacuum chamber

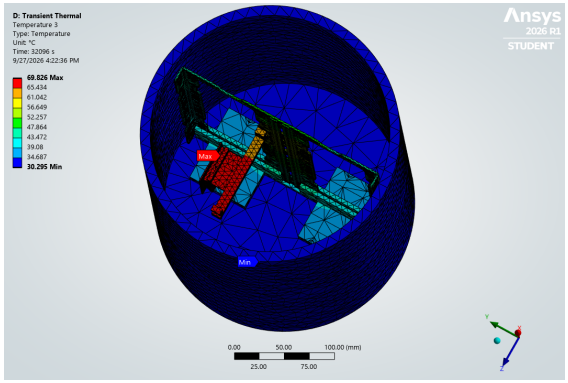


(b) temperature evolution across the Jetson module

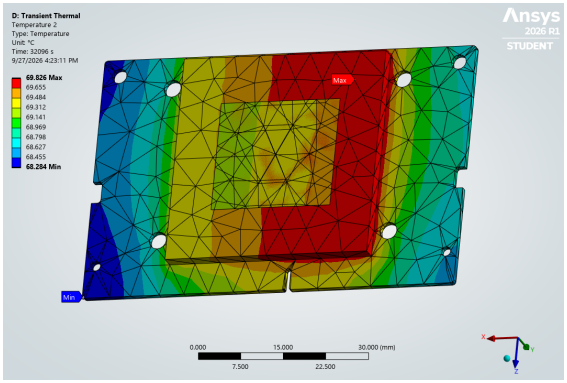
Figure 4.3 Steady state thermal simulation of configuration B

Also, we performed a transient thermal analysis of 32096 seconds (8 hours and 55 minutes) to capture the thermal evolution matching the experimental duration to compare them in section 4.3.2.

Reaching 69.83 $°C$ within 32096 seconds (figure 4.4) in configuration B, compared to 81.76 $°C$ within 1700 seconds in configuration A, is a significant improvement and indicates that the conduction path to the structure improved thermal management.



(a) Configuration B meshed geometry inside the vacuum chamber

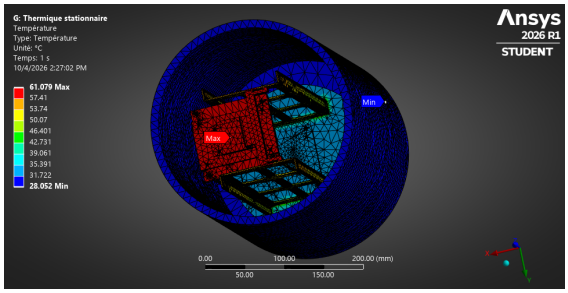


(b) temperature evolution across the Jetson module

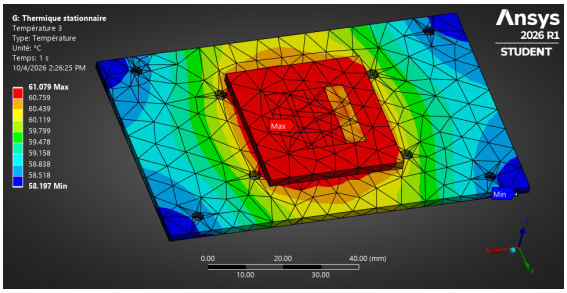
Figure 4.4 Transient thermal simulation of configuration B within 32096 [s] inside the vacuum chamber

4.2.3 configuration C

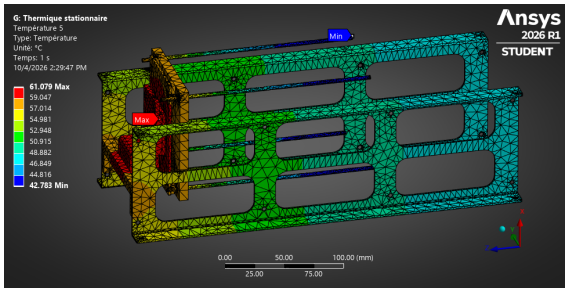
The plots (figure 4.5) show the steady-state thermal simulation results for configuration C. This simulation showed a maximum NVIDIA Jetson Orin Nano temperature of 61.079 $°C$ (figure 4.5b) at steady state. The rods showed temperatures ranging from 42.783 $°C$ to 57.594 $°C$, with higher values near the NVIDIA Jetson and the plate (figure 4.5d) and a minimum value at the far-end extremity of the rods.



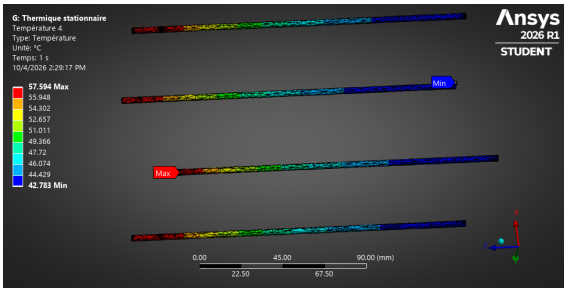
(a) Steady-state simulation results of Configuration C inside the vacuum chamber



(b) Temperature variation across the Jetson module model (SoC0 + PCB)



(c) temperature variation through the CubeSat structures and the rods

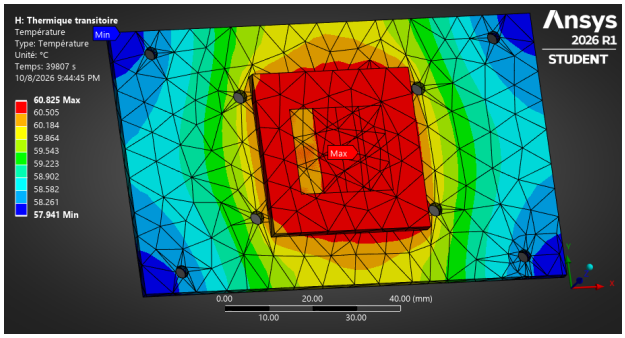


(d) Temperature evolution across the CubeSat's rods with a maximum value in proximity to the Jetson plate and a minimum value at the rods' far ends

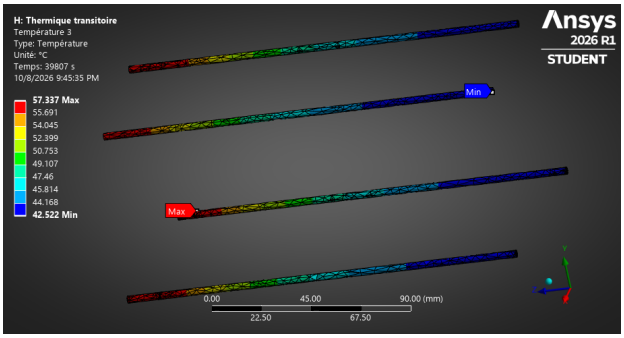
Figure 4.5 Steady- state thermal results of configuration C

We cannot compare this steady-state result (61.079 $°C$) after adding the second CubeSat structure with Configuration B (81.171 $°C$), because the NVIDIA Jetson power in this configuration is roughly 1.17 W, equivalent to $3.1e^{-4}$ W/mm^3 internal heat generation, less than that of Configuration B. However, reducing the power by 1.17 W led to a 20.292 $°C$ drop in temperature.

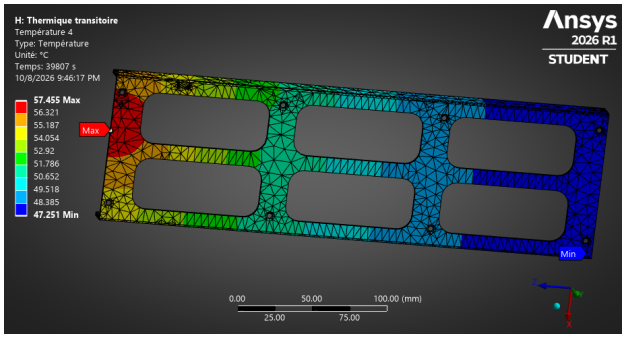
In the transient thermal analysis for this configuration (figure 4.6), we obtained, within 39087 seconds (roughly 11 hours and 27 minutes), a temperature of 60.825 $°C$ (figure 4.6a) at the NVIDIA Jetson Orin Nano/NX, with a margin of 0.254 $°C$ below the steady-state value, and the rods' temperature varied between 42.52 $°C$ and 57.34 $°C$ (figure 4.6b). Across the CubeSat structure, the temperature varied between 47.25 $°C$ and 57.46 $°C$ (Figure 4.6c), and for the Jetson plate, it varied between 57.86 $°C$ and 58.09 $°C$ after 39807 seconds (Figure 4.6d).



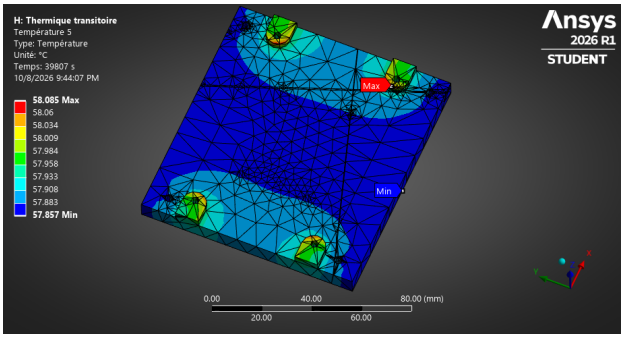
(a) Temperature achieved by the NVIDIA Jetson Orn Nano within 39807 seconds



(b) Temperature achieved by the rods within 39807 seconds



(c) Temperature achieved by the CubeSat structure within 39807 seconds



(d) Temperature achieved by the Jetson plate within 39807 seconds

Figure 4.6 Transient thermal simulation within 39807 seconds in configuration C

The curves (Figure 4.7) demonstrate the temperature evolution across the SoC0, the rods, the Jetson plate, and one CubeSat structure in configuration C during a 39807-second transient thermal analysis. The SoC0 temperature increased from 38.65 $°C$ to 60.83 $°C$. Meanwhile, the rods and the CubeSat structure increased from 36.39 $°C$ to 57.34 $°C$ and from 36.36 $°C$ to 57.46 $°C$, respectively, at the hottest part near the NVIDIA Jetson and heat spreader subassembly. The Jetson plate temperature also increased from 36.75 $°C$ to 58.09 $°C$. These results show that in the conduction path, the mounting positions of the NVIDIA Jetson to the A5083 Jetson plate are the parts that heat up by 0.69 $°C$ more than the CubeSat structure and the rods, and it is 2.74 $°C$ less than SoC0.

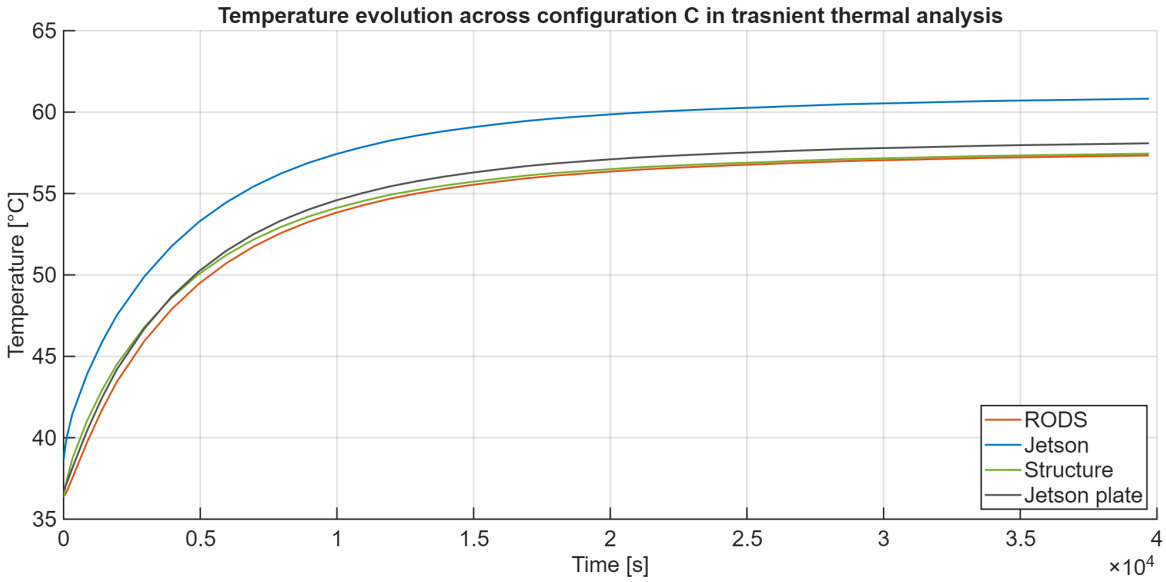


Figure 4.7 Temperature evolution in the SoC0, the rods, the CubeSat structure, and the Jetson plate in Configuration C during 39807 seconds transient thermal analysis

Table 4.2 Steady-state and transient thermal simulation results for configurations A, B, and C.

		Steady-State	Transient thermal	
		Temperature [°C]	**Duration [s]**	**Temperature [°C]**
Configuration A		–	1700	81.58
Configuration B		81.17	32096	69.83
Configuration C	SoC0	61.08	39087	60.83
	Rods	42.78–55.59	39087	42.52–57.34
	CubeSat structure	47.68–57.72	39087	47.25–57.46
	A5083 Jetson plate	58.11–58.34	39087	57.86–58.09

This table (table 4.2) summarizes our steady-state and transient thermal results for configurations A, B, and C.

To reach about 81 $^{\circ}C$, the NVIDIA Jetson Orin Nano with the heat spreader alone needs only 1700 seconds in configuration A. This value is the steady-state temperature with a conduction path to the CubeSat structure in configuration B.

In configuration B, 32096 seconds are not enough to reach thermal equilibrium (81.17 $^{\circ}C$). Instead, the steady-state temperature takes longer than 32096 seconds to be reached.

Adding the second CubeSat structure in configuration C, the four rods and the Jetson plate, provides additional conduction paths, and using a thermal interface material TIM with a thermal conductivity k=13.9 $\frac{W}{m.K}$ decreases the steady-state temperature by approximately 20 $^{\circ}C$ from 81.17 $^{\circ}C$ in configuration B to 61.08 $^{\circ}C$ in configuration C.

Within 39087 seconds, SoC0 reached 60.83 $^{\circ}C$, which is 0.25 $^{\circ}C$ less than steady-state, indicating that the experiment that we ran in the vacuum chamber with this same configuration reached roughly the thermal equilibrium (section 4.3.3).

The steady-state temperature decrease across the configurations was expected, as conduction heat transfer reduced it.

4.3 Experiments vs. simulations

In this section, we present our experimental results and compare them with our simulations across the different configurations.

The NVIDIA Jetson has multiple thermal zones, including GPU, SoC2, SoC0, T_j, CPU, etc., each reaching a different temperature at the same time, with T_j always showing a tendency to overheat more than the other thermal zones. For simplicity, we considered the temperature evolution at SoC0, which is part of the silicon chip area that we simulated and is normally 2–3 $^{\circ}C$ less than the T_j temperature.

Note: To assess the error between the simulated and the experimental results, we computed the maximum absolute error (equation 4.1)

$$\max_i \left[T_{\text{sim}}(t_i) - T_{\text{meas}}(t_i)\right] \tag{4.1}$$

Also, we calculated the peak error percentage (equation 4.2)

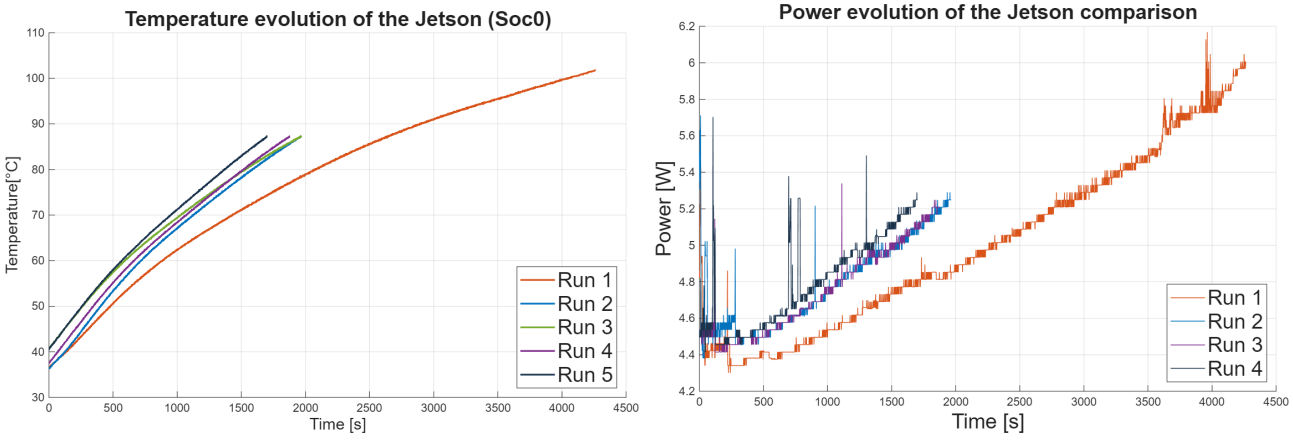
$$\frac{T_{meas,max} - T_{sim,max}}{T_{meas,max}} \times 100 \tag{4.2}$$

4.3.1 Configuration A

We repeated this experiment 5 times to compare the results of temperature and power evolution effectively (Figure 4.8), and we plotted the curves of SoC0 and VDD-in using MATLAB scripts "ConfigAruns.m" and "PowerA.m" [30]. The following table illustrates the initial and final temperature of the SoC0 in each run of configuration A (table 4.3).

Table 4.3 Temperature range and duration of the experimental runs

Run no.	T_{initial} $[^{\circ}C]$	T_{final} $[^{\circ}C]$	Duration [s]
1	36.50	101.80	4263
2	36.20	87.20	1964
3	40.63	87.28	1967
4	37.59	87.25	1878
5	40.53	87.16	1700



(a) Comparison of the Temperature evolution of the NVIDIA Jetson (Soc0) across different runs

(b) Comparison of the Power evolution of the NVIDIA Jetson across different runs

Figure 4.8 SoC0 temperature and power evolution of the NVIDIA Jetson Orin Nano in configuration A

Table 4.4 Cutoff slope of the temperature evolution plots in the last $\Delta t = 133[s]$

	Run 1	Run 2	Run 3	Run 4	Run 5
Cutoff slope [$^{\circ}C$/s]	0.016	0.020	0.013	0.019	0.021

Run 5 tends to heat up faster than the other four runs, which heat up as follows: Run 2 > Run 4 > Run 1 > Run 3 according to the cutoff slope values (table 4.4) (Figure 4.8).

In the first run, the shutdown temperature was still 104.5 $^{\circ}C$, which explains its longer duration compared to the four other runs and it lacked the (PU) packaging foam, relying only on a plastic insulating bag.

The lack of convection during the initial transient step in the temperature evolution curves of runs 2-5 led to a steeper early temperature rise, as we began those runs directly inside the vacuum chamber, as mentioned in section 3.4.

The decrease in thermal conductivity from k= 13.9 [W/m.k] to a value of 4.185 [W/m.K] caused an increased resistance in heat circulation at the SoC0 and heat spreader levels, which explains the higher temperature of the last run compared to the previous ones and reflects the shorter time of the run (Run 5) compared to the third one (Run 3) though they both started roughly at the same temperature, as the heat finds it hard to escape the chip area due to the low thermal conductivity of the TIM used, so the heat stays trapped and leads to a faster shutdown. Power also fluctuated across runs, showing different behavior in each experiment. In fact, temperature and power are coupled. That's why the thermal path determines die temperature for a given power, and Jetson's Dynamic Voltage and Frequency Scaling (DVFS) and thermal-throttling policy adjust power in response to temperature [37].

From these plots (Figure 4.8), radiating heat from the heat spreader without a conduction path to another structure prevents steady-state operation and leads to an unwanted shutdown at $T_{shutdown}$. As a result, we added another conduction path to assess the behavior of the NVIDIA Jetson Orin Nano (section 4.3.2).

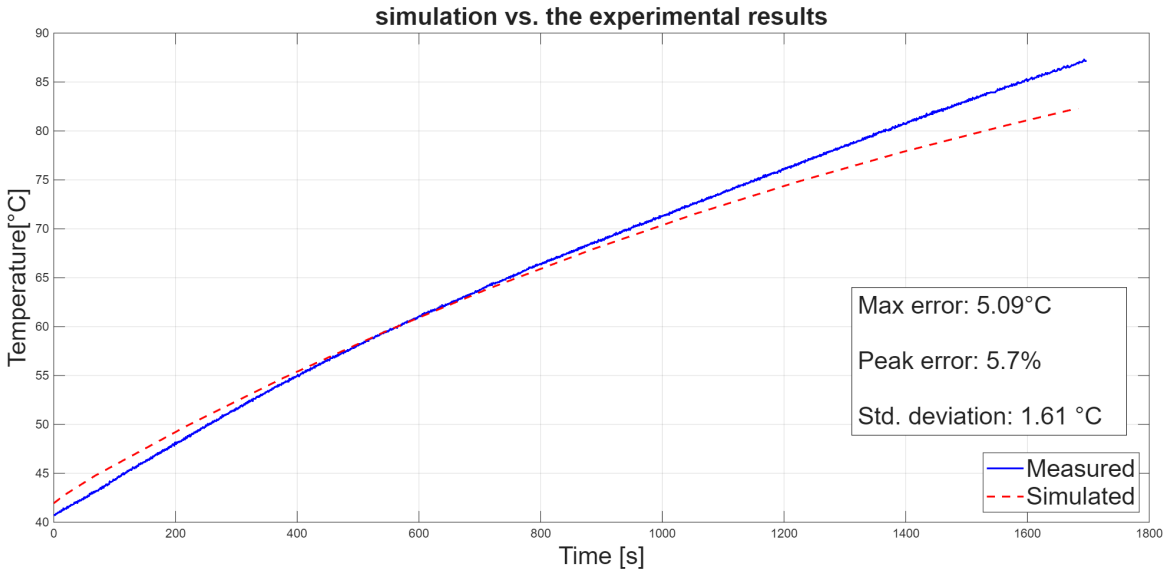


Figure 4.9 Simulated vs. measured SoC0 temperature of configuration A

From the MATLAB script "errorA.m" [30], the results of the comparison showed a maximum absolute error of 5.09 $^{\circ}C$, a peak error of 5.7% , and a standard deviation of 1.61 $^{\circ}C$ (figure 4.9).

From the plot (Figure 4.9), the simulation reproduces roughly the same characteristic shape as the experimental results. Nevertheless, the measured temperature was higher than the simulation at the end of the run, indicating that the simulation underestimates the temperature evolution.

This discrepancy was expected, as multiple experimental factors could contribute to the variation, namely the CNC finish of the heat spreader, which influences the emissivity value and the power bank's temperature rise. From the simulation side, we used a simplified CAD model of the NVIDIA Jetson Orin Nano.

For this reason, we considered the comparison only at the SoC0 level, as the other thermal zones were not modeled in the CAD file provided by NVIDIA. We also considered an effective emissivity, which we retrieved by trial and error.

Note: We documented the temperature and power evolution of the different runs in Excel files "heatpipe222", "exp1 heatpipe jetson", "exp2 heatpipe jetson", "exp3heatpipejetson", and "exp4heatpipejetson" [30].

Also, across all the runs, the shutdown temperature didn't reach 104.5 $°C$ or 90 $°C$. That goes back to the fact that our T_J reaches the shutdown temperature faster than the other thermal zones, leading to the shutdown behavior even though the other parts didn't reach it yet.

4.3.2 Configuration B

We repeated this experiment 4 times to check the accuracy of the results (Table 4.5).

Table 4.5 SoC0 Temperature range and duration of the experimental runs

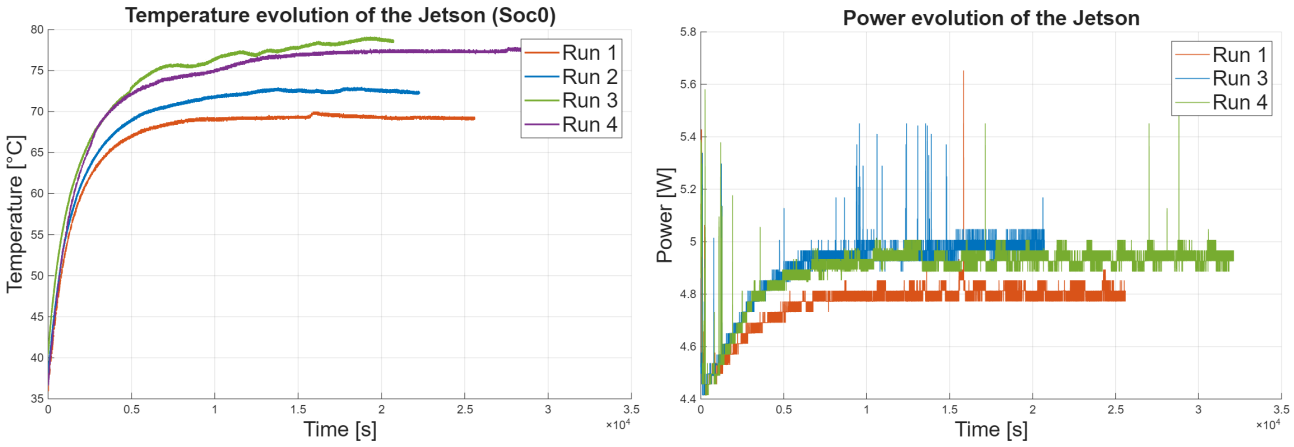
Run no.	T_{initial} $[°C]$	T_{final} $[°C]$	Duration [s]
1	36	69.1	25590
2	38	72.22	22567
3	40.50	78.72	20981
4	36.6	77.4	32096

The experiments differed in their durations due to the shutdown of the power bank. In the last run, we started the experiment with a fully charged power bank and kept it running until it shuts down (table 4.5).

We plotted the SoC0 temperature evolution and VDD-in plots (Figure 4.10) using the MATLAB scripts "ConfigBruns.m" and "PowerB.m" [30], respectively.

The three runs: Run 1, Run 3, and Run 4, showed different power behavior, as shown in the plots (Figure 4.10b). Their average power is 4.76 W, 4.89 W, and 4.89 W, respectively.

From the graphs, Run 1 reaches the lowest power and the lowest steady-state temperature (Figure 4.10a). In Run 2, we didn't register the power evolution mistakenly.



(a) Comparison of the Temperature evolution of the Jetson (SoC0) across different runs

(b) Comparison of VDD-In (Power) evolution of the Jetson across different runs

Figure 4.10 SoC0 temperature and the VDD-In (power) evolution plots

The four runs showed different heating behaviors, which reflect a combination of factors that varied slightly across the different runs. The steady-state temperature is mainly determined by electrical power consumption, while the thermal conductivity of the die-to-spreader TIM has only a secondary influence when the heat spreader-structure interface and the radiating area are taken into account. Despite the 1/3 transition in the thermal conductivity value of the thermal interface material, the roughly similar steady-state temperatures of the SoC0 in Runs 3 and 4 are due to the thermal path (equation 4.3)

$$R_{total} = R_{\text{die}\rightarrow\text{Heat spreader}} + R_{\text{heat spreader}\rightarrow\text{Structure}} + R_{\text{structure}\rightarrow\text{environment}} \tag{4.3}$$

For a TIM layer: $R_{TIM} = \dfrac{l}{K_{TIM}} + R1 + R2$, such that l is the thickness of the TIM layer.

Literature on TIMs shows that contact resistances can dominate the total interface resistance. Consequently, improving a single TIM's thermal conductivity has a limited influence when other resistances are large [83] [91].

Temperature and power are coupled, that's why the thermal path determines the die temperature for a given power, and the NVIDIA Jetson's DVFS and thermal throttling policy adjusts power in response to temperature [37].

Comparing the simulation and the experimental results via a MATLAB script "errorB.m" [30], we got a maximum absolute error of 9.08 $^{\circ}C$, a peak error of 10.2%, , and a standard deviation of 2.38 $^{\circ}C$ (figure 4.11).

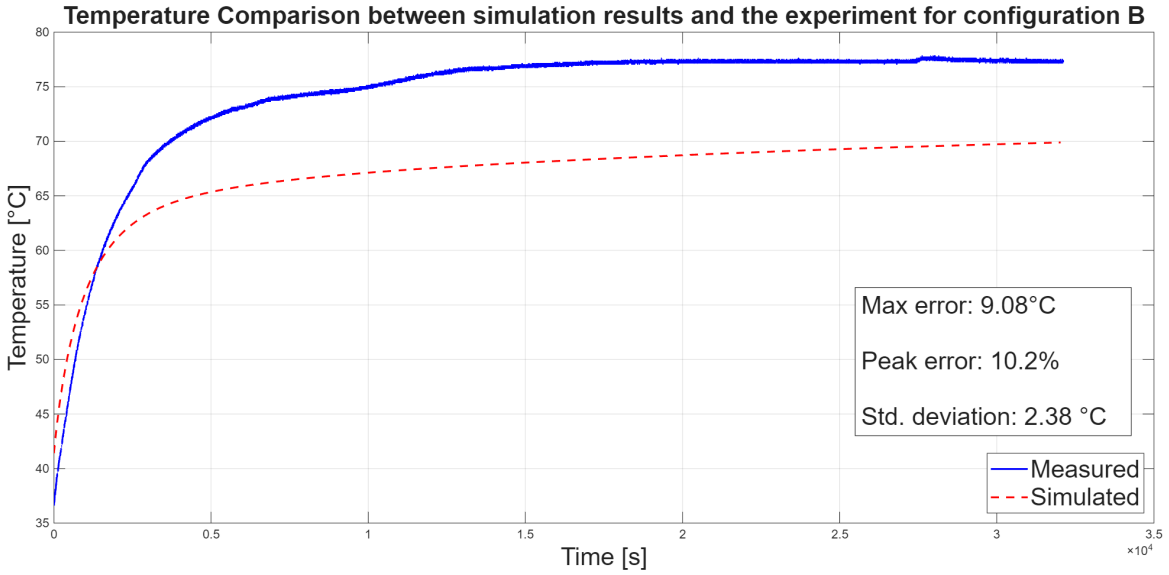


Figure 4.11 Simulated vs. measured SoC0 temperature of configuration B within 32096 seconds comparison

A mechanical factor that affected the thermal behavior in these Runs was how we mounted the Cu 2.0060 heat spreader relative to the CubeSat aluminum structure. During the different test cycles, the threaded mounting part of the copper heat spreader wore out, making it harder to insert the screw and achieve a secure mechanical connection between the heat spreader and the CubeSat structure. As a result, in the last runs, the copper heat spreader was not clamped to the structure as firmly as in Runs 1 and 2, increasing thermal contact resistance at this interface and reducing conductive heat transfer. In Runs 3 and 4, we held the heat spreader in place with Kapton tape instead of having a direct screw joint. This arrangement gave a much lower clamping pressure and a less reliable thermal interface than the bolted connections employed in the first and second runs. Consequently, poor mechanical contact at the heat spreader-structure interface in Runs 3 and 4 led to the observed rise in steady-state temperature.

The plot (figure 4.11) shows a simulated transient thermal analysis result of 69.83 $^{\circ}C$ after 32096 seconds, corresponding to 8 hours and 55 minutes, indicating that in configuration B, the simulated steady-state temperature of 81.17 $^{\circ}C$ (figure 4.3a) takes longer than 32096 seconds to be reached. In contrast, in

Run 4, we remark that the temperature plot stabilized at 77.4 $°C$ within 32096 seconds as well and differs by 3.77 $°C$ from the steady-state simulated temperature result and by 7.51 $°C$ from the transient analysis one achieved after 32096 seconds. In this case, the simulation is also underestimating the temperature evolution in the transient thermal. This could be due to an imperfect conduction path in the experiment, as mentioned at the beginning of this section, the power bank temperature increase, the CNC finish of the structure and the heat spreader, the simplified CAD model of the NVIDIA Jetson module, and the effective emissivities used.

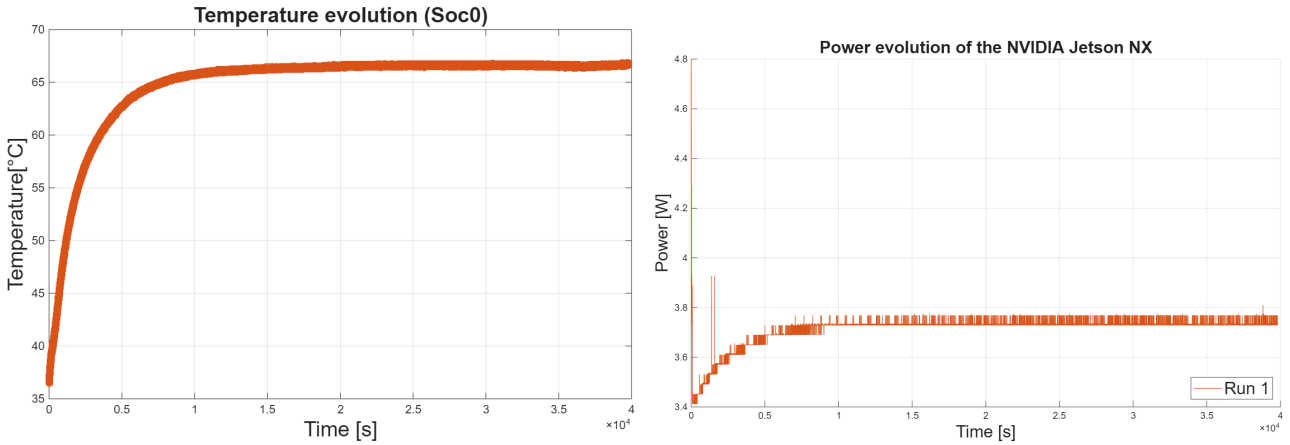
Before simulating 32096 seconds, we simulated 10000 s, 15000 s, and 20000 s. They gave a steady-state temperature of 67.14 $°C$, 68.08 $°C$, and 68.75 $°C$, respectively. Each time, by adding 5000s to the previous result, we assess an increase of 0.940 $°C$, 0.67 $°C$, and 0.88 $°C$ in the temperature, which does not indicate a fixed step temperature increase. This result won't show us exactly how much time it would take to reach the steady-state temperature in the simulation. As the simulation duration increased, the peak error decreased by about 1% each time.

Note: We documented the temperature and power evolution of the different runs in Excel files "structure444", "exp1+structure", "exp2+structure", and "structure final" [30].

4.3.3 Configuration C

This experiment took 11 hours, 13 minutes, and 32 seconds, and the NVIDIA Jetson temperature increased from 36.47 $°C$ to 66.75 $°C$ (figure 4.12). For the rod's temperature, we recorded only 7 hours. This experimental setup was not ideal because the heat spreader's threaded holes wore off, and the 2 CubeSat structures and the Jetson plate support may have added a conduction path between them.

To plot the SoC0 temperature evolution, VDD-in, and rod temperature curves, we used MATLAB scripts " ConfigCruns. m ", " PowerC.m ", and " rods. m " [30].



(a) Temperature evolution of the NVIDIA Jetson Orin nano NX SoC0

(b) Power evolution of the NVIDIA Jetson NX

Figure 4.12 NVIDIA Jetson Orin Nano/NX SoC0 and power evolution in configuration C

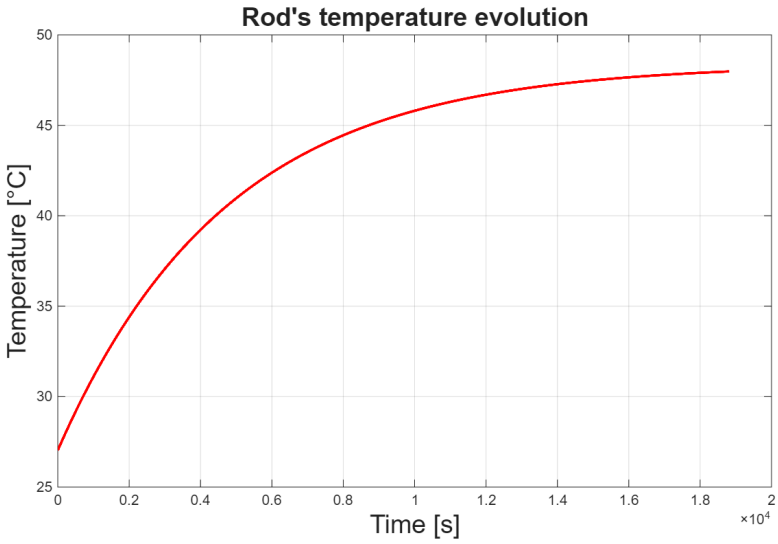


Figure 4.13 One rod temperature evolution

From these plots, the rod temperature did not reach thermal equilibrium (figure 4.13). The rod reached a maximum temperature of 48 $°C$, with a margin of 18.75 $°C$ compared to the maximum temperature of 66.75 $°C$ of the NVIDIA Jetson Orin Nano reached after 7 hours, as the rod's temperature is only recorded for seven hours. This indicates that the remaining heat is dissipated by conduction to the CubeSat structure and the Jetson plate, as well as by radiation.

Also, the NVIDIA Jetson Orin Nano/NX model dissipated less power than the NVIDIA Jetson 8GB. In fact, the average power calculated in this experiment was 3.71 W, and the steady-state power is approximately 3.73 W (Figure 4.12).

In configuration C, we used a power bank with a 25000 mAh nominal capacity and 92.5 Wh nominal energy to power NVIDIA Jetson Orin Nano/NX, which dissipated an average power of 3.71 W for a 39807-second duration. Whereas, in configuration B, even though we didn't reach the shutdown temperature, the experiment terminated after 32096 seconds due to a less powerful power bank with a nominal capacity of 20000 mAh and a nominal energy of 74 Wh powering the NVIDIA Jetson Orin Nano/NX, dissipating an average power of 4.89 W, which led to a faster shutdown of the Power bank. Using a more powerful battery for configuration C and lower average power for the NVIDIA Jetson explains its longer duration compared to configuration B.

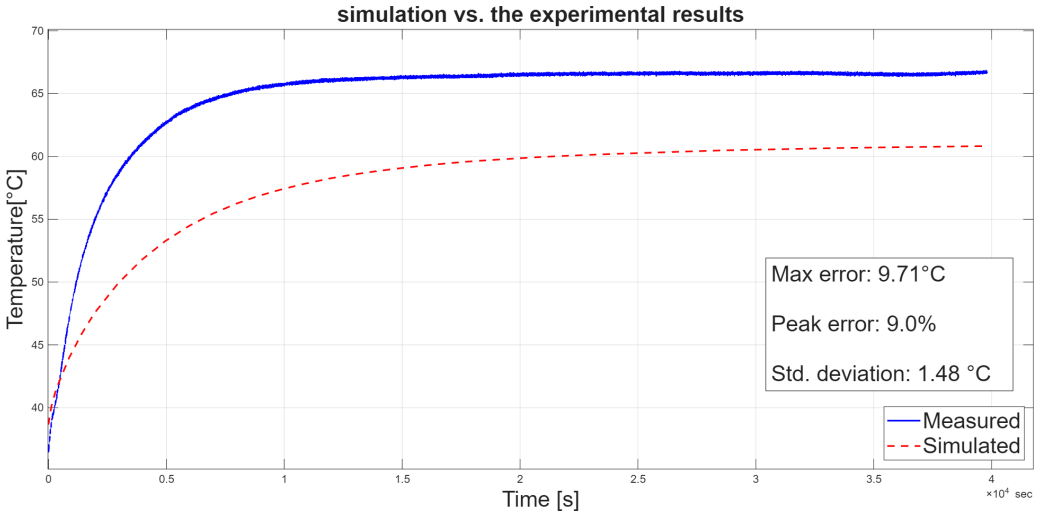


Figure 4.14 Simulated vs. measured SoC0 temperature of configuration C within 39807 seconds

Comparing this experimental result of 66.75 $°C$ with the simulation's steady-state temperature of 61.079 $°C$ at SoC0 (figure 4.14), we find that the simulation underestimates the result by 5.671 $°C$. In a transient thermal simulation of 39807 seconds, we obtained a maximum temperature of 60.83 $°C$, which is 5.92 $°C$ less than the experimental results. Overall, from the plot and a MATLAB script "errorC.m" [30], we got a maximum error of 9.71 $°C$, a peak error of 9.0%, and a standard deviation of 1.48 $°C$.

This variation could be due to the simplified CAD model of the NVIDIA Jetson Orin Nano and poor contact between the heat spreader and the CubeSat structure, which makes the experimental results higher than the simulation that assumes perfect contact and the effective emissivity values assigned to the components.

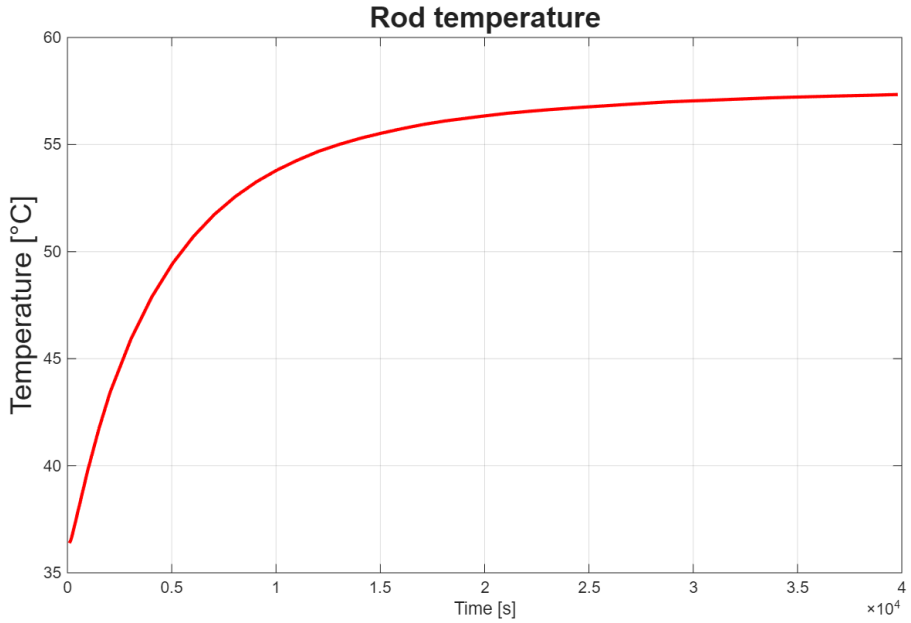


Figure 4.15 Simulation result of one rod temperature evolution

In the simulation, after 18000 seconds, the rods achieved a maximum temperature of 56.34 $°C$, which is 8.34 $°C$ more than the experimental result (Figure 4.15). This result may be due to poor contact pressure between the thermocouple tip and the rod.

Note: We documented the temperature and power evolution of this run in the Excel file "ConfigurationC" [30].

The results show that the NVIDIA Jetson Orin Nano exceeds 60 $°C$, even though orbital simulation predicted 40.475 $°C$. This variation does not mean our module will violate the operating limits in orbit, because in our vacuum chamber heat is rejected to the chamber walls at 25 $°C$, whereas in orbit the radiative sink is deep space at 3K.

In orbit, the satellite is fully assembled, including all the solar panels, structures, other payloads, and longer rods, whereas in our experiment we used shorter ones.

4.4 Vibrational analysis

In this section, we retrieved the static structure, modal, random vibrations, and harmonic response results to investigate whether our components would survive vibrations.

4.4.1 Static structure

In all configurations, the maximum Von Mises stress in the heat spreader and the Jetson plate didn't exceed the prescribed yield strength of each material (table 3.3), with safety margins of 1.61–3.05 and 13.66–37.87, respectively (tables 4.6, 4.7, 4.8, 4.9, and 4.10).

In all configurations, the maximum Von Mises stress in the heat spreader was concentrated at the mounting position to the NVIDIA Jetson glsPCB, which is treated as a flexible part. On the other hand, the Jetson plate has the maximum Von Mises stress concentrated at the mounting positions of the Jetson plate-CubeSat structure (Appendix A).

This is primarily due to the 657.47 N clamping force of the bolts. Moreover, all configurations showed insensitivity to acceleration load direction, as the stress results for each component fell on the same stress-value plateau within each configuration.

Table 4.6 Configuration 1's static structure maximum Von Mises stress and maximum total deformations of the heat spreader and the Jetson plate

	Heat Spreader			Jetson plate		
	stress [MPa]	Safety factor	total deformation [mm]	stress [MPa]	Safety factor	total deformation [mm]
(+X)	77.62	2.32	0.06	14.41	14.22	0.06
(-X)	77.33	2.33	0.06	14.07	14.57	0.06
(+Y)	77.44	2.32	0.06	15.004	13.66	0.06
(-Y)	77.51	2.32	0.06	14.16	14.48	0.06
(+Z)	77.45	2.32	0.003	14.15	14.49	0.002
(-Z)	77.50	2.32	0.003	14.32	14.32	0.002

In configuration 1 (table 4.6), the von Mises stress of the heat spreader and the Jetson plate varied between 77.33-77.62 MPa and 14.15-15.004 MPa, respectively. Both components showed the same deformation tendency in $\pm X$ and $\pm Y$ of 0.06 mm, and they differed by 0.001 mm in the Z direction, with the heat spreader reaching 0.003 mm and the Jetson plate 0.002 mm. The subassembly (Jetson plate + NVIDIA Jetson + Heat spreader) was attached at the furthest point from the support, reflecting a higher effective lever arm and hence greater deflection. This mounting topology produced the highest total deformation for both components compared with configurations 2, 3, 4, and 5. This result does not reflect a real deformation in the Subassembly. Instead, it is the subassembly moving together with the CubeSat structure under the imposed acceleration (figure 4.16).

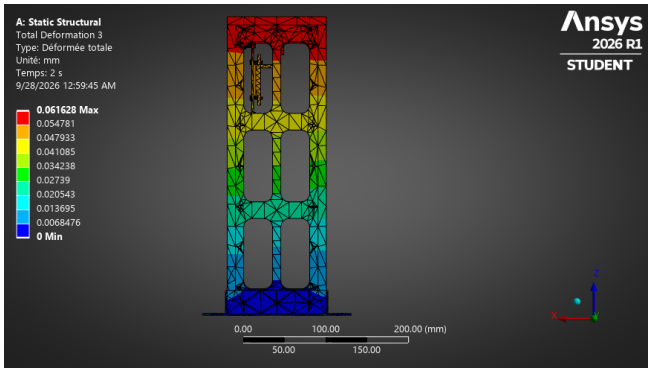


Figure 4.16 : total deformation of the CubeSat structure

Table 4.7 Configuration 2's static structure maximum Von Mises stress and maximum total deformations of the heat spreader and the Jetson plate

	Heat Spreader			Jetson plate		
	stress [MPa]	Safety factor	Maximum total deformation [mm]	stress [MPa]	Safety factor	total deformation [mm]
(+X)	77.34	2.33	0.01	14.61	14.03	0.02
(-X)	77.61	2.32	0.01	13.79	14.87	0.02
(+Y)	77.51	2.32	0.01	14.04	14.60	0.01
(-Y)	77.44	2.32	0.01	14.45	14.19	0.01
(+Z)	77.46	2.32	0.002	14.25	14.39	0.001
(-Z)	77.49	2.32	0.002	14.02	14.62	0.001

In configuration 2 (Table 4.7), relocating the subassembly closer to the support while keeping the same boundary conditions and joint-point relationships leaves the stress essentially unchanged, in the range of 77.34-77.61 MPa and 13.79-14.61 MPa for the heat spreader and Jetson plate, respectively. Compared to the first configuration, the deformation decreases significantly, reaching a range of 0.002-0.01 mm and 0.001-0.02 mm for the heat spreader and the Jetson plate, respectively, reflecting a shorter bending arm due to the reduced distance with respect to the support.

Table 4.8 Configuration 3's static structure maximum Von Mises stress and maximum total deformations of the heat spreader and the Jetson plate

	Heat Spreader			Jetson plate		
	Mises stress [MPa]	Safety factor	total deformation [mm]	stress [MPa]	Safety factor	total deformation [mm]
(+X)	77.69	2.32	0.01	5.57	36.80	0.01
(-X)	77.63	2.32	0.01	5.44	37.68	0.01
(+Y)	77.48	2.32	0.005	5.53	37.07	0.001
(-Y)	77.84	2.31	0.005	5.47	37.48	0.001
(+Z)	77.46	2.32	0.004	5.56	36.87	0.003
(-Z)	77.86	2.31	0.004	5.44	37.68	0.003

In configuration 3 (table 4.8), the structure is supported horizontally, and the subassembly is attached to the frame parallel to the one supported, routing the load through bending. This behavior reflects an essentially unchanged von Mises stress for the heat spreader, ranging between 77.46 MPa and 77.86 MPa, whereas it drops for the Jetson plate to values between 5.44 MPa and 5.57 MPa.

Regarding the deformation, the heat spreader had a deformation between 0.004–0.01 mm, whereas the Jetson plate deformation ranged from 0.001 mm to 0.01 mm.

Table 4.9 Configuration 4's static structure maximum Von Mises stress and maximum total deformations of the heat spreader and the Jetson plate

	Heat Spreader			Jetson plate		
	stress [MPa]	Safety factor	total deformation [mm]	stress [MPa]	Safety factor	total deformation [mm]
(+X)	58.98	3.05	0.01	12.15	16.87	0.01
(-X)	59.27	3.04	0.01	12.83	15.98	0.01
(+Y)	59.25	3.04	0.002	12.23	16.76	0.001
(-Y)	59.004	3.05	0.002	12.74	16.09	0.001
(+Z)	59.07	3.05	0.002	12.43	16.49	0.002
(-Z)	59.19	3.04	0.002	12.69	16.15	0.002

In configuration 4 (table 4.9), the heat spreader has the lowest von Mises stress of all configurations. A possible cause is that the perpendicular subassembly position engages the PCB mounting position less severely than the other configurations, reflecting a better-distributed load path, which gives a stress range

of 58.98 MPa to 59.27 MPa. Whereas the Jetson plate showed a stress between 12.15 MPa and 12.83 MPa.

At the deformation level, we get the same deformation in ±X and ±Z for the heat spreader and the Jetson plate, and they differ only in ±Y, reaching 0.002 and 0.001 mm, respectively.

Table 4.10 Configuration 5's static structure maximum Von Mises stress and maximum total deformations of the heat spreader and the Jetson plate

	Heat Spreader			Jetson plate		
	stress [MPa]	Safety factor	total deformation [mm]	stress [MPa]	Safety factor	total deformation [mm]
(+X)	111.58	1.61	0.002	6.56	31.25	0.001
(-X)	111.91	1.61	0.002	6.62	30.97	0.001
(+Y)	111.91	1.61	0.004	6.57	31.02	0.001
(-Y)	111.58	1.61	0.004	6.61	31.01	0.001
(+Z)	111.79	1.61	0.002	6.59	31.11	0.001
(-Z)	111.7	1.61	0.002	6.59	31.11	0.001

In configuration 5 (Table 4.10), the heat spreader achieves the highest von Mises stress compared to all other configurations, in the range of 111.70 MPA and 111.91 MPA. Whereas the jetson plate reaches a value between 6.56 and 6.62 MPa. Regarding deformation, we got 0.002 in $\pm X$ and $\pm Z$ and a value of 0.004 mm in $\pm Y$ for the heat spreader and a constant value of 0.001 mm for the Jetson plate across all directions.

Across all configurations, the Cu 2.0060 heat spreader always achieved higher stress than the Jetson plate.

The heat spreader has a stiffness mismatch, as only 2 points are bolted to the rigid structure, whereas the other 4 mounting points are mounted to a flexible component, namely the Jetson PCB. In fact, the load transfer between two clamped components is highly dependent on the relative stiffness of the fastener and the clamped components [41] [67]. The Cu 2.0060 heat spreader has a higher Young's modulus of $E_{(Cu,2.0060)} = 127000MPa$ compared to the PCB, which has an in-plane Young's modulus of 16900 MPa and an out-of-plane Young's modulus of 7400 MPa. This stiffness disparity creates a non-uniform stress distribution [53]. Nevertheless, the A5083 Jetson plate is attached to the Aluminum EN AW 6082 T6 CubeSat structure via 4 mounting points, which distribute the forces more evenly and reduce stress. In other words, the Young's moduli are roughly equal at 68000 MPa and 70000 MPa, respectively, which minimizes the stiffness mismatch. Also, the 657.47 N bolt pretension contributes to local stress at the mounting positions.

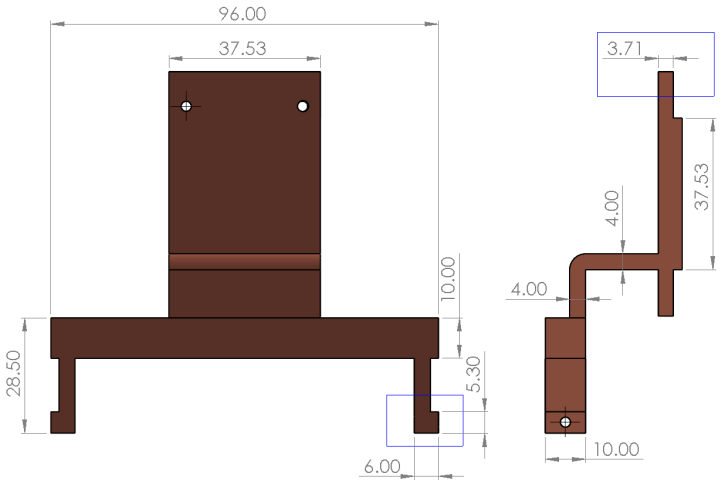
Looking more closely at the deformation figures in Appendix A, the peak deformation for the heat spreader occurs at the 3.71 mm part in configurations 2, 3, 4, and 5, and at the 6 mm thickness in configuration 1 (Figure 4.17). Whereas, for the Jetson plate, it happens at the 10.85 mm thickness in configurations 1, 2, and 3 and at the 7.60 mm thickness in configurations 4 and 5.

Although $E_{(Cu,2.0060)} > E_{A5083}$, the deformation of the heat spreader is equal to or larger than that of the Jetson plate (except in configuration 2 in the launch directions $\pm X$), driven essentially by the geometric stiffness difference, namely its lower bending stiffness.

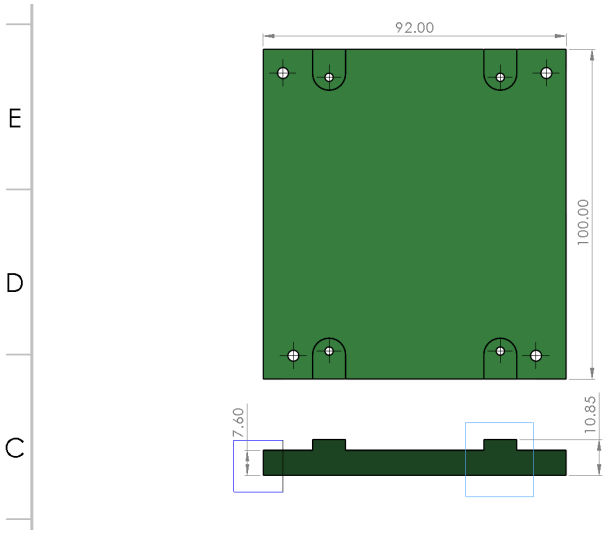
The bending stiffness is proportional to EI, where $I \propto t^3$ [47] such that I denotes the second moment of inertia and t: is the thickness of the piece.

Since in both thickness cases across the different configurations $t_{(heatspreader)} < t_{(jetsonplate)}$, this results in a lower moment of inertia, as the geometry outweighs the high Young's modulus of Cu 2.0060, which leads to lower EI for the heat spreader. Hence, the Jetson plate is geometrically stiffer than the heat spreader.

Due to the low A5083 Young's modulus, a larger thickness was of interest to boost the bending stiffness EI, improve the vibrational behavior, and decrease the potential deformation.



(a) 2D drawing of the Cu 2.0060 heat spreader



(b) 2D drawing of the A5083 Jetson plate

Figure 4.17 2D drawings of our CNC-machined components

4.4.2 Modal analysis

From modal analysis, we retrieved the natural frequencies, the cumulative effective mass fractions, and the mode shapes. We performed six modal analyses for each configuration. The resulting natural frequencies and mode shapes across all runs per configuration were nearly identical, differing only at the decimal level, confirming that the directional quasi-static load does not impose a stress-stiffening asymmetry in the modal response.

Consequently, in this results section, we considered only one run per configuration for modal results instead of retrieving 6 depending on the axial acceleration direction, as we did for the static structure.

This meant we performed only one random vibration analysis considering PSD-G in the X-, Y-, and Z-axial directions and one harmonic response for each configuration by applying a sine sweep in the X-, Y-, and Z-axes.

In the 5 configurations, we set the search up to 30 modes in the 0-2000 Hz range. The results show that the structure has fewer than 30 modes below 2000 Hz, as the launcher SSMS VEGA-C User's manual specifies 2000 Hz as the limit for PSD-G loads for small spacecraft. Hence, capturing modes up to 2000 Hz is sufficient to capture dynamically relevant behavior under VEGA-C launch conditions. Notably, the first natural frequencies in each configuration exceeded the 115 Hz limit, indicating no dynamic cou-

pling between the spacecraft and the launcher. In this table, we arranged the frequencies found in each configuration (table 4.11)

Table 4.11 Natural frequencies (f_n in Hz) for all configurations in the range of 0–2000 Hz

Mode	Configuration 1	Configuration 2	Configuration 3	Configuration 4	Configuration 5
1	244.48	277.29	808.76	848.62	872.0
2	254.59	300.18	973.27	1110.6	1073.3
3	558.48	609.34	1080.5	1144.7	1104.1
4	1167.8	1174.0	1289.3	1435.8	1358.2
5	1136.8	1281.3	1521.6	1480.7	1633.6
6	1379.7	1414.6	1815.3	1856.1	1891.8
7	1386.8	1457.8	1966.9	1928.5	1912.8
8	1596.8	1582.8	–	–	–
9	1987.0	1886.0	–	–	–

The results (Table 4.11) show that the first frequency in each configuration exceeds 115 Hz stated by the SSMS manual, which prevents resonance failure.

4.4.3 Modal participation factor and effective mass

To evaluate the structure's dynamic response, we retrieved these results for the X, Y, and Z axes. In our SSMS manual, they didn't require retrieving the participation factor for the rotational degrees of freedom RX, RY, and RZ, since they specify only a 115 Hz requirement in the linear orthogonal axes. Consequently, we omitted them from our Modal analysis results section.

From the ANSYS participation factor calculation tables, we considered the frequency, cumulative mass fraction, and ratio of effective mass to total mass for each mode (tables 4.12 4.13 4.14 4.15 4.16).

The ratio of effective mass to total mass indicates the participation of mode i's effective mass relative to the structure's total mass. However, the cumulative mass fraction is normalized across the extracted modes, that's why it reaches 1 (100%) regardless of whether we extracted enough modes in an adequate range.

We also retrieved the total deformation of the most dominant mode shapes in each configuration. The remaining plots are shown in the Appendix [A].

The retrieved mode shapes are multiplied by a stochastic factor, so the magnitudes shown in the plots do not reflect a real physical displacement. Instead, it illustrates the deformation pattern [13].

Configuration 1

In configuration 1, the initial 2 modes, 244.48 Hz and 254.59 Hz, are the fundamental lateral modes in the X- and Y-axes, with 63.6% and 64.4% of the total mass, respectively. The dominant axial mode in the Z-axis occurs at 1368.82 Hz with 91.5% effective mass. The cumulative effective mass reaches 76.41%, 76.85%, and 94.65% in the X-, Y-, and Z-axes, respectively, of the total mass (table 4.12).

Table 4.12 Participation factor calculation in X, Y, and Z directions for configuration 1

Mode	Frequency (Hz)	Cumulative mass fraction			Ratio eff. mass / total mass		
		X	Y	Z	X	Y	Z
1	244.48	0.83	4.08E-02	5.37E-05	0.64	3.13E-02	5.09E-05
2	254.59	0.87	0.88	1.02E-04	3.12E-02	0.64	4.53E-05
3	558.48	0.87	0.88	1.07E-04	3.37E-05	1.60E-03	4.74E-05
4	1167.82	0.92	0.88	8.65E-02	3.41E-01	2.97E-06	8.09E-02
5	1368.82	0.92	0.88	0.97	3.40E-02	4.32E-04	0.91
6	1379.66	1.00	0.88	0.99	5.86E-02	9.40E-04	1.15E-02
7	1386.81	1.00	0.97	1.00	5.64E-04	0.65	1.21E-02
8	1596.76	1.00	1.00	1.00	1.54E-04	2.34E-02	7.69E-05
9	1986.96	1.00	1.00	1.00	1.62E-05	3.26E-03	5.57E-05
	Sum				**0.76**	**0.77**	**0.95**

In this configuration and across all the axes X, Y, and Z, mode 5 with frequency 1368.82 Hz is the most dominant, giving the following mode shape (figure 4.18)

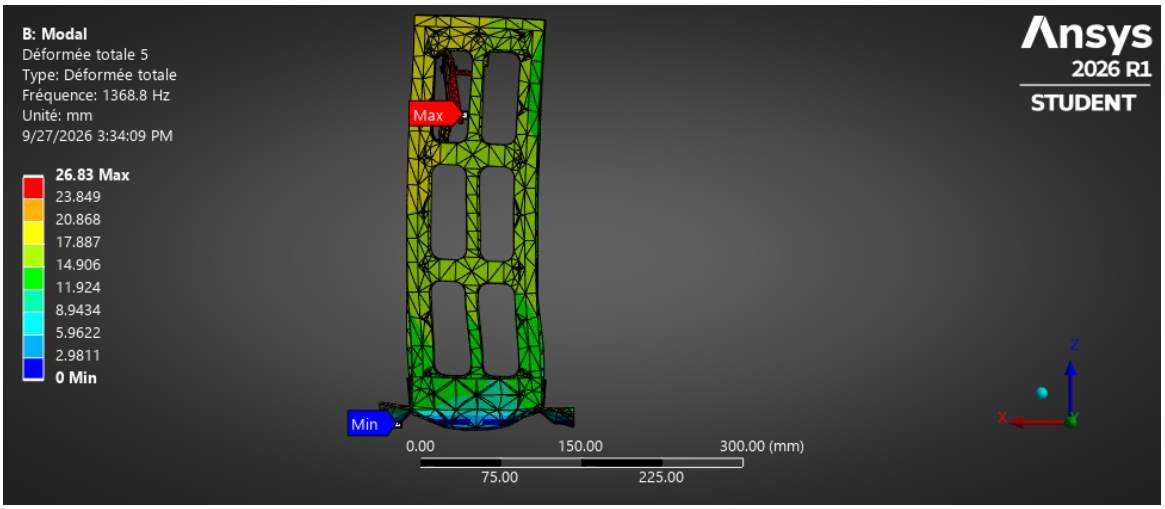


Figure 4.18 Total deformation in mode 5 at the frequency 1368.82 Hz for the CubeSat structure in configuration 1

This plot (figure 4.18) indicates that the displacement is concentrated at the heat spreader and the Jetson plate proximity.

Configuration 2

In configuration 2, the initial 2 modes, 277.29 Hz and 300.18 Hz, are the fundamental lateral modes in the X- and Y-axes, with 57.17% and 54.53% of the total mass, respectively. Whereas the dominant axial mode in the Z-axis occurs at a frequency of 1457.84 Hz with 94.68% effective mass. The cumulative effective mass reaches 74.44%, 72.22%, and 97.29% in the X-, Y-, and Z-axes, respectively, of the total mass (table 4.13).

Table 4.13 Configuration 2 participation factor calculation in X, Y, and Z directions

Mode	Frequency (Hz)	Cumulative mass fraction			Ratio eff. mass / total mass		
		X	Y	Z	X	Y	Z
1	277.29	0.77	9.07E-03	7.05E-04	0.57	6.55E-03	6.86E-04
2	300.18	0.78	0.76	1.35E-03	6.73E-03	0.55	6.28E-04
3	609.35	0.78	0.77	1.36E-03	1.80E-05	1.09E-02	1.33E-05
4	1173.94	0.88	0.77	4.55E-02	8.01E-02	2.55E-06	4.29E-02
5	1281.34	1.00	0.77	1.48E-01	8.58E-02	4.98E-06	1.00E-02
6	1414.61	1.00	0.87	2.52E-01	1.05E-05	7.50E-02	1.01E-02
7	1457.84	1.00	0.87	1.00	5.50E-07	4.37E-04	0.95
8	1582.77	1.00	0.97	1.00	3.08E-04	7.55E-02	1.47E-03
9	1886.00	1.00	1.00	1.00	1.63E-05	1.83E-02	1.18E-05
	Sum				**0.74**	**0.72**	**0.97**

In this configuration and across all the axes X, Y, and Z, mode 7 with frequency 1457.84 Hz is the most dominant, giving the following mode shape (figure 4.19)

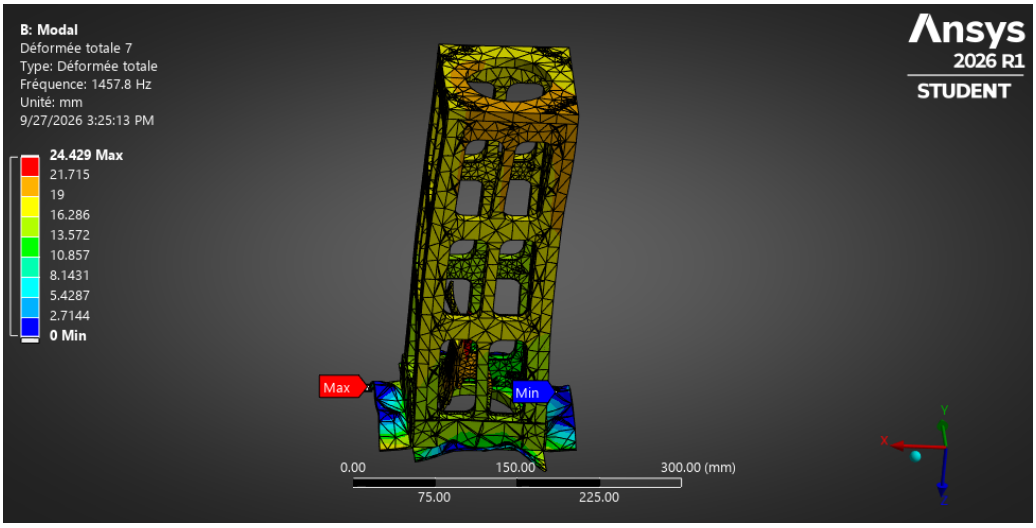


Figure 4.19 Total deformation in mode 7 at the frequency 1457.84 Hz for the CubeSat structure in configuration 2

This plot (figure 4.19) indicates that the displacement is concentrated at the Support location.

Configuration 3

In configuration 3, modes 1 and 5, with frequencies of 808.76 Hz and 1521.56 Hz, respectively, are the fundamental lateral modes in the X- and Y-axes, with 58.13% and 61.99% of the total mass, respectively. Whereas the dominant axial mode in the Z-axis occurs at a frequency of 973.275 Hz with 53.50% effective mass. The cumulative effective mass reaches 59.16%, 83.21%, and 61.68% in the X-, Y-, and Z-axes, respectively, of the total mass (table 4.14).

Table 4.14 Configuration 3 participation factor calculation (X, Y, Z directions)

Mode	Frequency (Hz)	Cumulative mass fraction			Ratio eff. mass / total mass		
		X	Y	Z	X	Y	Z
1	808.76	0.98	5.60E-06	1.45E-03	0.58	4.67E-06	8.93E-04
2	973.28	0.98	3.64E-03	0.87	8.49E-04	3.02E-02	0.54
3	1080.54	0.98	0.25	0.88	1.49E-04	0.21	6.82E-03
4	1289.35	1.00	0.25	0.88	9.30E-03	3.33E-04	2.87E-04
5	1521.56	1.00	1.00	0.88	3.12E-05	0.62	4.17E-06
6	1815.33	1.00	1.00	1.00	9.13E-06	4.15E-05	6.89E-02
7	1966.85	1.00	1.00	1.00	2.31E-05	6.81E-04	4.91E-03
Sum					**0.59**	**0.83**	**0.62**

In this configuration and across all the axes X, Y, and Z, mode 5 with frequency 1521.56 Hz is the most dominant, giving the following mode shape (figure 4.20):

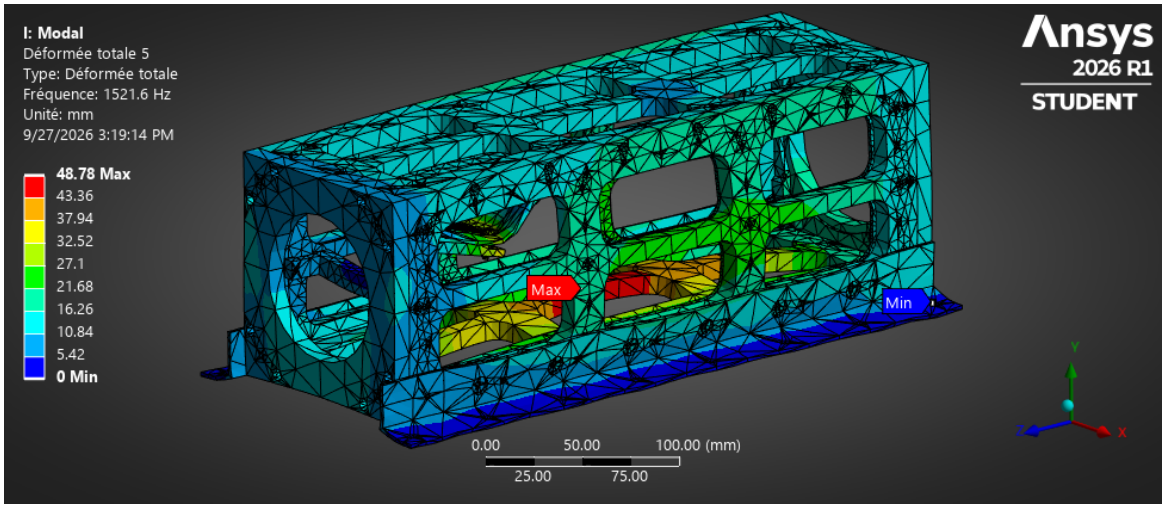


Figure 4.20 Total deformation in mode 5 at the frequency 1521.56 Hz for the CubeSat structure in configuration 3

This plot (figure 4.20) indicates that the displacement is concentrated at the lower frame near the L-bracket supports.

Configuration 4

In configuration 4, modes 1 and 5, with frequencies of 808.76 Hz and 1521.56 Hz, respectively, are the fundamental lateral modes in the X- and Y-axes, with 55.68% and 63.06% of the total mass, respectively. Whereas the dominant axial mode in Z-axis occurs at the second mode and a frequency of 1110.60 Hz with 43.74% effective mass. The cumulative effective mass reaches 56.46%, 86.09%, and 55.51% in the X-, Y-, and Z-axes, respectively, of the total mass (table 4.15).

Table 4.15 Configuration 4 participation factor calculation (X, Y, Z directions)

Mode	Frequency (Hz)	Cumulative mass fraction			Ratio eff. mass / total mass		
		X	Y	Z	X	Y	Z
1	848.61	0.99	1.56E-03	7.17E-03	0.56	1.34E-03	3.98E-03
2	1110.60	1.00	8.75E-03	0.80	2.27E-03	6.19E-03	0.44
3	1144.67	1.00	8.98E-02	0.80	2.87E-06	2.03E-03	4.50E-02
4	1435.78	1.00	0.23	0.80	8.21E-04	0.19	4.19E-04
5	1480.74	1.00	0.96	0.83	7.57E-04	0.63	0.13
6	1856.13	1.00	1.00	0.98	4.82E-03	3.28E-02	8.31E-02
7	1928.49	1.00	1.00	1.00	3.51E-02	1.74E-04	1.28E-02
Sum					**0.56**	**0.86**	**0.56**

In this configuration and across all the axes X, Y, and Z, mode 5 with frequency 1480.74 Hz is the most dominant, giving the following mode shape (figure 4.21):

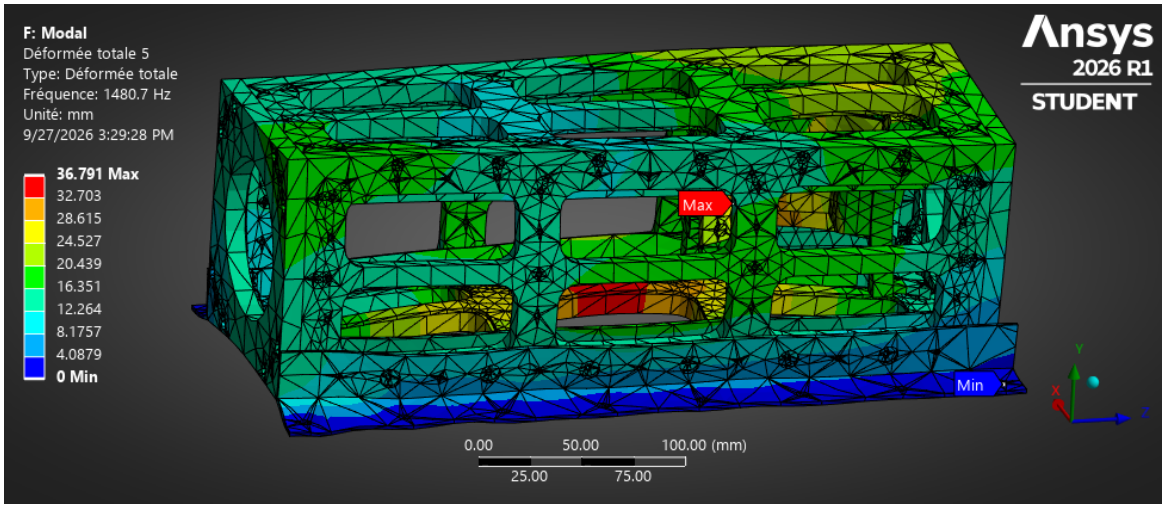


Figure 4.21 Total deformation in mode 5 at the frequency 1480.74 Hz for the CubeSat structure in configuration 4

This plot (figure 4.21) indicates that the displacement is concentrated at the lower frame near the L-bracket supports.

Configuration 5

In configuration 5, modes 1 and 5, with frequencies of 871.996 Hz and 1633.60 Hz, respectively, are the fundamental lateral modes in the X- and Y-axes, with 49.47% and 65.55% of the total mass, respectively. Whereas the dominant axial mode in the Z-axis occurs at the second mode and has a frequency of 1073.28 Hz with 35.59% effective mass. The cumulative effective mass reaches 56.46%, 86.09%, and 55.51% in the X-, Y-, and Z-axes, respectively, of the total mass (Table 4.16).

Table 4.16 Configuration 5 Participation factor calculation (X, Y, Z directions)

Mode	Frequency (Hz)	Cumulative mass fraction			Ratio eff. mass / total mass		
		X	Y	Z	X	Y	Z
1	871.996	0.95	9.96E-07	9.44E-04	0.49	8.29E-07	4.93E-04
2	1073.28	0.95	6.43E-02	0.68	3.26E-04	5.35E-02	0.36
3	1104.07	0.95	0.19	0.91	1.50E-04	0.11	0.12
4	1358.25	0.96	0.19	0.91	4.80E-03	5.61E-04	2.33E-04
5	1633.60	0.96	0.98	0.92	3.36E-06	0.66	6.71E-02
6	1891.78	1.00	0.98	0.93	1.96E-02	2.58E-03	4.50E-03
7	1912.84	1.00	1.00	1.00	1.43E-02	1.63E-02	0.36
Sum					**0.52**	**0.83**	**0.52**

In this configuration and across all the axes X, Y, and Z, mode 5 with frequency 1633.60 Hz is the most dominant, giving the following mode shape (figure 4.22):

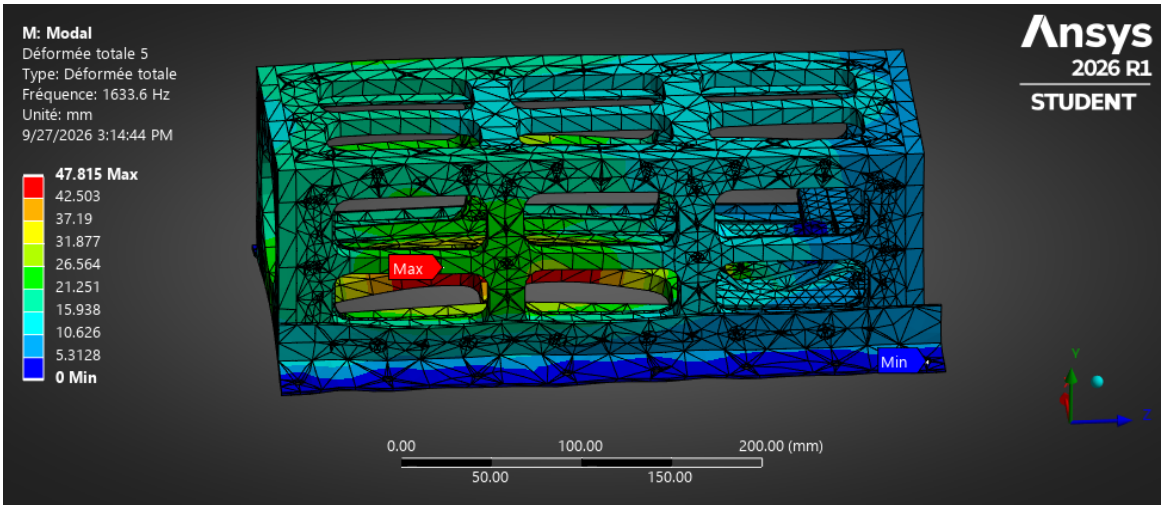


Figure 4.22 Total deformation in mode 5 at the frequency 1633.60 Hz for the CubeSat structure in configuration 5

This plot (figure 4.22)indicates that the displacement is concentrated at the lower frame near the L-bracket supports.

In all 5 configurations, the cumulative effective mass in the X-, Y-, and Z-axes didn't reach 85–90%, as stated in common guidelines [77]. In this case, ANSYS recommends increasing the number of modes to capture additional mass participation, which is not our case, as we retrieved 30 modes and they were not completely covered in the 0-2000 Hz [6]. The remaining mass is associated with modes above 2000 Hz.

Nevertheless, NASA-STD-5002B states that: "The total modal effective mass for a direction is the sum of all contributions of all modes in that direction. For typical payload/spacecraft, these modal effective mass totals are often above 80 percent, with a goal of being greater than 90 percent. For smaller, stiffer payload/spacecraft, the totals can be significantly less or even zero if no modes are retained" [21].

4.4.4 Random vibrations

In this section, we retrieved the maximum 3-sigma Von Mises stresses and maximum 3-Sigma deformations, which estimate the peak response expected during random vibrations of the Heat spreader and the Jetson plate.

Table 4.17 First Configuration's random vibration results (3 Sigma Von Mises stress and directional deformations) after applying PSD-G in X-, Y-, and Z- axes

	Heat Spreader				Jetson plate			
	stress [MPa]	Directional deformation [mm]			stress [MPa]	Directional deformation [mm]		
		X	Y	Z		X	Y	Z
X axis	58.17	0.73	0.19	0.03	65.61	0.71	0.19	0.08
Y axis	28.31	0.20	0.68	0.07	65.90	0.20	0.66	0.08
Z axis	27.99	0.04	0.01	0.05	47.06	0.008	0.006	0.04

In the **first** configuration (table 4.17), the peak 3-sigma stress of the heat spreader was 58.17 MPa under PSD-G on the X-axis, with a safety factor of 3.09 relative to the yield strength of 180 MPa. However, for the Jetson plate, the peak 3-sigma stress occurred under PSD-G on the Y-axis with a value of 65.90 MPa, with a safety margin of 3.11 relative to the 205 MPa yield strength. In configuration 1, the maximum 3-sigma deformations of the heat spreader and the Jetson plate were 0.73 mm and 0.71 mm, respectively, in the X direction under PSD-G applied in the X-axis.

Table 4.18 Second Configuration's random vibration results (3-sigma Von Mises stress and directional deformations) after applying PSD-G in X-, Y-, and Z- axes

	Heat Spreader				Jetson plate			
	stress [MPa]	Directional deformation [mm]			stress [MPa]	Directional deformation [mm]		
		X	Y	Z		X	Y	Z
X axis	80.75	0.16	0.02	0.02	129.54	0.24	0.03	0.06
Y axis	48.60	0.03	0.14	0.06	101.48	0.04	0.18	0.08
Z axis	31.85	0.03	0.004	0.04	32.91	0.01	0.002	0.03

In the **second** configuration (table 4.18), the peak 3-sigma stress of the heat spreader and the Jetson plate was 80.75 MPa and 129.54 MPa, with a safety factor of 2.23 and 1.58 relative to the yield strength of 180 MPa and 205 MPa, respectively, under PSD-G in the X-axis.

The maximum 3-sigma deformation for the heat spreader and the Jetson plate both occur under PSD-G in the X-axis, with deformations of 0.16 mm and 0.24 mm, respectively, in the X-axis direction.

Table 4.19 Third Configuration's random vibration results (3-sigma Von Mises stress and directional deformations) after applying PSD-G in X-, Y-, and Z- axes

	Heat Spreader				Jetson plate			
	stress [MPa]	Directional deformation [mm]			stress [MPa]	Directional deformation [mm]		
		X	Y	Z		X	Y	Z
X axis	44.95	0.10	0.03	0.01	9.62	0.11	0.02	0.01
Y axis	77.78	0.01	0.13	0.02	25.23	0.001	0.02	0.01
Z axis	32.36	0.01	0.07	0.07	17.38	0.01	0.02	0.06

In the **third** configuration (Table 4.19), the peak 3-sigma stress of the heat spreader and the Jetson plate was 77.78 MPa and 25.23 MPa, with a safety factor of 2.31 and 8.13 relative to the yield strength of 180 MPa and 205 MPa, respectively, under PSD-G in the Y-axis.

The maximum 3 Sigma deformation achieved for the heat spreader is 0.13 mm in the Y direction under PSD-G applied in the Y-axis. On the other hand, the Jetson case achieved a maximum 3-sigma directional deformation of 0.11 mm in the X-axis direction under PSD-G in the X-axis.

Table 4.20 Fourth Configuration's random vibration results (3-sigma Von Mises stress and directional deformations) after applying PSD-G in X-, Y-, and Z- axes

	Heat Spreader				Jetson plate			
	stress [MPa]	Directional deformation [mm]			stress [MPa]	Directional deformation [mm]		
		X	Y	Z		X	Y	Z
X axis	78.95	0.10	0.02	0.01	48.75	0.1	0.01	0.01
Y axis	43.24	0.05	0.04	0.01	68.21	0.02	0.03	0.01
Z axis	19.91	0.02	0.03	0.04	34.80	0.01	0.03	0.04

In the **fourth** configuration (table 4.20), the peak 3-sigma stress of the heat spreader was 78.95 MPa under PSD-G on the X-axis, with a safety factor of 2.28 relative to the yield strength of 180 MPa. However, for the Jetson plate, the peak 3-sigma stress occurred under PSD-G in the Y-axis with a value of 68.21 MPa, with a safety margin of 3.01 relative to the 205 MPa yield strength. In configuration 4, the maximum 3-sigma deformation achieved for the heat spreader is 0.10 mm in the X direction under PSD-G applied in the X-axis. On the other hand, the Jetson case achieved a maximum 3-sigma directional deformation of 0.1 mm in the X-axis direction under PSD-G in the X-axis.

Table 4.21 Fifth Configuration's random vibration results (3-sigma equivalent Von Mises stress and directional deformations) after applying PSD-G in X-, Y-, and Z- axes

	Heat Spreader				Jetson plate			
	stress [MPa]	Directional deformation [mm]			stress [MPa]	Directional deformation [mm]		
		X	Y	Z		X	Y	Z
X axis	40.45	0.04	0.01	0.003	7.88	0.001	0.01	0.001
Y axis	81.71	0.01	0.10	0.02	21.45	0.003	0.03	0.003
Z axis	78.05	0.01	0.11	0.04	19.61	0.001	0.01	0.01

In the **fifth** configuration (Table 4.21), the peak 3-sigma stress of the heat spreader and the Jetson plate was 81.71 MPa and 22.45 MPa, with safety factors of 2.20 and 9.13 relative to the yield strength of 180 MPa and 205 MPa, respectively, under PSD-G in the Y-axis. In configuration 5, the maximum 3-sigma deformation reached for the heat spreader is 0.11 mm in the Z direction under PSD-G applied in the Y-axis and Z-axis, respectively. On the other hand, the Jetson plate achieved a maximum 3-sigma directional deformation of 0.01 mm in the X- and Y-axis directions under PSD-G in the X-axis and in the Y- and Z-axis directions under PSD-G in the Z-axis, respectively.

In all configurations, the heat spreader and the Jetson plate 3-sigma stresses are below the yield strengths of 180 MPa and 205 MPa, respectively.

Under PSD-G excitations, we see 3 stress concentration mechanism patterns:

• **Mechanism 1**: **Free contact corner concentration**

This pattern is consistent with the free end or the bolted part of the heat spreader touching the CubeSat structure and/or the Jetson plate corner touching the Structure. Geometrically, a corner introduces a sudden change in contact geometry, leading to high stress accumulation[82]. Furthermore, at such points,

stress could be a singularity [34], which is worth investigating if it exceeds our margins, and it is mesh-dependent.

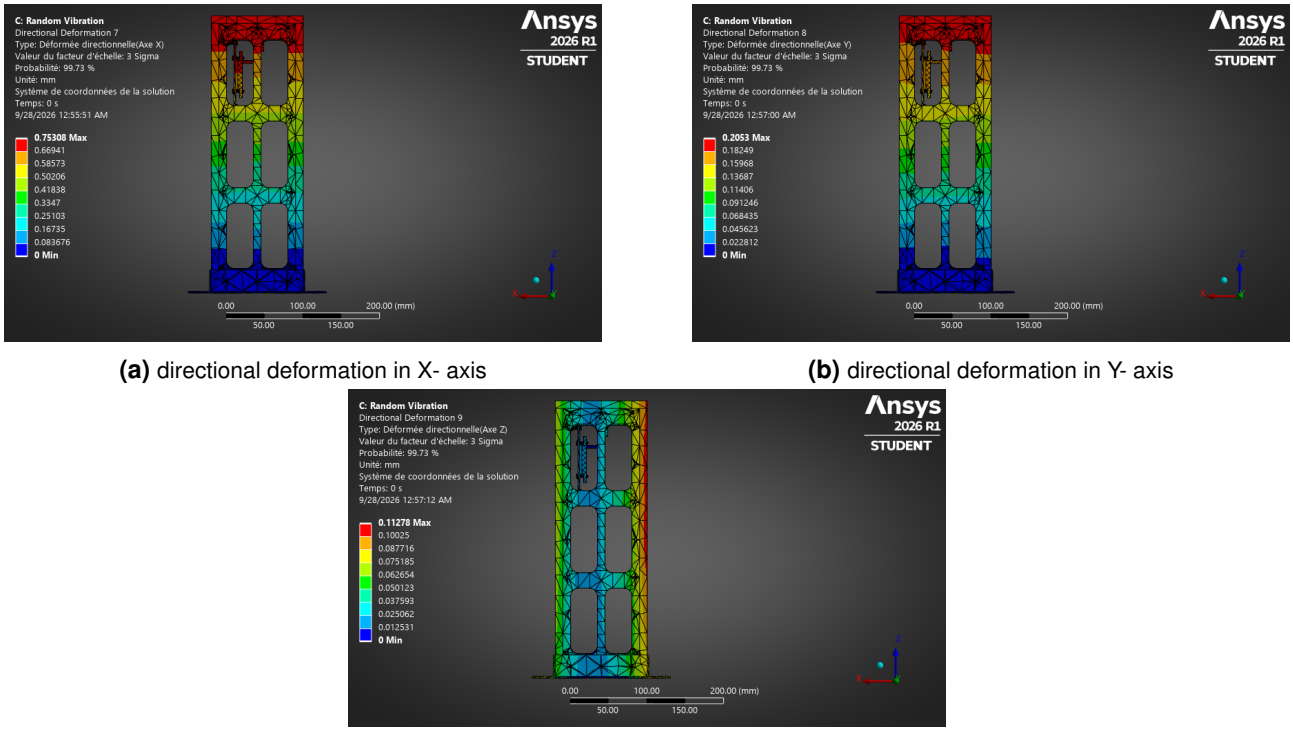
• **Mechanism 2**: **Spacer standoff (spacer acting as a lever arm)**

This causes a bending moment due to a force acting at a distance from the mounting point. In this case, the 3-Sigma Von Mises stress concentrates at the mounting point in the Jetson plate, since the spacer acts as a stiff lever arm that transmits stress to this point.

• **Mechanism 3**: **Geometric stress riser**

It occurs at the end corner of the L-bend and is present only in configuration 1 under PSD-G excitation in the Z-axis [33].

Regarding the 3-sigma directional deformation, the highest directional deformation occurs in configuration 1 in the X-axis direction under PSD-G excitation in the X-axis, with a value of 0.73 mm (figure 4.23).



(a) directional deformation in X- axis

(b) directional deformation in Y- axis

(c) directional deformation in Z- axis

Figure 4.23 3 Sigma directional deformation under PSD-G excitation in the X-axis direction in configuration 1

From the plots (Figure 4.23), the maximum free-end deflection in the X-axis is 0.75 mm, with margins of 0.02 mm and 0.04 mm for the heat spreader (0.73 mm) and the Jeston plate (0.71 mm), respectively. This indicates that deformation of the subassembly components is driven by rigid-body motion of the CubeSat tip rather than flexibility in both components. The same applies to the Y-axis directional deformation under PSD-X excitation.

For the 0.68 mm Y-axis deformation under PSD-G in the Y-axis, we follow the same methodology.

In configuration 2, the directional deformation is reduced as the subassembly moves closer to the support.

For all other configurations, the directional deformation was in the range 0.001–0.11 mm.

4.4.5 Harmonic response

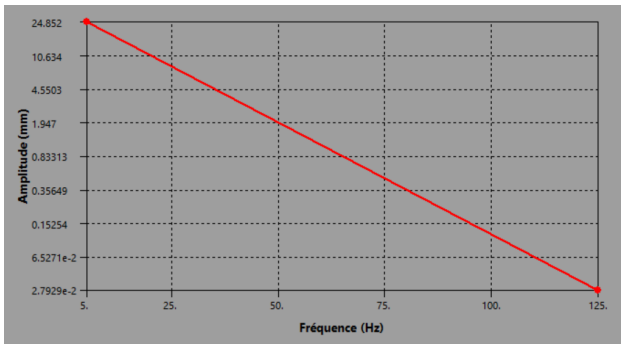
In the harmonic response, we applied sine sweep accelerations independently in the X-, Y-, and Z-axis directions. We retrieved the frequency-response directional deformation of the heat spreader and the Jetson plate, and the maximum over-frequency stresses for both. The directional deformation curves showed a linear decrease behavior for both components with different magnitudes.

Notably, the maximum directional deformation always occurred along the axis to which the sine sweep was applied.

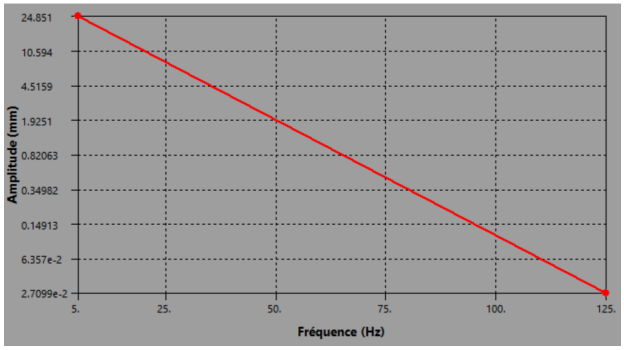
Note: In this section, for frequency-response directional deformation, we show only the plots for configuration 1 under a sine sweep along the X-axis, since all other configurations followed the same linear decreasing behavior with different amplitudes. We also didn't retrieve the phase angle plots because SSMS didn't define them as acceptance criteria for our components.

First configuration

First, we applied our sine sweep in the X- axis direction and we retrieved the directional deformation of the heat spreader and the Jetson case. We found that our maximum directional deformation occurs in the X-axis, with a value of 24.85 mm (Figure 4.24). The highest value of the directional deformation in the Y-axis for them are $1.78e^{-004}$ mm and $1.63e^{-004}$ mm (Figure 4.25), and they achieved values of $1.18e^{-004}$ mm and $9.88e^{-004}$ mm, respectively, in the Z-axis (Figure 4.26).

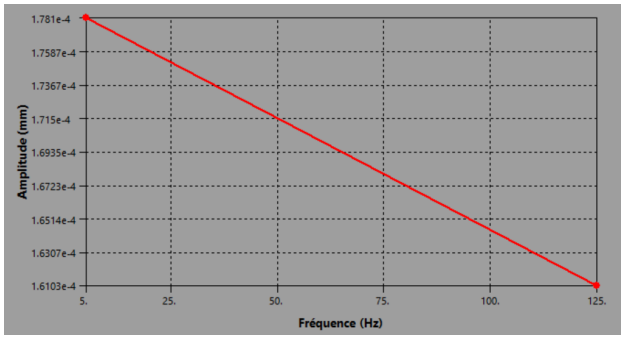


(a) Frequency response of the X-directional deformation of the Heat spreader after being subjected to a sine sweep acceleration in the X- axis direction

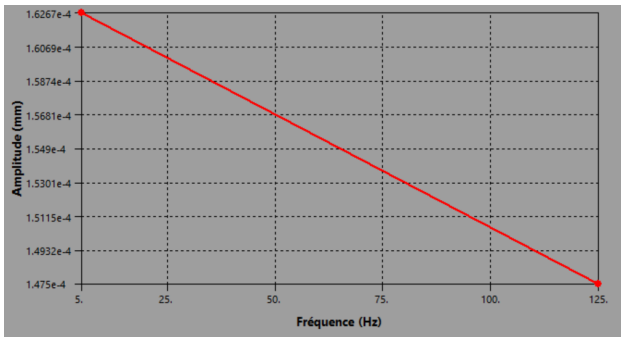


(b) Frequency response of the X-directional deformation of the Jetson plate after being subjected to a sine sweep acceleration in the X- axis direction

Figure 4.24 Frequency response of the X-directional deformation of the heat spreader Jetson plate after being subjected to a sine sweep acceleration in the X-axis direction in configuration 1

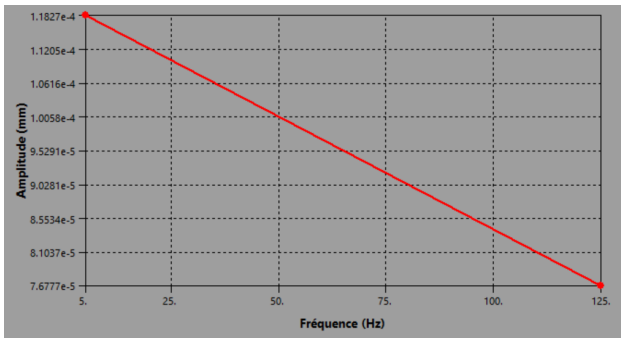


(a) Frequency response of the Y-directional deformation of the Heat spreader after being subjected to a sine sweep acceleration in the X-axis direction

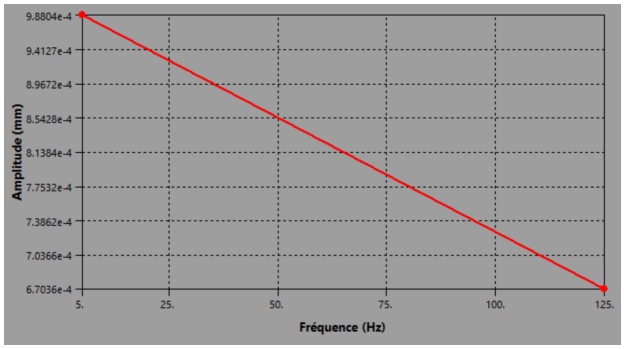


(b) Frequency response of the Y-directional deformation of the Jetson plate after being subjected to a sine sweep acceleration in the X-axis direction

Figure 4.25 Frequency response of the Y-directional deformation of the Heat spreader and the Jetson plate after being subjected to a sine sweep acceleration in the X-axis direction in configuration 1



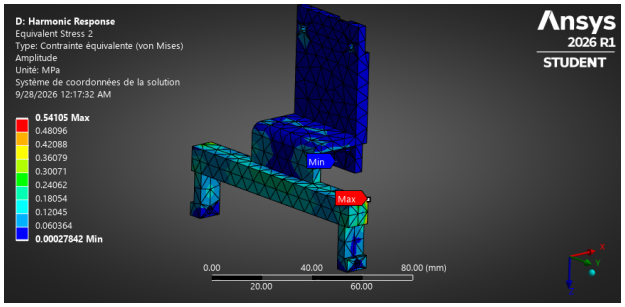
(a) Frequency response of the Z-directional deformation of the Heat spreader after being subjected to a sine sweep acceleration in the X-axis direction



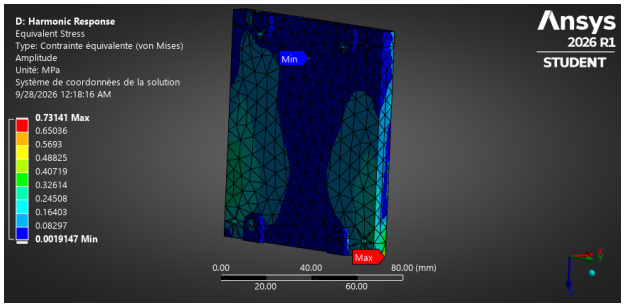
(b) Frequency response of the Z- directional deformation of the Jetson plate after being subjected to a sine sweep acceleration in the X- axis direction

Figure 4.26 Frequency response of the Z- directional deformation of the heat spreader and Jetson plate after being subjected to a sine sweep acceleration in the X-axis direction in configuration 1

For the maximum over-frequency stress in the 5-125 Hz range, the heat spreader and the Jetson plate remained well below the yield strength, with maximum over-frequency stresses of 0.54 MPa and 0.73 MPa, respectively (figure 4.27).



(a) Maximum over frequency equivalent von Mises stress of the Jetson plate

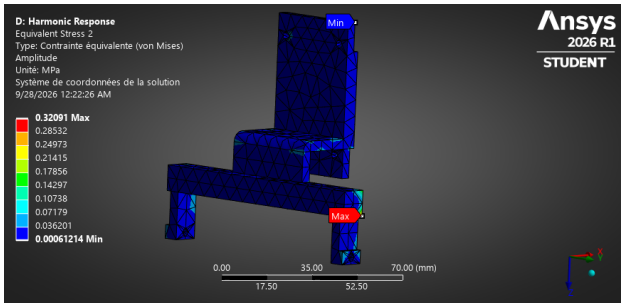


(b) Maximum over frequency equivalent von Mises stress of the heat spreader

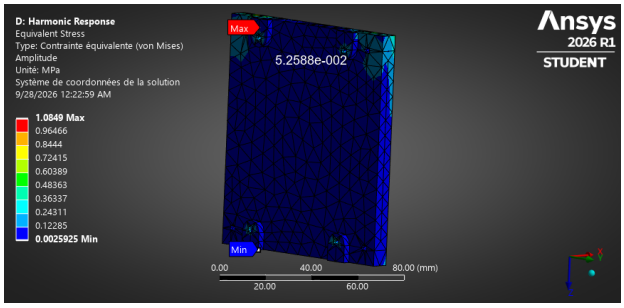
Figure 4.27 Harmonic response directional deformation of the jetson plate and the heat spreader under X- axis sine sweep in configuration 1

For the sine sweep applied along the Y-axis, the heat spreader and the Jetson plate showed a Y-directional deformation of 24.85 mm. The directional deformation in the X-axis for them is $1.87e^{-004}$ mm and $1.74e^{-004}$ mm, and they achieved values of $4.83e^{-005}$ mm and $1.16e^{-005}$ mm, respectively, in the Z-axis.

In this case, the maximum over frequency stress in the 5-125 Hz range for the heat spreader and the Jetson plate remained below the yield strength and achieved a maximum over frequency stress of 0.32 MPa and 1.08 MPa, respectively (figure 4.28).



(a) Maximum over frequency equivalent von Mises stress of the Jetson plate

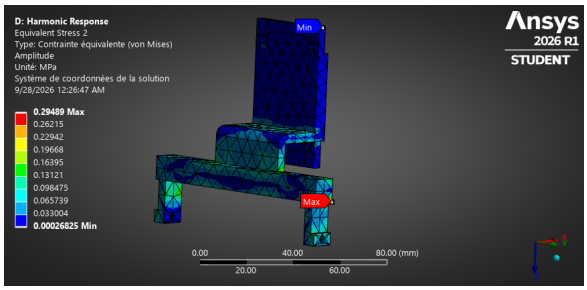


(b) Maximum over frequency equivalent von Mises stress of the heat spreader

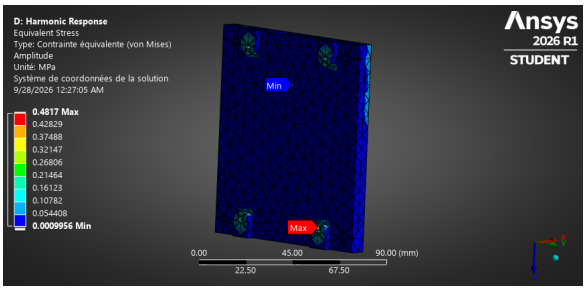
Figure 4.28 Harmonic response directional deformation of the jetson plate and the heat spreader under Y-axis sine sweep in configuration 1

For a sine sweep applied along the Z-axis, the heat spreader and the Jetson plate showed a Z-directional deformation of 24.84 mm. The directional deformation in the X-axis for them is $9.69e^{-005}$ mm and $1.09e^{-004}$ mm, and they achieved values of $5.95e^{-005}$ mm and $5.44e^{-005}$ mm, respectively, in the Y-axis.

Moreover, the maximum over-frequency stress in the 5-125 Hz range for the heat spreader and the Jetson plate remained below the yield strength and achieved a maximum over-frequency stress of 0.30 MPa and 0.48 MPa, respectively (Figure 4.29).



(a) Maximum over frequency equivalent von Mises stress of the Jetson plate

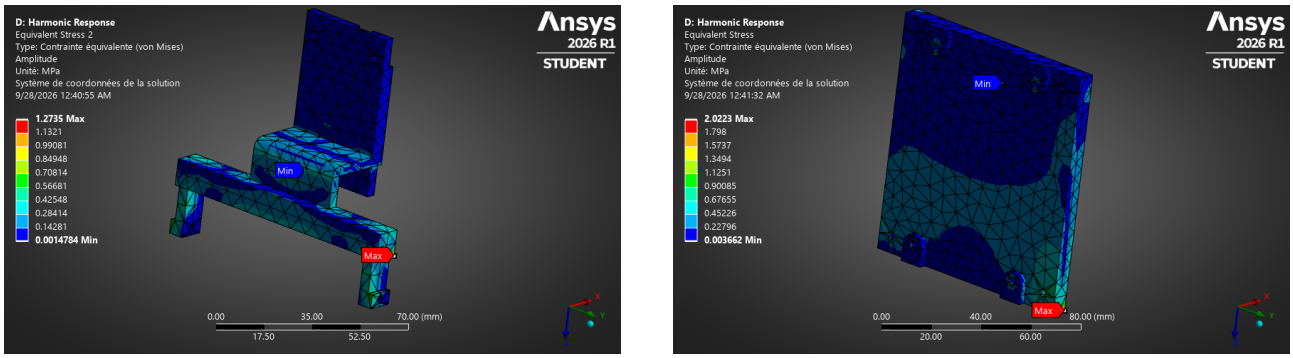


(b) Maximum over frequency equivalent von Mises stress of the heat spreader

Figure 4.29 Harmonic response directional deformation of the Jetson plate and the heat spreader under Z-axis sine sweep in configuration 1

Second configuration

For a sine sweep applied along the X-axis, the heat spreader and the Jetson plate showed an X-directional deformation of 24.84 mm. The directional deformation in the Y-axis for them is $3.67e^{-005}$ mm and $4.02e^{-005}$ mm, and they achieved values of $2.28e^{-005}$ mm and $6.74e^{-004}$ mm, respectively, in the Z-axis. For the maximum over-frequency stress in the 5-125 Hz range, the heat spreader and the Jetson plate remained below the yield strength, with maximum over-frequency stresses of 1.27 MPa and 2.02 MPa, respectively (figure 4.30).

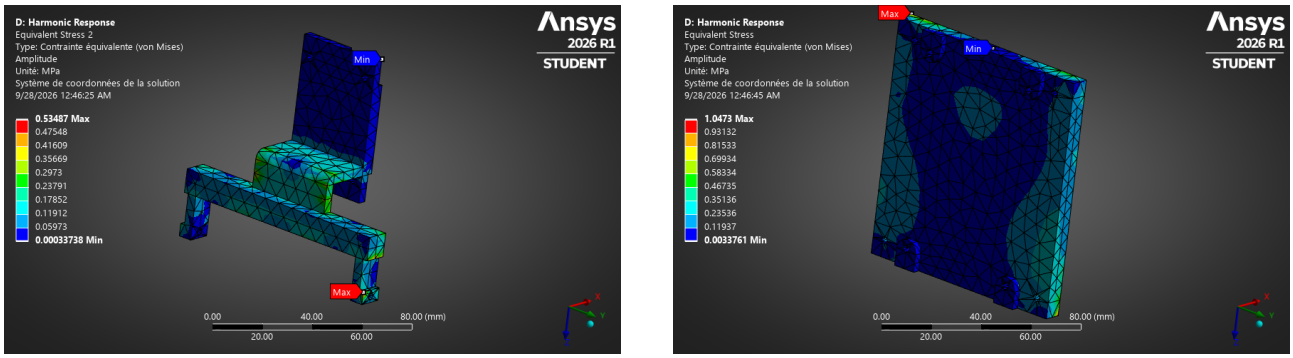


(a) Maximum over frequency equivalent von Mises stress of the Jetson plate

(b) Maximum over frequency equivalent von Mises stress of the heat spreader

Figure 4.30 Harmonic response directional deformation of the Jetson plate and the heat spreader under X-axis sine sweep in configuration 2

For the sine sweep applied along the Y-axis, the heat spreader and the Jetson plate showed a Y-directional deformation of 24.84 mm. The directional deformation in the X-axis for them is $4.39e^{-005}$ mm and 3.1 e-005 mm, and they achieved values of $2.87e^{-005}$ and $9.43e^{-006}$, respectively, in the Z-axis. In this case, the maximum over-frequency stress in the 5-125 Hz range for the heat spreader and the Jetson plate remained below the yield strength and achieved a maximum over-frequency stress of 0.53 MPa and 1.05 MPa, respectively (Figure 4.31).

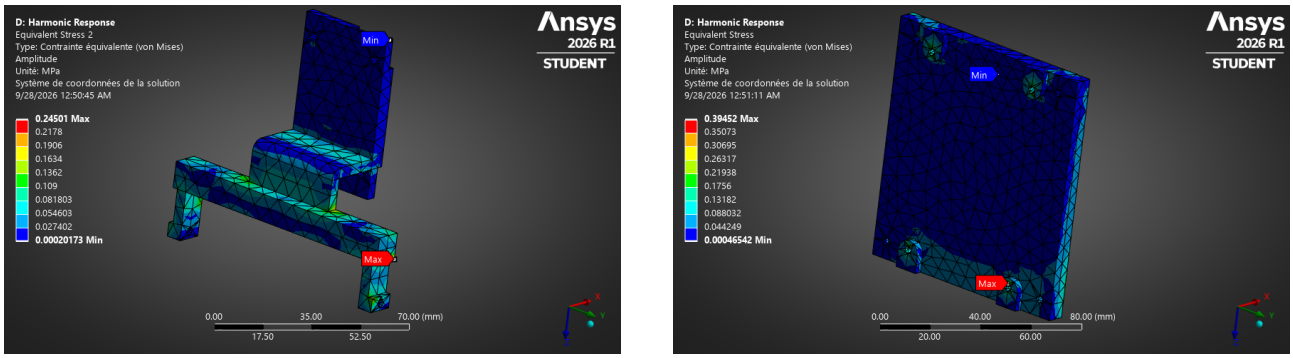


(a) Maximum over frequency equivalent von Mises stress of the Jetson plate

(b) Maximum over frequency equivalent von Mises stress of the heat spreader

Figure 4.31 Harmonic response directional deformation of the jetson plate and the heat spreader under Y-axis sine sweep in configuration 2

For a sine sweep applied along the Z-axis, the heat spreader and the Jetson plate showed a Z-directional deformation of 24.84 mm. The directional deformation in the X-axis for them is $3.72e^{-005}$ mm and $1.06e^{-005}$ mm, and they achieved values of $1.43e^{-005}$ and $1.20e^{-005}$ mm, respectively, in the Y-axis. Moreover, the maximum over-frequency stress in the 5-125 Hz range for the heat spreader and the Jetson plate remained below the yield strength and achieved a maximum over-frequency stress of 0.25 MPa and 0.39 MPa, respectively (Figure 4.32).



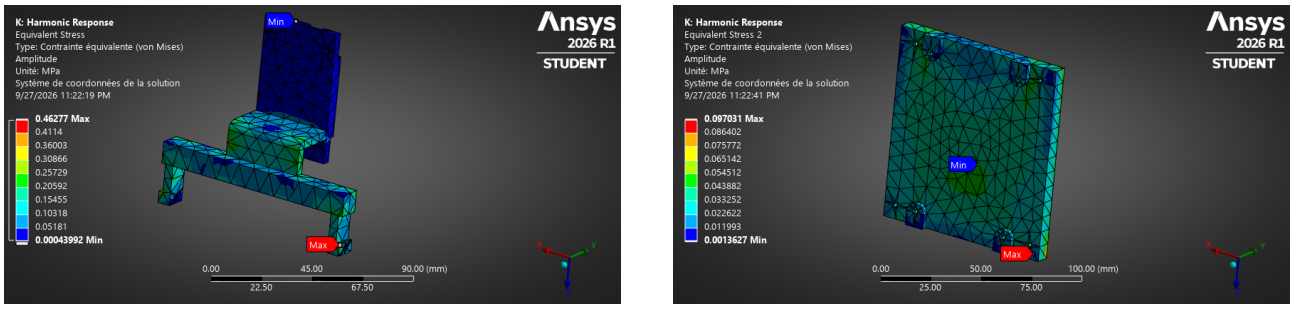
(a) Maximum over frequency equivalent von Mises stress of the Jetson plate

(b) Maximum over frequency equivalent von Mises stress of the heat spreader

Figure 4.32 Harmonic response directional deformation of the Jetson plate and the heat spreader under Z-axis sine sweep in configuration 2

Third Configuration

For a sine sweep applied along the X-axis, the heat spreader and the Jetson plate showed an X-directional deformation of 24.84 mm. The directional deformation in the Y-axis for them is $7.18e^{-006}$ mm and $2.75e^{-006}$ mm, and they achieved values of $7.76e^{-006}$ and $1.10e^{-005}$, respectively, in the Z-axis. Regarding the maximum over-frequency stress in the 5-125 Hz range, the heat spreader and the Jetson plate remained below the yield strength and achieved a maximum over-frequency stress of 0.46 MPa and 0.097 MPa, respectively (figure 4.33).

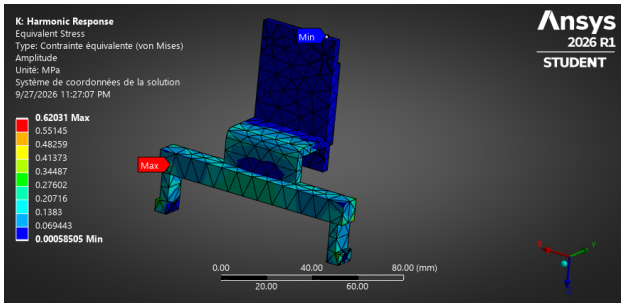


(a) Maximum over frequency equivalent von Mises stress of the Jetson plate

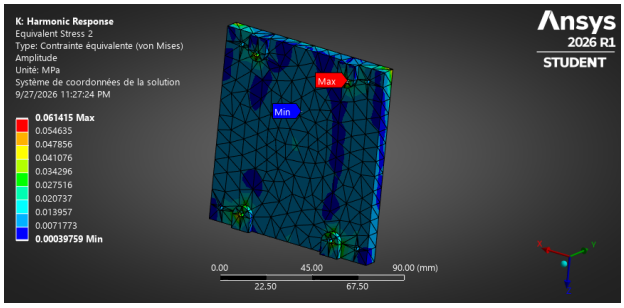
(b) Maximum over frequency equivalent von Mises stress of the heat spreader

Figure 4.33 Harmonic response directional deformation of the Jetson plate and the heat spreader under X- axis sine sweep in configuration 3

For a sine sweep applied along the Y-axis, the heat spreader and the Jetson plate showed a Y-directional deformation of 24.84 mm. The directional deformation in the X-axis for them is $4.46e^{-006}$ mm and $3.23e^{-006}$ mm, and they achieved values of $2.86e^{-005}$ and $1.19e^{-005,}$ respectively, in the Z-axis. Regarding the maximum over-frequency stress in the 5-125 Hz range, the heat spreader and the Jetson plate remained below the yield strength and achieved a maximum over-frequency stress of 0.62 MPa and 0.06 MPa, respectively (figure 4.34).



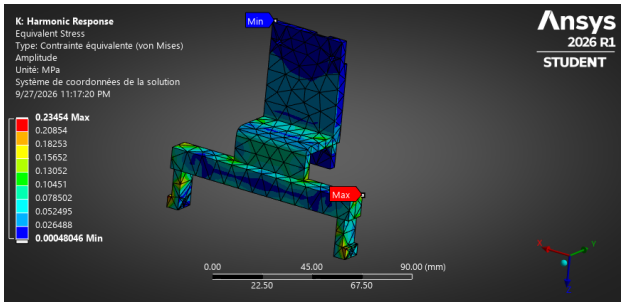
(a) Maximum over frequency equivalent von Mises stress of the Jetson plate



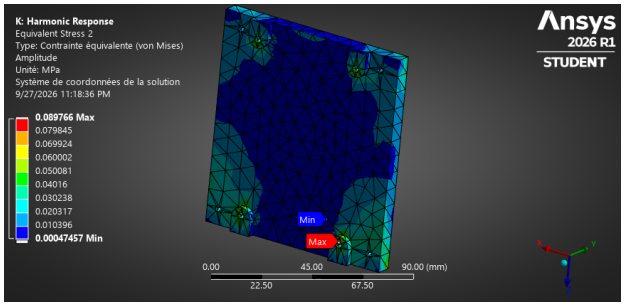
(b) Maximum over frequency equivalent von Mises stress of the heat spreader

Figure 4.34 Harmonic response directional deformation of the Jetson plate and the heat spreader under Y- axis sine sweep in configuration 3

For a sine sweep applied along the Z-axis, the heat spreader and the Jetson plate showed a Z-directional deformation of 24.84 mm. The directional deformation in the X-axis for them is $5.97e^{-006}$ mm and $1.02e^{-005}$ mm, and they achieved values of $2.98e^{-005}$ and $6.81e^{-005}$, respectively, in the Y-axis (figure 4.35).



(a) Maximum over frequency equivalent von Mises stress of the Jetson plate



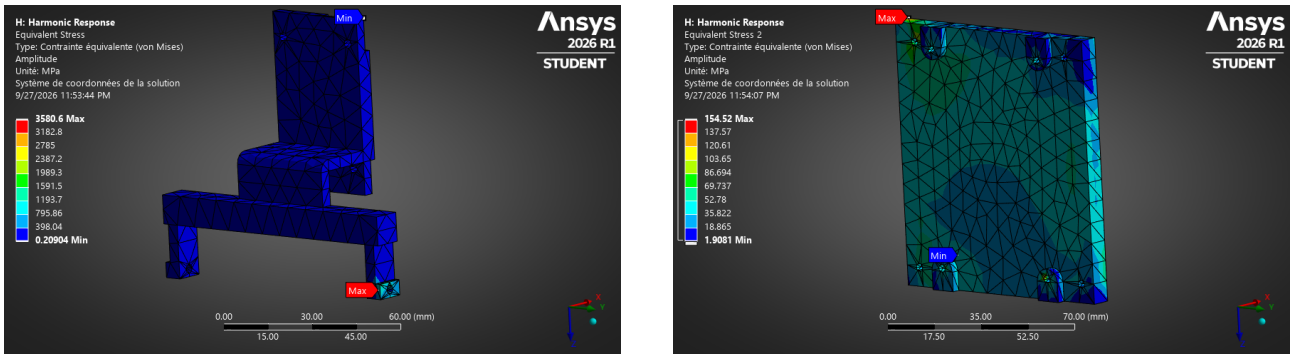
(b) Maximum over frequency equivalent von Mises stress of the heat spreader

Figure 4.35 Harmonic response maximum over-frequency Von Mises stress of the jetson plate and the heat spreader under Z-axis sine sweep in configuration 3

Regarding the maximum over-frequency stress in the 5-125 Hz range for the heat spreader and the Jetson plate, it remained below the yield strength and achieved a maximum over-frequency stress of 0.23 MPa and 0.09 MPa, respectively.

Fourth Configuration

For a sine sweep along the X-axis, the heat spreader and the Jetson plate showed X-directional deformations of 24.84 mm and 24.82 mm, respectively. The directional deformation in the Y-axis for them is 0.17 mm and $3.72e^{-002}$ mm, and they achieved values of $3.72e^{-002}$ mm and $8.36e^{-004}$ mm, respectively, in the Z-axis. Regarding the maximum over-frequency Von Mises stress in the 5-125 Hz range, the heat spreader and the Jetson plate achieved a maximum over-frequency stress of 3580.6 MPa and 154.52 MPa, respectively (Figure 4.36).



(a) Maximum over frequency equivalent von Mises stress of the Jetson plate

(b) Maximum over frequency equivalent von Mises stress of the heat spreader

Figure 4.36 Harmonic response maximum over-frequency Von Mises stress of the jetson plate and the heat spreader under X- axis sine sweep in configuration 4

Those high stresses are proven to be artifacts by retrieving the maximum over frequencies of normal and shear stresses and calculating the Von Mises stress analytically (table 4.22)

Table 4.22 Normal and shear stresses of the heat spreader and the Jetson plate under X-axis sine sweep in configuration 4

Maximum stress over	Normal stress			Shear stress		
frequencies	X	Y	Z	XY	YZ	XZ
Heat spreader	0.15	7.48	1.19	9.73	7.51	0.16
Jetson plate	0.76	1.01	2.82	0.87	0.65	0.22

Calculating the Von Mises stress analytically using the following formula (equation 4.4)

$$\sigma_{VM} = \sqrt{0.5\left[(\sigma_x - \sigma_y)^2 + (\sigma_y - \sigma_z)^2 + (\sigma_x - \sigma_z)^2\right] + 3\left(\tau_{xy}^2 + \tau_{yz}^2 + \tau_{xz}^2\right)} \tag{4.4}$$

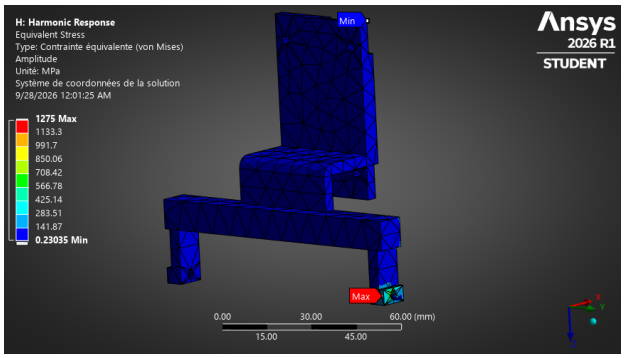
We got a maximum Von Mises stress of 22.37 MPa and 2.73 MPa for the heat spreader and the Jetson case, respectively (Table 4.23).

Table 4.23 Calculated Von Mises stress of the Heat spreader and the Jetson plate under X-axis sine sweep in configuration 4

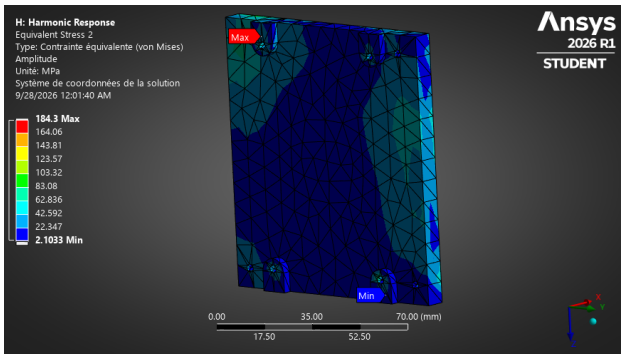
	σ_{VM} [MPa]
Heat spreader	22.37
Jetson plate	2.73

For a sine sweep applied along the Y-axis, the heat spreader and the Jetson plate showed Y-directional deformations of 24.78 mm and 24.85 mm, respectively. The directional deformation in the X-axis for them are 0.33 mm and 0.25 mm, and they achieved values of 0.15 mm and $9.53e^{-002}$ mm, respectively, in the

Z-axis. Regarding the maximum over-frequency stress in the 5-125 Hz range, the heat spreader and the Jetson plate remained below the yield strength and achieved a maximum over-frequency stress of 1275 MPa and 184.3 MPa, respectively (figure 4.37).



(a) Maximum over frequency equivalent von Mises stress of the Jetson plate



(b) Maximum over frequency equivalent von Mises stress of the heat spreader

Figure 4.37 Harmonic response maximum over-frequency Von Mises stress of the jetson plate and the heat spreader under Y- axis sine sweep in configuration 4

Following the same formula (equation 4.4), we retrieved the frequency response normal and shear stresses (table 4.24)

Table 4.24 Normal and shear stresses of the heat spreader and the Jetson plate under Y-axis sine sweep

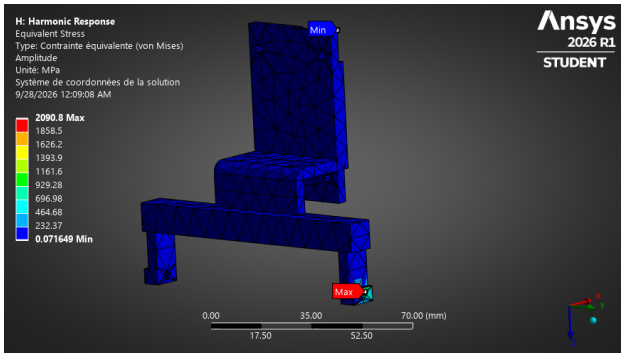
Maximum stress over	Normal stress [MPa]			Shear stress [MPa]		
frequencies	X	Y	Z	XY	YZ	XZ
Heat spreader	0.46	1.60	4.21	2.47	0.40	0.21
Jetson plate	0.87	8.01e−02	1.82	7.51e−02	4.68	0.33

Following the same formula (equation 4.4), we get a von Mises stress of 5.48 MPa and 8.27 MPa for the heat spreader and the Jetson plate, respectively. These values show a decent margin to the yield strength of both materials (table 4.25).

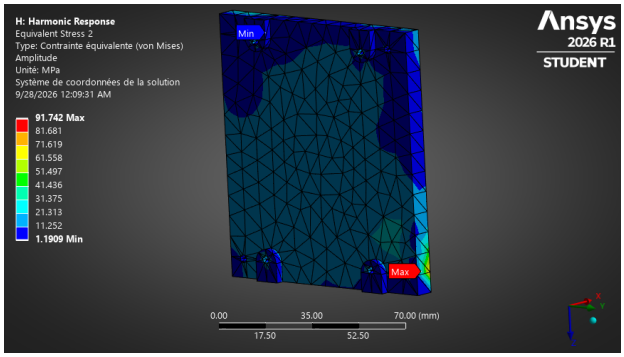
Table 4.25 Calculated maximum over-frequency Von Mises stress of the Heat spreader and the Jetson plate under Y-axis sine sweep

Component	σ_{VM}
Heat spreader	5.48
Jetson plate	8.27

For a sine sweep along the Z-axis, the heat spreader and Jetson plate showed Z-directional deformations of 24.82 mm and 24.84 mm, respectively. The directional deformation in the X-axis for them is $1.70e^{-002}$ mm and $2.12e^{-003}$ mm, and they achieved values of 1.01e-002 mm and 1.03e-002 mm, respectively, in the Y-axis. Moreover, the maximum over-frequency stress in the 5-125 Hz range for the heat spreader and the Jetson plate remained below the yield strength, reaching 2090.8 MPa and 91.74 MPa, respectively (Figure 4.38).



(a) Maximum over frequency equivalent von Mises stress of the Jetson plate



(b) Maximum over frequency equivalent von Mises stress of the heat spreader

Figure 4.38 Harmonic response maximum over-frequency Von Mises stress of the jetson plate and the heat spreader under Z-axis sine sweep in configuration 4

Following the same formula (equation 4.4), we retrieved the frequency response normal and shear stresses (table 4.26)

Table 4.26 Normal and shear stresses of the heat spreader and the Jetson plate under Z-axis sine sweep in configuration 4

Maximum stress over	Normal stress			Shear stress		
frequencies	X	Y	Z	XY	YZ	XZ
Heat spreader	3.64	7.37	5.70	6.57e−03	0.95	0.32
Jetson plate	0.27	1.83	0.70	6.22e−02	2.22	4.99e−02

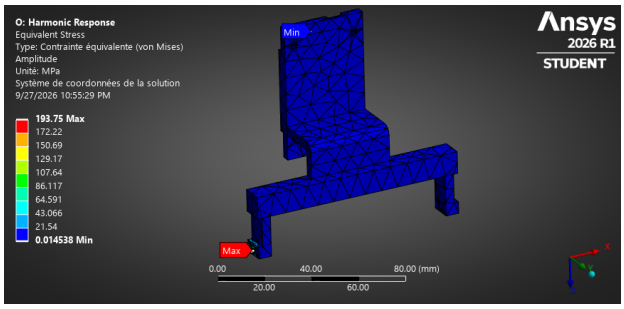
Table 4.27 Calculated maximum over-frequency von Mises stress of the Heat spreader and the Jetson plate in configuration 4 under Z-axis sine sweep

Component	σ_{VM}
Heat spreader	3.67
Jetson plate	4.09

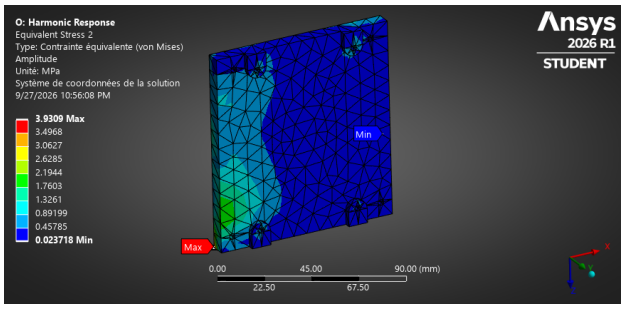
Following the same formula (equation 4.4), we get a von Mises stress of 3.67 MPa and 4.09 MPa for the heat spreader and the Jetson plate, respectively (table 4.27).

Fifth Configuration

For a sine sweep applied along the X-axis, the heat spreader and the Jetson plate showed an X-directional deformation of 24.84 mm. The directional deformation in the Y-axis for them is $4.77e^{-003}$ mm and $1.13e^{-003}$ mm, and they achieved values of $6.35e^{-003}$ mm and $1.08e^{-003}$ mm, respectively, in the Z-axis. For the maximum over-frequency stress in the 5-125 Hz range, the heat spreader and the Jetson plate remained below the yield strength, with maximum over-frequency Von Mises stresses of 193.75 MPa and 3.93 MPa, respectively (figure 4.39).



(a) Maximum over frequency equivalent von Mises stress of the Jetson plate



(b) Maximum over frequency equivalent von Mises stress of the heat spreader

Figure 4.39 Harmonic response maximum over-frequency Von Mises stress of the jetson plate and the heat spreader under X-axis sine sweep in configuration 5

We retrieved the frequency response normal and shear stresses (Table 4.28)

Table 4.28 Maximum over-frequency normal and shear stresses of the heat spreader and the Jetson plate under X-axis sine sweep in configuration 5

Maximum stress over frequencies	**Normal stress**			**Shear stress**		
	X	Y	Z	XY	YZ	XZ
Heat spreader	9.15e-02	1.00e-02	0.12	5.91e-03	1.53e-02	4.92e-02
Jetson plate	0.18	0.19	0.30	3.58e-02	9.63e-02	0.27

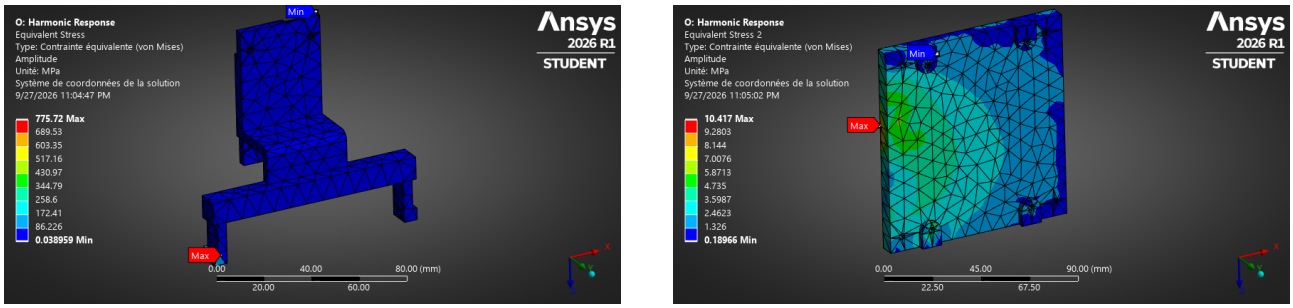
Following the same rule (equation 4.4), we get those values (table 4.29)

Table 4.29 Calculated maximum over-frequency Von Mises stress of the Heat spreader and the Jetson plate under X-axis sine sweep in configuration 5

Component	σ_{VM}
Heat spreader	0.14
Jetson plate	0.51

The von Mises stress in the Heat spreader and the Jetson plate is 0.14 MPa and 0.51 MPa, respectively, with a large margin relative to their yield strength (Table 4.29).

For a sine sweep along the Y-axis, the heat spreader and the Jetson plate showed Y-directional deformations of 24.88 mm and 24.86 mm, respectively. The directional deformation in the X-axis for them is $1.04e^{-002}$ mm and $9.04e^{-004}$ mm, and they achieved values of $3.17e^{-003}$ and $1.08e^{-003}$, respectively, in the Z-axis. Regarding the maximum over-frequency stress in the 5-125 Hz range, the heat spreader and the Jetson plate remained below the yield strength and achieved a maximum over-frequency stress of 775.72 MPa and 10.42 MPa, respectively (Figure 4.40)



(a) Maximum over frequency equivalent von Mises stress of the Jetson plate

(b) Maximum over frequency equivalent von Mises stress of the heat spreader

Figure 4.40 Harmonic response maximum over-frequency Von Mises stress of the jetson plate and the heat spreader under Y- axis sine sweep in configuration 5

We retrieved the frequency response of normal and shear stresses (Table 4.30)

Table 4.30 Maximum over-frequency normal and shear stresses of the heat spreader and the Jetson plate under Y-axis sine sweep in configuration 5

Maximum stress over frequencies	**Normal stress**			**Shear stress**		
	X	Y	Z	XY	YZ	XZ
Heat spreader	0.83	6.31e-02	2.70e-02	2.95e-02	2.22e-02	0.17
Jetson plate	0.50	0.19	0.93	1.68	0.19	0.25

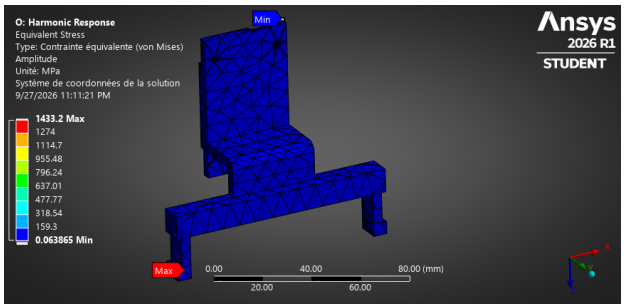
Following the same rule (equation 4.4), we get those values (table 4.31)

Table 4.31 Calculated maximum over-frequency Von Mises stress of the Heat spreader and the Jetson plate in configuration 5 under sine sweep in the Y-axis

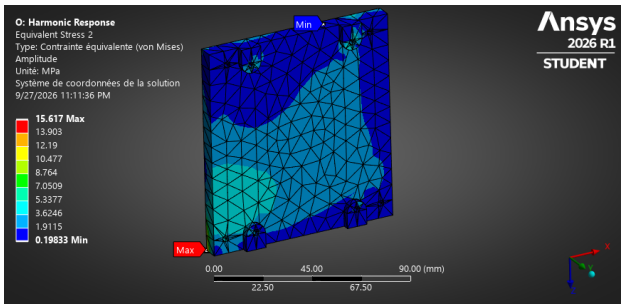
Component	σ_{VM}
Heat spreader	0.84
Jetson plate	3.03

The Heat spreader and the Jetson plate achieved a Von Mises stress of 0.84 MPa and 3.03 MPa, respectively, with a great margin relative to their yield strength (Table 4.31).

For a sine sweep along the Z-axis, the heat spreader and the Jetson plate showed Z-directional deformations of 24.82 mm and 24.84 mm, respectively. The directional deformation in the X-axis for them is $1.70e^{-002}$ mm and $2.12e^{-003}$ mm, and they achieved values of $1.01e^{-002}$ and $1.03e^{-002}$, respectively, in the Y-axis. Furthermore, the maximum over-frequency stress in the 5-125 Hz range for the heat spreader and the Jetson plate remained below the yield strength and achieved a maximum over-frequency stress of 1433.2 MPa and 15.62 MPa, respectively (Figure 4.41).



(a) Maximum over frequency equivalent von Mises stress of the Jetson plate



(b) Maximum over frequency equivalent von Mises stress of the heat spreader

Figure 4.41 Harmonic response maximum over-frequency Von Mises stress of the jetson plate and the heat spreader under Z- axis sine sweep in configuration 5

We retrieved the frequency response normal and shear stresses (Table 4.32)

Table 4.32 Maximum over-frequency normal and shear stresses of the heat spreader and the Jetson plate under Z-axis sine sweep in configuration 5

Maximum stress over frequencies	**Normal stress**			**Shear stress**		
	X	Y	Z	XY	YZ	XZ
Heat spreader	0.11	4.63e-04	0.48	2.57e-03	3.82e-02	0.50
Jetson plate	2.60	1.75	1.04	0.96	0.12	0.90

Table 4.33 Calculated maximum over-frequency von Mises stress of the Heat spreader and the Jetson plate in configuration 5 under Z-axis sine sweep

Component	σ_{VM}
Heat spreader	0.97
Jetson plate	2.65

The von Mises stress achieved by the Heat spreader and the Jetson plate is 0.97 MPa and 2.65 MPa, respectively, with a great margin relative to their yield strength (Table 4.33).

Remark

In all the configurations, the peak stress of the heat spreader occurred at the corner touching the 3U testing CubeSat structure. Whereas for the Jetson plate it occurred at the mounting position of the PCB and the corner touching the CubeSat structure as well.

Across all configurations, the modal analysis showed a minimum first natural frequency of 244 Hz. Harmonic response analysis in the range of 5-125 Hz showed a peak deformation at a very low frequency of 5 Hz in the direction of the applied acceleration. In the harmonic analysis

$$a = (2\pi f)^2 u \tag{4.5}$$

leading to

$$u = \frac{a}{(2\pi f)^2} \tag{4.6}$$

such that:

a: Acceleration

f: frequency

u: displacement

For an applied acceleration, the displacement u decreases as frequency increases. This explains the high value at the excitation axis level, since the acceleration drops from 2.5g to 1.25 g across the frequency range. This result does not reflect excessive elastic deformation in the evaluated parts. Instead, it is associated with the imposed base excitation. In other words, it reflects the whole assembly moving together, not local deformation.

Furthermore, the maximum stresses across frequencies that we retrieved directly from the simulations or calculated were below the yield strength of the different components' materials.

Analyzing across all the configurations, the worst-case Von Mises stress for the Cu 2.0060 heat spreader in the static structure analysis occurred in configuration 5, with a value of 111.91 MPa, and the lowest value was 59.004 MPa with a safety factor of 3.05 to the 180 MPa yield strength in configuration 4 in the (-Y)-axis launch direction, where the Jetson plate achieved a value of 12.74 MPa with a safety margin of 16.09 to the yield strength 205 MPa. In the same configuration and launch direction, the Cu 2.0060 heat spreader and the A5083 Jetson plate achieved maximum total deformations of 0.002 mm and 0.001 mm, respectively. At the 3-sigma directional deformation level, both components reached values between 0.01 mm and 0.1 mm. For random vibrations in the same configuration, the heat spreader achieved values between 19.91 MPa and 78.95 MPa. Meanwhile, the Jetson plate achieved values between 34.80 MPa and 68.21 MPa. In modal analysis, all configurations achieved frequencies above 115 Hz, and configuration 4 achieved a first frequency of 848.62 MPa.

Although configuration 4 performed better than the other configurations, we cannot state that the launch load on the (-Y) axis is the most favorable, because EventSat is a 6U CubeSat mission and 3U and 6U CubeSats differ in mass distribution. We ran our vibrational simulations on a 3U test CubeSat to examine the local behavior of the heat spreader and the Jetson plate, specifically their mounting positions and structural integrity.

5 Future work and Conclusion

As future work, we still need to conduct the vibrational experiments in the Oberpfaffenhofen facility, as operational problems prevented us from testing the manufactured components: the Cu 2.0060 heat spreader and A5083 Jetson plate.

Also, in vibrational analysis, we still need to perform a fatigue analysis under random vibrations to assess how long the components could survive. It is important to note, that we do not know the temper of the A5083 aluminum for the Jetson plate. Instead, we took a conservative value from an older thesis.

As a mechanical limitation which was due to the repeated mounting and dismounting of the heat spreader to the CubeSat structure, we observed thread wear at this position, while the threads remained intact at the mounting position of the heat spreader-NVIDIA Jetson PCB, which may be due to a poor finish at the damaged position. This condition introduced uncertainty in our experimental results because the heat spreader had poor contact with the CubeSat structure. A temporary solution is to enlarge the current holes, mount an M3 threaded hollow cylinder in the same positions, and secure it by soldering. For future applications, we recommend manufacturing the same model again, to avoid mounting and dismounting in future tests, and to prevent stress accumulation, it would be better to replace the sharp corners with fillets.

Also, it would be beneficial to repeat the Configuration A, B, and C experiments inside a high-vacuum chamber reaching a pressure of 10^{-7} Torr, since the BacoENG provides only a rough vacuum.

The Jetson plate was originally designed to be mounted in the 3U testing CubeSat structure. For this reason, we added threaded holes to fix it directly into the structure. However, in the 6U CubeSat, the Jetson plate mounting holes should not be threaded, as it will mount directly to the threaded rails.

The current mounting holes fit the 3U testing CubeSat structure made for vibrational testing.

Since the distance between the rails in the 3U testing CubeSat structure differs than the one in the 6U structure, it is possible to add 4 unthreaded holes to the Jetson plate without the need to manufacture a new one.

Another limitation is the lack of mechanical data for the NVIDIA Jetson Orin Nano, as it is not publicly available. As a result, we need to test the NVIDIA Jetson Orin Nano directly in Oberpfaffenhofen to assess its vibration resistance, even though one of the checked studies proved that the NVIDIA Jetson stayed intact under the vibrations [54].

Finally, in the thermal simulations, we did not consider the screws and the structural ring at the end of the rods in configuration C, which was used to fix the rod in the experiment, and we considered an average power from the performed experiments and an ambient temperature of 25 $^{\circ}C$. We also considered a TIM thickness of 0.5 mm, which might vary in real life. We retrieved the emissivities of the heat spreader and the Jetson plate by trial and error. In future experiments, it would be great to use an emissometer to assess the emissivity of both manufactured components, which would reduce simulation error compared to the experimental results. We also estimated the conductance values analytically. We applied the Yovanovich plastic contact correlation (equation 3.23) without validating it, making the assumed value uncertain. For the external natural convection, we used average heat transfer coefficients without resolving an airflow model and ignored the forced convection imposed by the A/C. We also approximated the cylindrical chamber wall as a vertical plate. This makes the convection values uncertain.

Despite the cited limitations, this thesis addressed the thermal and physical qualification of a thermal management system.

The choice was between Cu 2.0060 and Aluminum A5083. Thermal and vibrational simulations confirmed that Cu 2.0060 is more adequate for our case. We conducted ANSYS orbital simulations, which confirmed that using A5083 would violate the thermal requirements by exceeding the operating limit by 4.02 $^{\circ}C$ after adding the safety margin.

However, the Cu 2.0060 showed a maximum steady-state temperature of 45.48 $^{\circ}C$ even after adding the safety margin.

Experiments in the Vacuum chamber confirmed that the NVIDIA Jetson Orin Nano reaches its shutdown temperature of 104.5 $^{\circ}C$ within 11 minutes without a thermal management system.

Adding a conductive path from the NVIDIA Jetson Orin Nano to the CubeSat structure improved its behavior.

Instead of shutting down after 11 minutes, adding the 2 testing 3U CubeSat structures produced a steady-state temperature of 66.75 $^{\circ}C$ within 11 hours.

To mount the NVIDIA Jetson Orin Nano and the heat spreader subassembly to the CubeSat structure, we manufactured a Jetson plate to hold the subassembly.

From a structural perspective, the Cu 2.0060 and A5083 Jetson plate showed acceptable vibrational behavior across the five different configurations.

In static structural simulations, neither component exceeded its yield strength limit and showed small deformations in the order of decimals.

In modal analysis, we got a minimum first frequency across all configurations of 244.48 Hz, which is comfortably above the 115 Hz stated by the SSMS VEGA-C user's manual.

In random vibration analysis, we evaluated the 3 Sigma Von Mises stress, which remained below the yield strength of both components, and the 3 Sigma deformations were also very small, except for those moving with the CubeSat structure under the applied loads and they remained below 1 mm.

In harmonic response, we got high stress over frequencies in configurations 4 and 5, which turned out to be numerical artifacts and an overall acceptable results.

In this thesis, we designed, simulated, and manufactured a thermal management solution for the NVIDIA Jetson Orin Nano that kept the module below the 60 $^{\circ}C$ threshold under orbital conditions.

Combined simulations proved that Cu 2.0060 and A5083 are suitable for the heat spreader and the Jetson plate, which we made as a support for our NVIDIA Jetson Orin Nano-heat spreader subassembly, respectively.

Also, experiments inside the BACOeng vacuum chamber showed that adding conductive thermal paths to the CubeSat structure improved thermal performance compared with relying on the heat spreader alone.

From a structural perspective, vibrational simulations showed that the results met our SSMS VEGA-C mechanical requirements.

Based on the simulation results and as a practical milestone, the heat spreader enables the use of COTS in a CubeSat application and, eventually, new technology, namely the Event camera.

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